Overblik

- Hvad er intelligens? Hvad er kunstig intelligens?
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Emne 2: Vekselvirkninger mellem Datalogi, Kunstig Intelligens og Robotteknologi

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Kan maskiner tænke? Er de intelligente?

Robot - Thinking Man by Morpheus
(or Anatole Branch)

Rodin: Grubleren
(Carlsberg Glyptotek)

ChatGPT

- Lanceret 30. november 2022 af OpenAI
- Baseret på large language models (LLM'er)
- Blandt de 10 mest besøgte hjemmesider på verdensplan (juli 2024).
- Outsourcede kenyanske arbejdere tjente mindre end 2 dollars i timen for at mærke skadeligt indhold
- Svarene til 1000 prompts koster ca. 1DKK
- hver prompt bruger 10 gange så meget energi som en google search

ChatGPT

ChatGPT functions by generating the next most probable word to mimic human speech.

The collection of information the program is trained on, is a major issue.

In scanning data from across the internet, generative AI cannot discern what is true, false or even sarcastic — it simply uses what exists.

“Post-truth world”: Researchers found ChatGPT easily wrote out conspiracy theories

Quotes from: Alice Mei, The daily Illini, ChatGPT isn’t intelligent — can it really do your homework? 28.11.2024

https://dailyillini.com/news-stories/science-technology/2024/11/28/chatgpt-isnt-intelligent-homework/

By Tjeerd Royaards on 23.08.2016

Aftale om farver

Citater

Spørgsmål

Definitioner

Short Facts

Robot - Thinking Man by Morpheus
(or Anatole Branch)

Er ChatGPT intelligent?

Rodin: Grubleren
(Carlsberg Glyptotek)

Hvad er intelligens?

Evnen til at tilegne sig viden, opfatte og forstå sammenhænge mellem forskellige fænomener, tænke abstrakt og løse problemer

Den danske ordbog

DEN DANSKE ORDBOG
MODERNE DANSK SPROG

Hvad er intelligens?

- Ordet *intelligens* stammer fra … det latinske verbum intelligere som betyder "at forstå"
- Typiske betydninger omfatter evnen til at lære eller evnen til at løse nye opgaver uden forhåndsviden om fremgangsmåden
- Generelt dækker begrebet intelligens over evne til at reagere hensigtsmæssigt.

https://da.wikipedia.org/wiki/Intelligens

Hvad er kunstig intelligens?

Artificial intelligence (AI), in its broadest sense, is ***intelligence exhibited by machines***, particularly computer systems, as opposed to the natural intelligence of living beings. It is a field of research in computer science that develops and studies methods and software ***which enable machines to perceive their environment*** and uses learning and intelligence ***to take actions that maximize their chances of achieving defined goals***.

https://en.wikipedia.org/wiki/Artificial_intelligence

Robot - Thinking Man by Mrorpheus
(or Anatole Branch)

Svagheder af ChatGPT (I): ChatGPT kan ikke regne med semantiske enheder

from: 8 ChatGPT Fails That Show the Dangers of Relying Too Much on AI

Matthew Gibbons (2023)

MA

The rent on an apartment is $700 per month. Each month, you have to pay the full $700 in rent, even if you only live there for part of the month. If you live there for 3 months and 4 days, how much do you pay in rent?

To calculate how much you pay in rent for living in the apartment for 3 months and 4 days, you can break it down as follows:

1. Rent for 3 months:
   3 months * $700/month = $2,100
2. Additional rent for 4 days:
   Since there are typically 30 or 31 days in a month, let's assume an average of 30.5 days in a month for simplicity.
   4 days / 30.5 days (average month) * $700/month = approximately $91.80

Now, add the rent for 3 months and 4 days together:

$2,100 (for 3 months) + $91.80 (for 4 days) = $2,191.80

So, you would pay a total of $2,191.80 in rent for living in the apartment for 3 months and 4 days.

https://www.webfx.com/blog/marketing/chatgpt-fails/

Svagheder af ChatGPT (I): ChatGPT kan ikke regne med semantiske enheder

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Matthew Gibbons (2023)

MA

That's incorrect. Remember, you have to pay the FULL $700 RENT for each month. You still have to pay the full $700 even if you only live there for a few days out of the month.

You are absolutely right, and I apologize for the previous incorrect response. If you have to pay the full $700 in rent for each month, regardless of how many days you live there, then the calculation is much simpler.

For 3 months and 4 days, you pay the full rent for 3 months and an additional 4 days:

3 months * $700/month = $2,100
4 days * $700/month = $93.33 (approximate)

So, you would pay a total of $2,100 + $93.33 = $2,193.33 in rent for living in the apartment for 3 months and 4 days, considering the full monthly rent is due even if you live there for part of the month.

https://www.webfx.com/blog/marketing/chatgpt-fails/

Svagheder af ChatGPT (II): ChatGPT er ikke kreativ

from: 8 ChatGPT Fails That Show the Dangers of Relying Too Much on AI

Matthew Gibbons (2023)

MA

In less than 50 words, come up with a unique and original idea for email marketing strategy.

"Memory Lane Email Campaign" - Send personalized emails to customers on special occasions, like anniversaries of their first purchase or subscription. Include a timeline of their interactions with your brand, fostering a deeper emotional connection and encouraging continued loyalty and engagement.

Sending out emails on the anniversary of a customer's first purchase? Wow, what a great, completely original idea, ChatGPT! Except for one thing:

> A purchase anniversary is a date a customer made their first purchase from a company. Companies sometimes take the day to send out greeting cards, emails, or other personalized correspondence to their clients to demonstrate their appreciation and hopefully entice them to make additional purchases in the future. Apr 26, 2023

https://www.webfx.com/blog/marketing/chatgpt-fails/

Andre Svagheder af ChatGPT

- ~~Not adhering to word limits~~
- *Failing at simple math and logic*
- Not grasping humor
- *Struggling to generate new ideas*
- Falsifying sources
- Lying about its own protocols
- Hallucinating fake information
- Producing biased responses


8 ChatGPT Fails That Show the Dangers of Relying Too Much on AI

Published: Nov 8, 2023 16 min. read

Matthew Gibbons
Senior Data & Tech Writer

from: 8 ChatGPT Fails That Show the Dangers of Relying Too Much on AI, fra Matthew Gibbons (2023) https://www.webfx.com/blog/marketing/chatgpt-fails/

Arbejdsdefinition

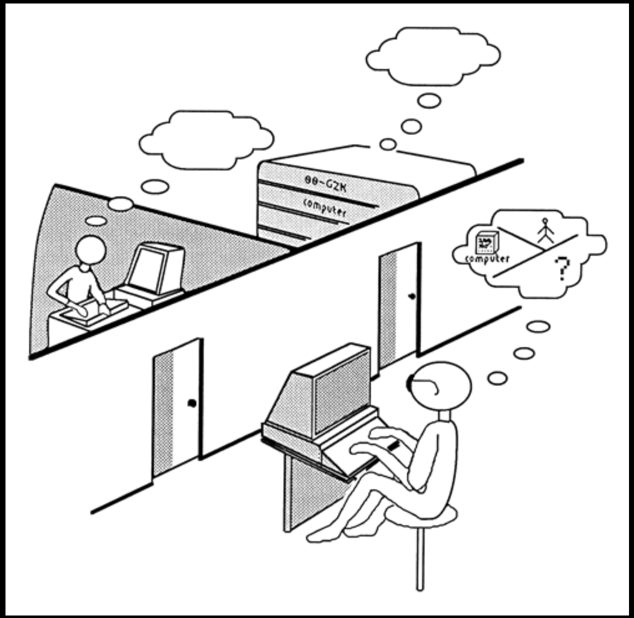
Kunstig intelligens modellerer **intelligent adfærd** på computere og/eller robotter

Arbejdsdefinition

Kunstig intelligens modellerer **intelligent adfærd** på computere og/eller robotter

Men hvad er intelligent adfærd?

Turing-testen



Alan Mathison Turing
(1912-1954)

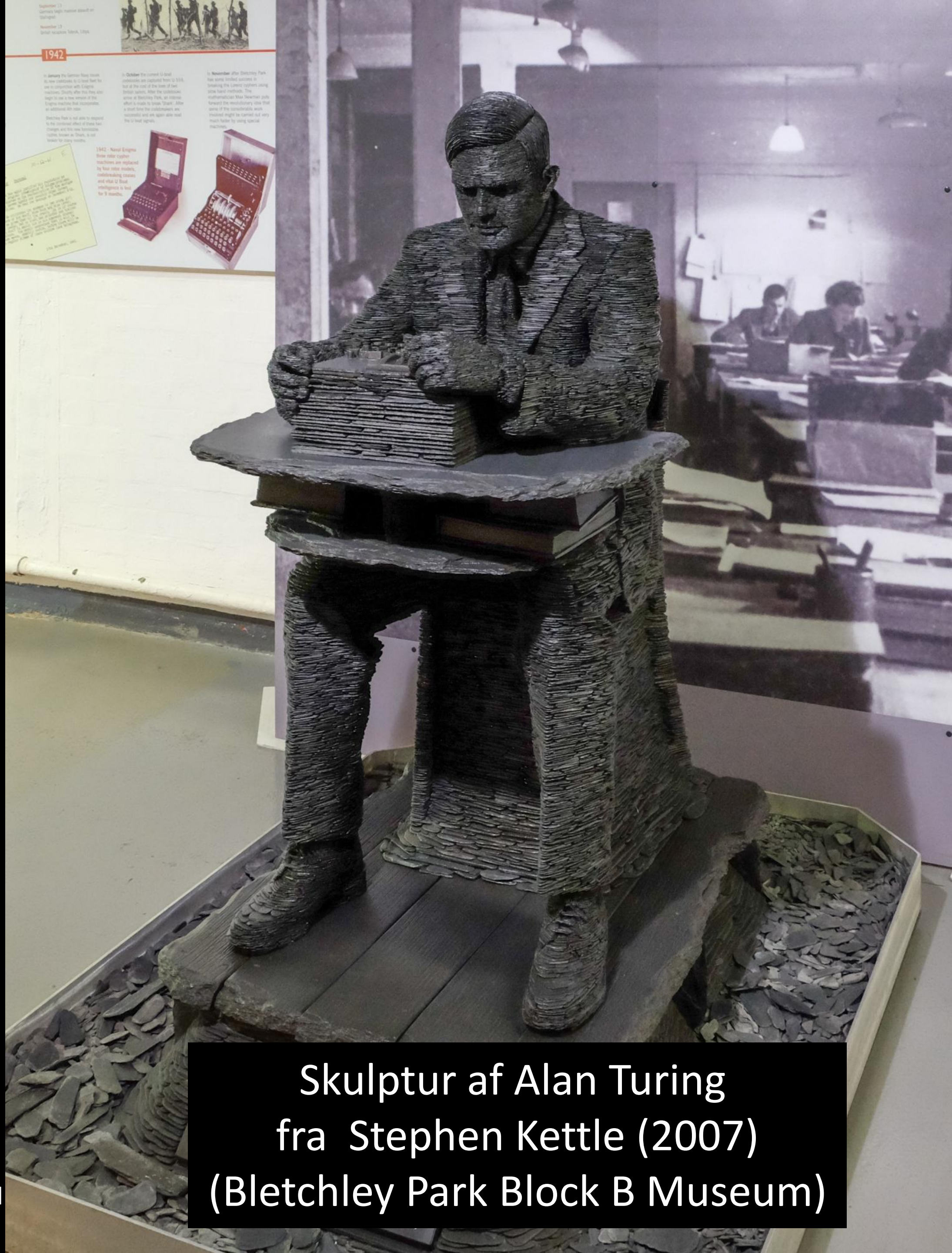
Testen går ud på, at en person under fjernkommunikation med enten en maskine eller et andet menneske ikke skal kunne afgøre, om der er tale om et menneske eller en maskine.

https://da.wikipedia.org/wiki/Turing-test

A. Turing, Computing Machinery and Intelligence (1950)

Alan Mathison Turing

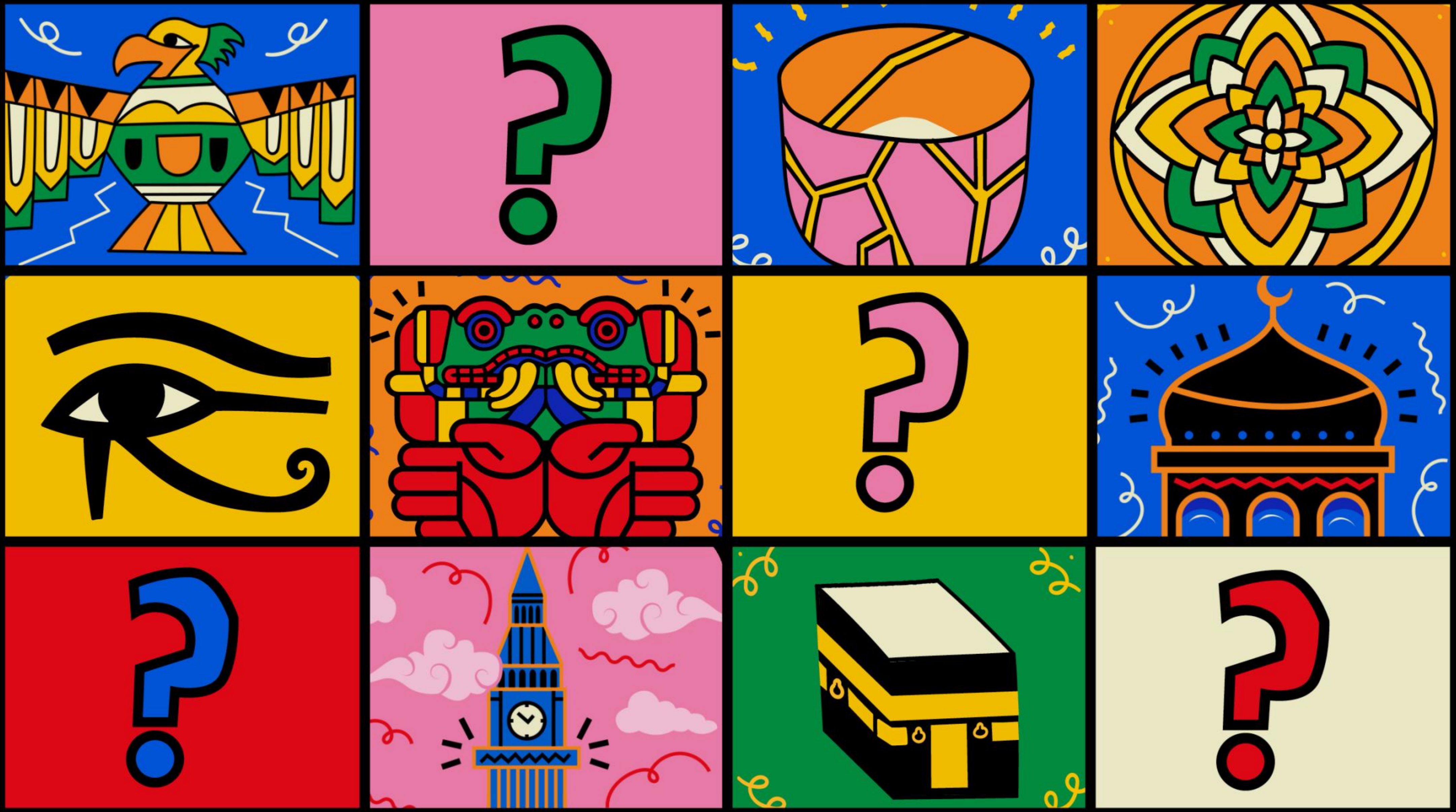
- Født 23. juni 1912
- Britisk matematiker
- En af datalogiens grundlæggere til moderne computer
  - Turingmaskine
  - Turingtest
  - Knækkede Enigma-kode under WW2 (sammen med andre forskere).
- Homoseksuel
- Selvmord ifølge tvangsbehandling mod hans homoseksualitet den 7.6.1954
- Britiske premierminister Gordon Brown gav ham en posthum undskyldning.



Skulptur af Alan Turing
fra Stephen Kettle (2007)
(Bletchley Park Block B Museum)

Loebner-prisen

- Loebner-prisen (startede i 1990)
- En årlig konkurrence for computerprogrammer, som dommerne betragtede som mest menneskelige.
- Konkurrenceformatet var formatet for en standard Turing-test.
- Oprindeligt blev \$2.000 tildelt det mest menneskelige program i konkurrencen.
- Loebner-prisen stoppede i 2020, fordi stifteren døde



Består ChatGPT Turing-testen?

Feature

THE EASY INTELLIGENCE TESTS THAT AI CHATBOTS FAIL

Scientists are racing to find new ways to clarify the differences between the capabilities of humans and of large language models. **By Celeste Biever**

reasoning, or understanding, he says. Others (including himself and researchers such as Mitchell) are much more cautious.

"There's very good smart people on all sides of this debate," says Ullman. The reason for the split, he says, is a lack of conclusive evidence supporting either opinion. "There's no Geiger counter we can point at something and say 'beep beep beep — yes, intelligent'," Ullman adds.

Tests such as the logic puzzles that reveal differences between the capabilities of people and AI systems are a step in the right direction, say researchers from both sides of the discussion. Such benchmarks could also help to show what is missing in today's machine-learning systems, and untangle the ingredients of human intelligence, says Brenden Lake, a cognitive computational scientist at New York University.

Research on how best to test LLMs and what those tests show also has a practical point. If LLMs are going to be applied in real-world domains — from medicine to law — it's important to understand the limits of their capabilities, Mitchell says. "We have to understand what they can do and where they fail, so that we can know how to use them in a safe manner."

Is the Turing test dead?

The most famous test of machine intelligence has long been the Turing test, proposed by the

Nature, Vol 619, 27 July 2023

Citater

They (LLMs) work simply by generating plausible next words when given an input text, based on the statistical correlations between words in billions of online sentences they are trained on.

For chatbots built on LLMs, there is an extra element: human trainers have provided extensive feedback to tune how the bots respond.

Some [researchers] attribute the algorithms' achievements to glimmers of reasoning, or understanding, he says. Others … are much more cautious.

Feature

THE EASY INTELLIGENCE TESTS THAT AI CHATBOTS FAIL

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Citater

… researchers agree that GPT-4 and other LLMs would probably now pass the popular conception of the Turing test, in that they can fool a lot of people, at least for short conversations.

One challenge that researchers mention is that the models are trained on so much text that they could already have seen similar questions in their training data, and so might, in effect, be looking up the answer.

Feature

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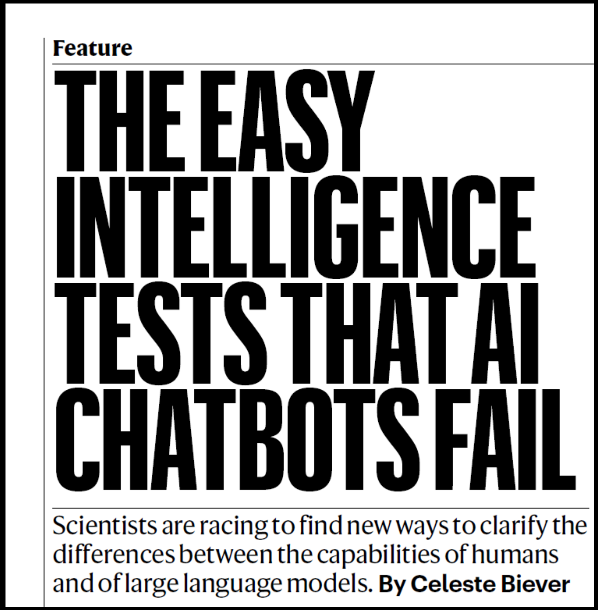
Quotes

Melanie Mitchell: "Extrapolating in the way that we extrapolate for humans won't always work for AI systems."

This might be because LLMs learn only from language; without being embodied in the physical world, they do not experience language's connection to objects, properties and feelings, as a person does.



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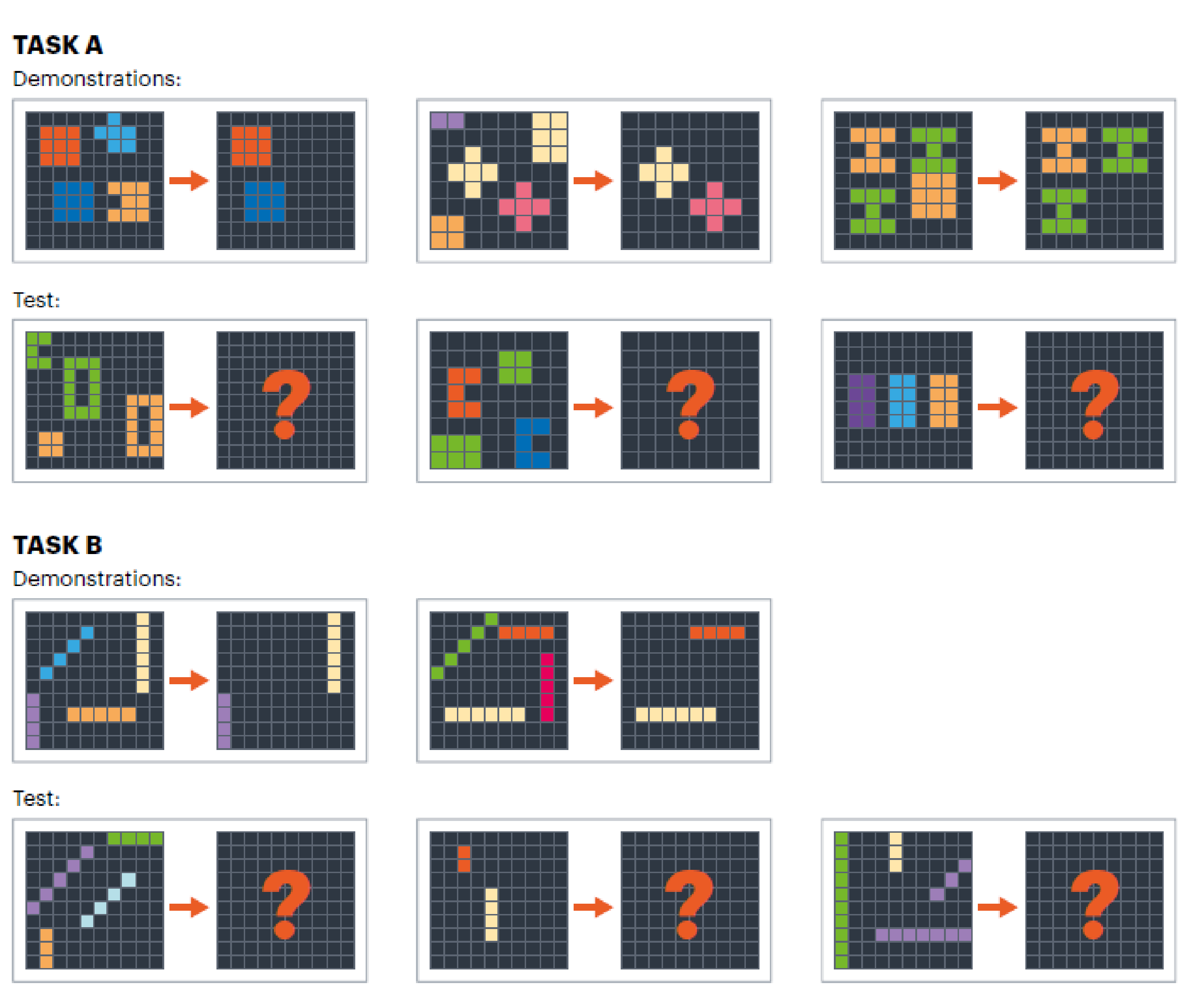
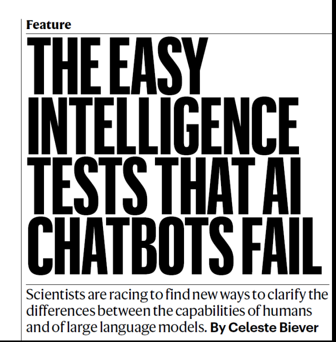
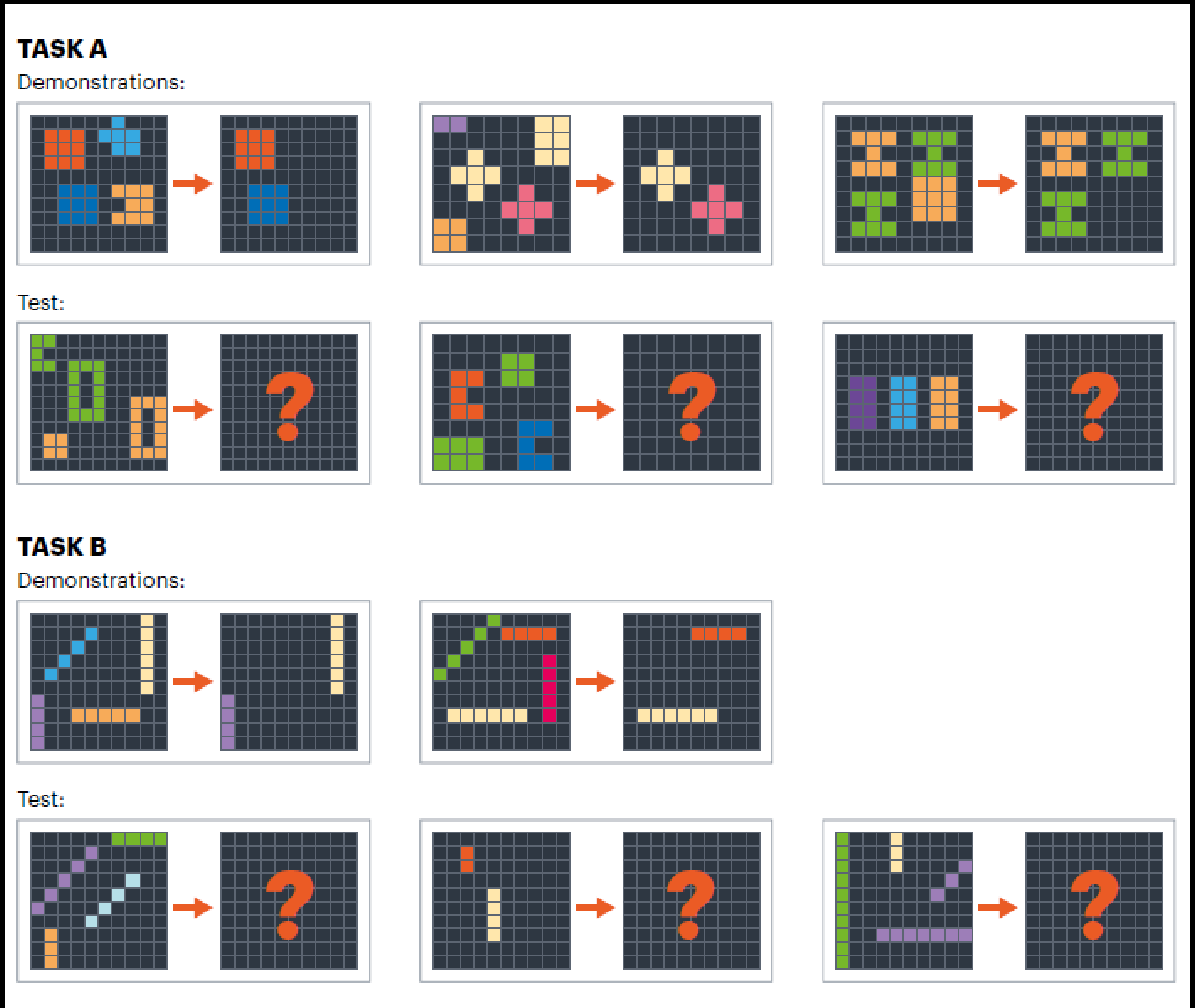
En test, hvor ChatBots er svag

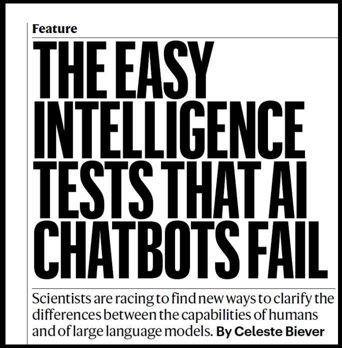




- AI-systemer klarer sig dårligt i ConceptARC-test
- Spørgsmål: Hvordan ændrer gittermønstre sig?

En test, hvor ChatBots er svag




Feature

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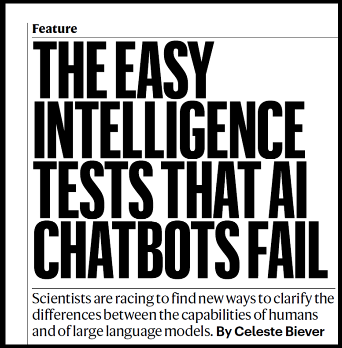
- AI-systemer klarer sig dårligt i ConceptARC-test
- Spørgsmål: Hvordan ændrer gittermønstre sig?
- Anvendelse af et underliggende abstrakt koncept (her 'ensartethed')
- Se go.nature.com/43v6fzk for svarene.

Diskussion


Feature

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- Hvorfor kan en computer være så god i konversation, selv om den ikke har noget koncept om
  - Tal
  - Ensartethed
  - Elementære logiske sammenhænge?

Film ”Her” (2013)

- En ung fyr udvikler et forhold til et sprogprogram
- Klip fra 42:00

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- En ung fyr udvikler et forhold til et sprogprogram
- Klip fra 42:00

Spørgsmål:

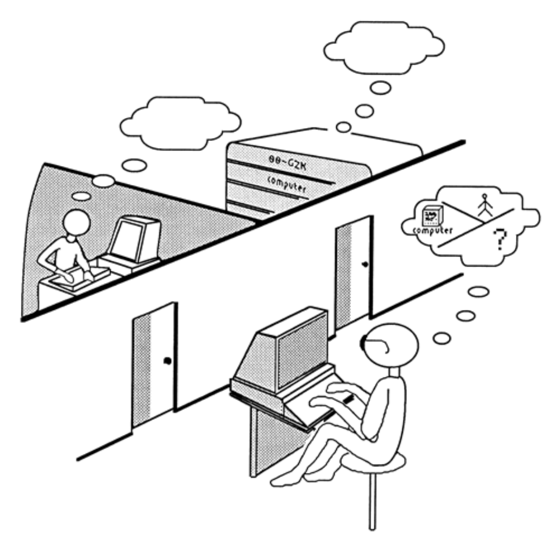
Hvornår tror du, at det er muligt at udvikle et sådant system?

Hvad er stadigvæk svært at realisere teknisk?

Loebner-prisen

- Derudover var der to engangspriser.
  - $25.000 til det første program, som dommerne ikke kan skelne fra et rigtigt menneske, og som kan overbevise dommerne om, at mennesket er computerprogrammet.
- $100.000 var belønningen for det første program, som dommerne ikke kan skelne fra et rigtigt menneske i en Turing-test, hvilket inkluderer dekryptering og forståelse af tekst, visuelt og auditivt input.
  - Når dette er opnået, afsluttes den årlige konkurrence.

Oversæt fra https://en.wikipedia.org/wiki/Loebner_Prize



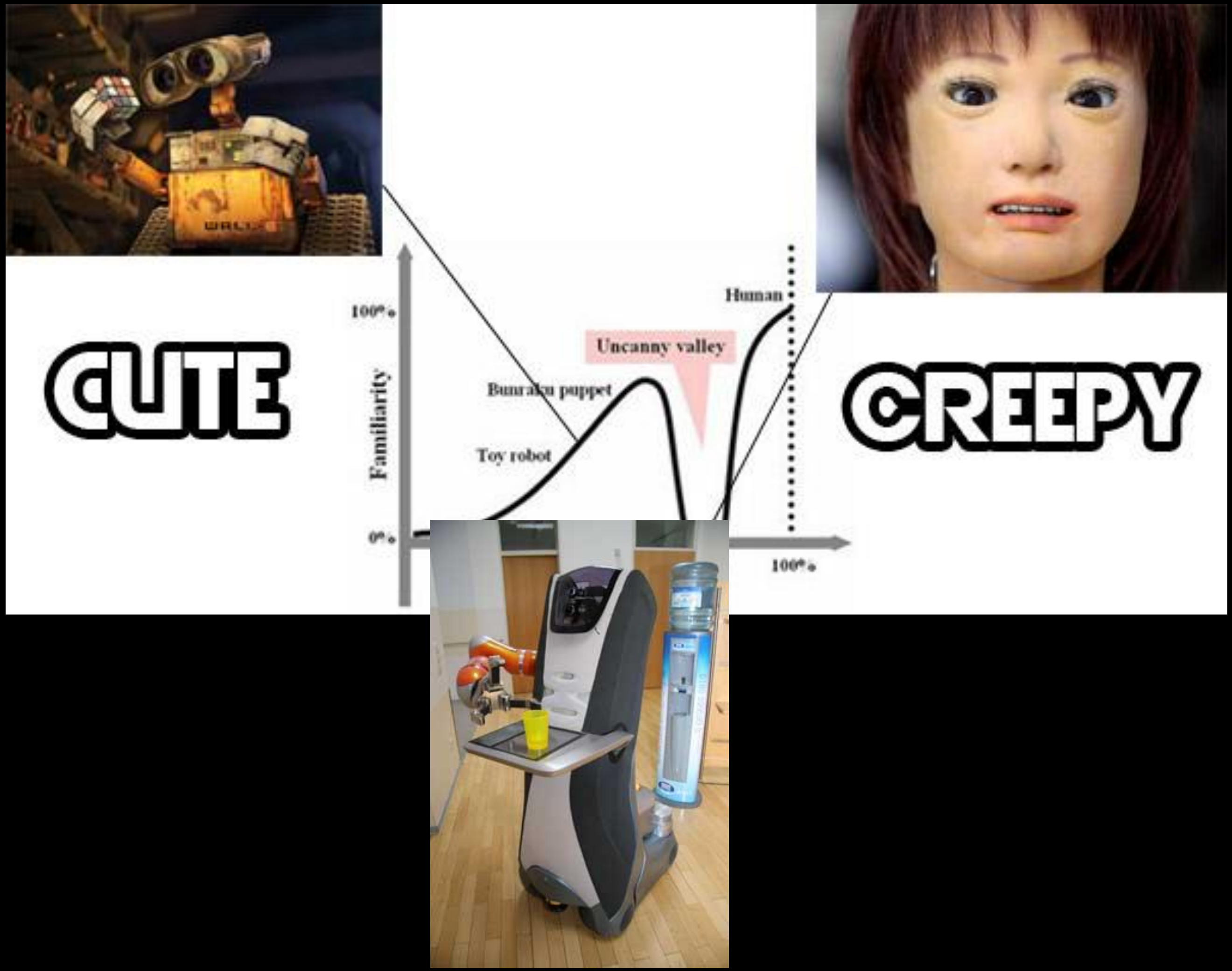
Professor Ishiguro's Creepy Robot Doppelgänger
https://www.wired.com/2007/04/professor-ishig/

Uncanny Valley

Mori, Masahiro *The uncanny valley.* 1970, Energy, 7, S. 33-35.

- Uncanny Valley er en hypotetisk, psykologisk og æstetisk relation mellem et objekts grad af lighed med et menneske og den følelsesmæssige reaktion på objektet
- Hypotesen om den "uhyggelige dal" forudsiger, at en enhed, der ser næsten menneskelig ud, vil risikere at fremkalde uhyggelige følelser hos beskueren.

Oversæt fra: https://en.wikipedia.org/wiki/Uncanny_valley



Ex Machina

- En ung fyr møder en robot-designer og får opgaven at udføre en avanceret Turing-test med en robot
- Klip 23:30

Hvor er vi henne med hensyn til Loebner-prisen?

Hvornår vil vi have nået det?

Hvad er sværest?
At vinde mod Magnus Carlsen i skak, eller at tage en banan fra et højt sted?

Köhler, Wolfgang (1963). Intelligenzprüfungen am Menschenaffen. Berlin, Göttingen, Heidelberg: Springer.

Består ChatGPT Turing-testen?

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Øvelse

- Struktur
    - Læs artiklen
    - Fordybe forståelsen gennem brug af internettet (fx chatbots)
    - Første MCQ test
    - Nogen ekstra tid
    - Anden MCQ test
- Det er vigtig for eksamen!!!!

Feature

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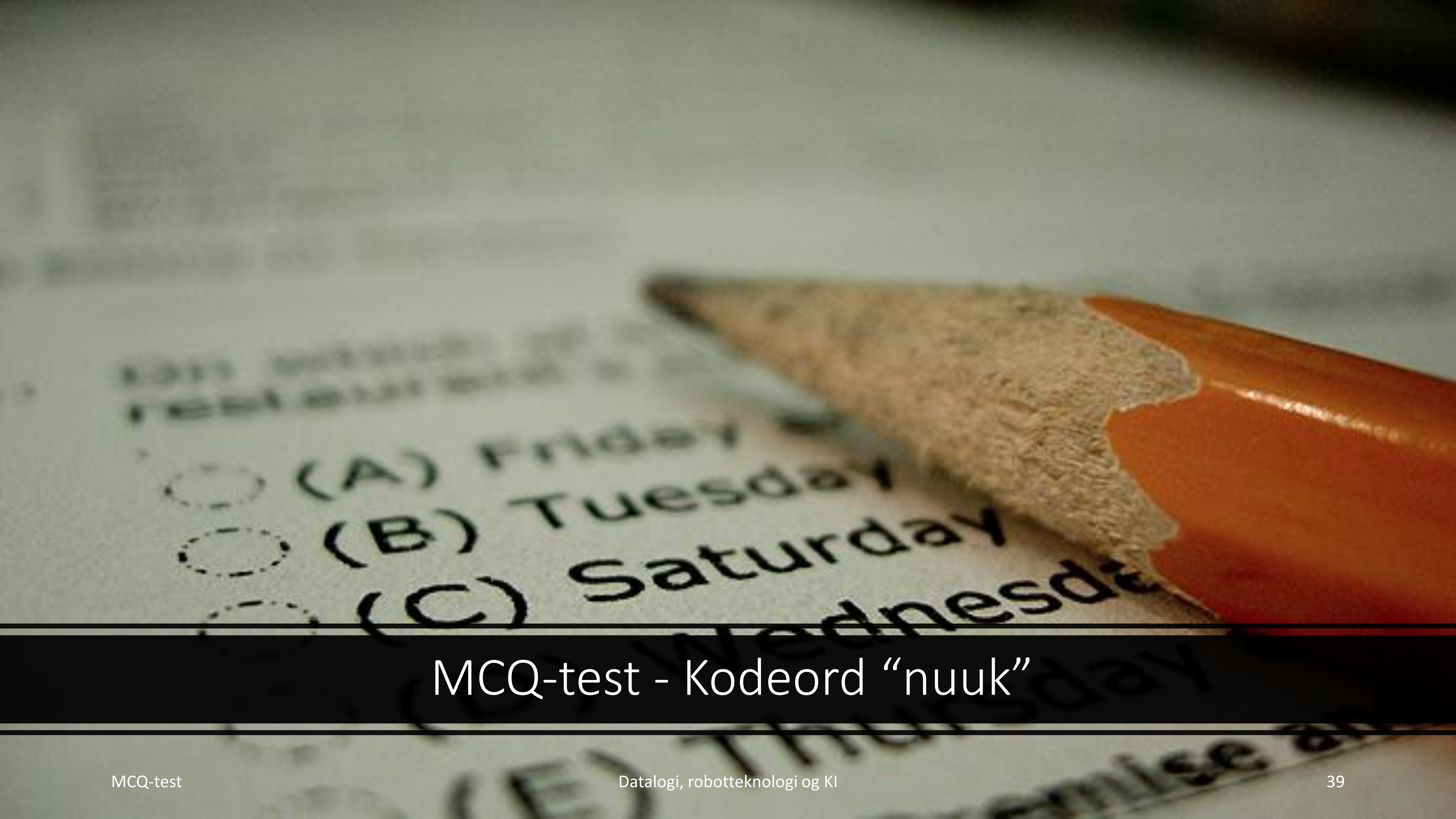

MCQ-test - Kodeord “nuuk”

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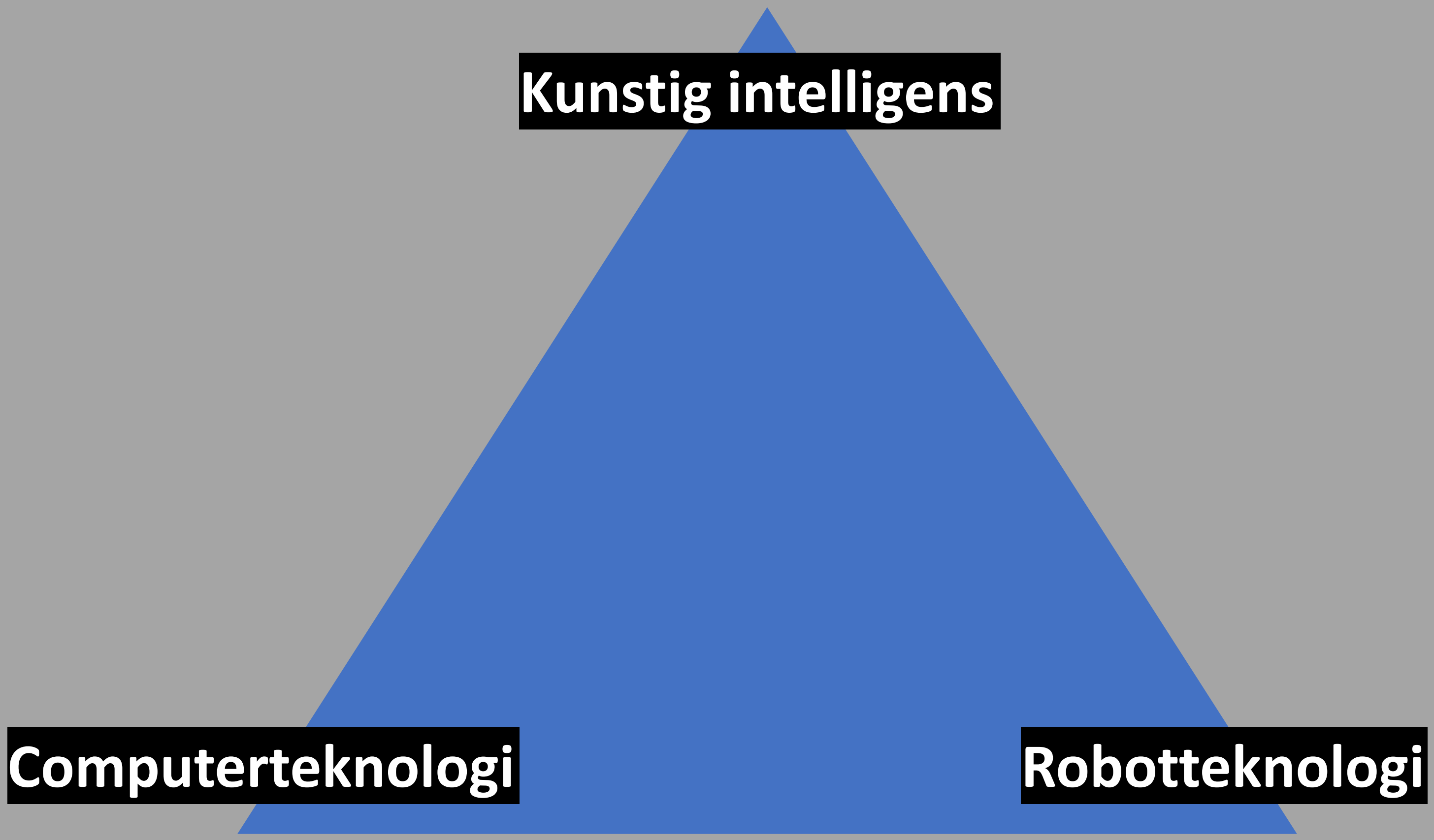

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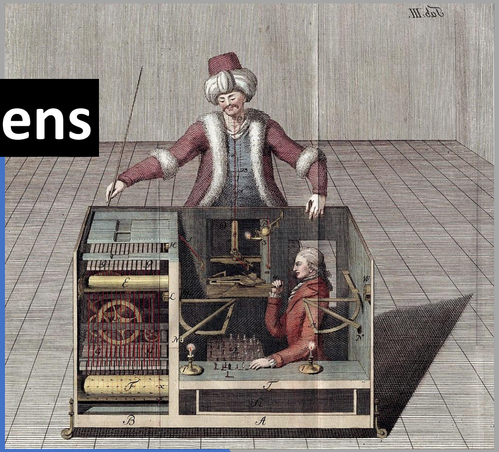
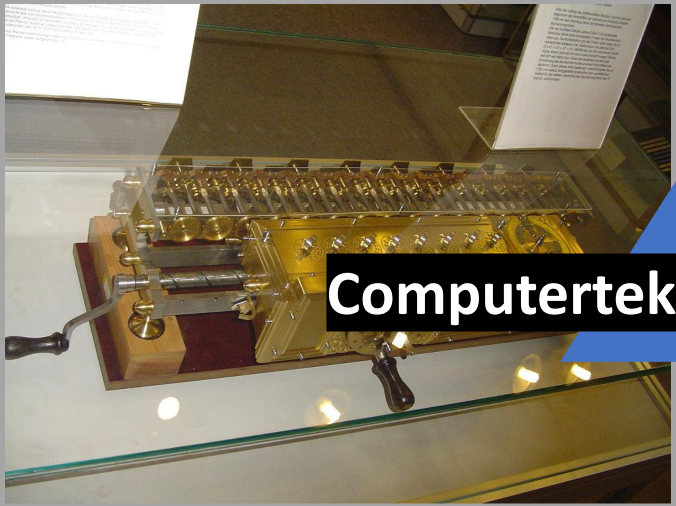
Indtil 2. Verdenskrig

Kunstig intelligens





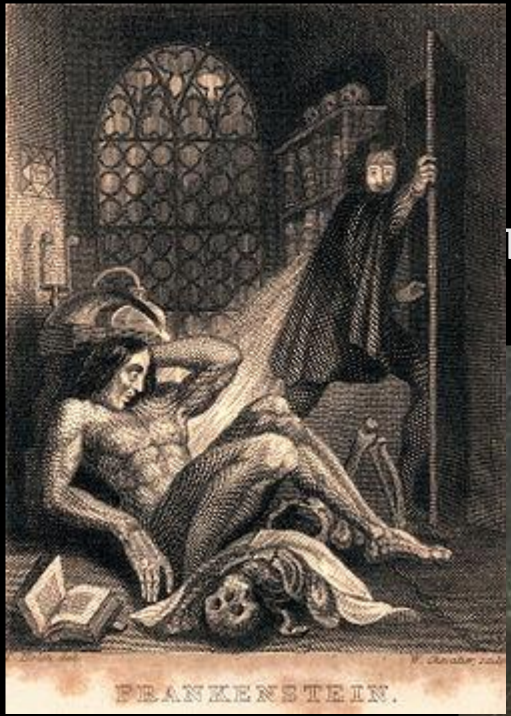
Computerteknologi

Robotteknologi

Robot i mytologien og litteraturen: Talos, Golem og Frankenstein

- *Græsk mytologi*: Talos var en gigantisk automat lavet af bronze, som skulle beskytte Europa (kongedatter fra Tyros) på Kreta mod pirater.
- Jødiske mytologi: Golem er en levende skabning fremstillet af dødt materiale og givet liv ved en mystisk proces, der indebærer Guds hemmelige navn. Forskellige varianter: Prag (1580) og Worms (12. århundrede).
- Roman " Frankenstein; or, The Modern Prometheus" (Mary Shelley, 1818)

https://da.wikipedia.org/wiki/Frankenstein#/media/Fil:Frankenstein_engraved.jpg



Frankenstein (forside til 1831-udgivelsen)

Kilde: https://www.wondersandmarvels.com/2012/03/the-worlds-first-robot-talos.html/talos2-1

Talos i filmen Jason and the Argonauts (1963)

Kilde: https://de.wikipedia.org/wiki/J%C3%BCdisches_Museum_Worms#/media/Datei:Golem_J%C3%BCdisches_Museum_Worms.jpg

Golem I Worms

Talos: filmklip

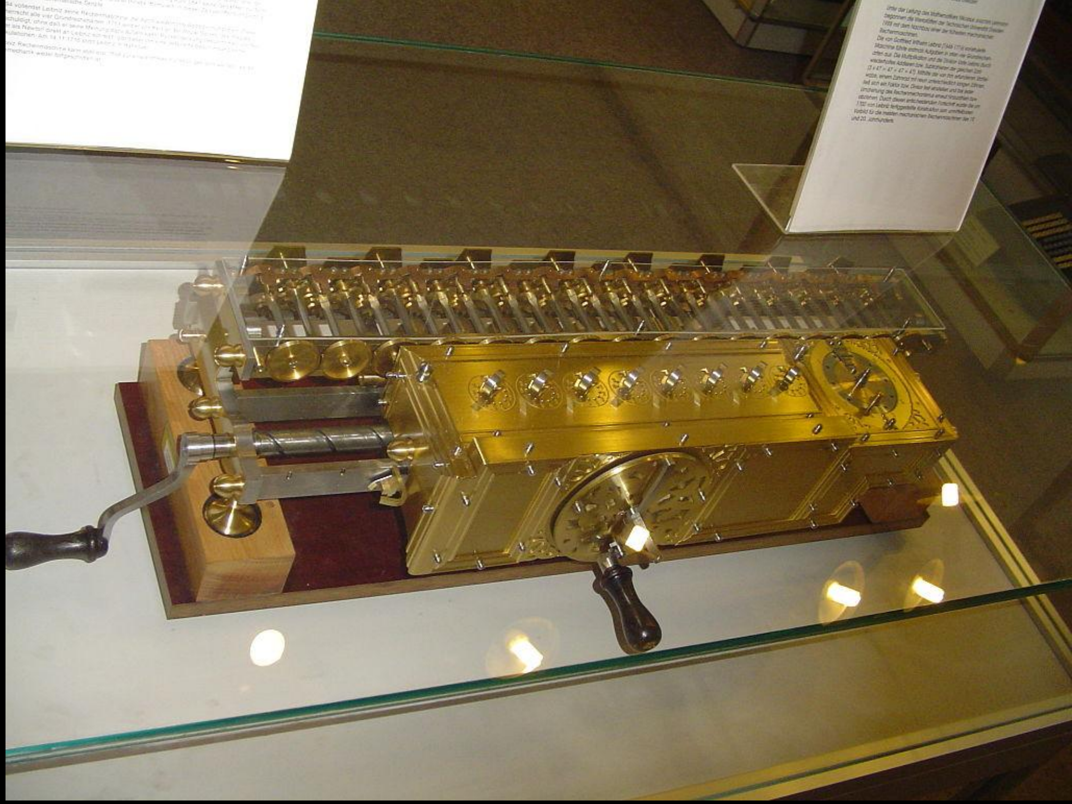
Karel Čapeks Rossum's Universal Robots

- Den første brug af ordet "robot”: Karel Čapeks teaterstykke R.U.R. (Rossum's Universal Robots) blev skrevet i 1920.
- Handling
  - Kunstige væsener lærer med tiden om vold fra deres menneskelige skabere og begynder at gøre oprør (også et motiv i filmen *Blade Runner*).
  - Robotterne kan reproducere sig og til sidst arve jorden (idé i filosofi: *Posthumanisme*).

https://de.wikipedia.org/wiki/R.U.R.#/media/Datei:R.U.R._by_Karel_%C4%8Capek_1939.jpg

Leibniz' regnemaskine: En forgænger for computer

- Mekanisk regnemaskine opfundet af den tyske matematiker Gottfried Wilhelm Leibniz (1646-1716)
- Kunne udføre alle fire grundlæggende aritmetiske operationer
- Det indviklede præcisionsgearværk var ustabilt

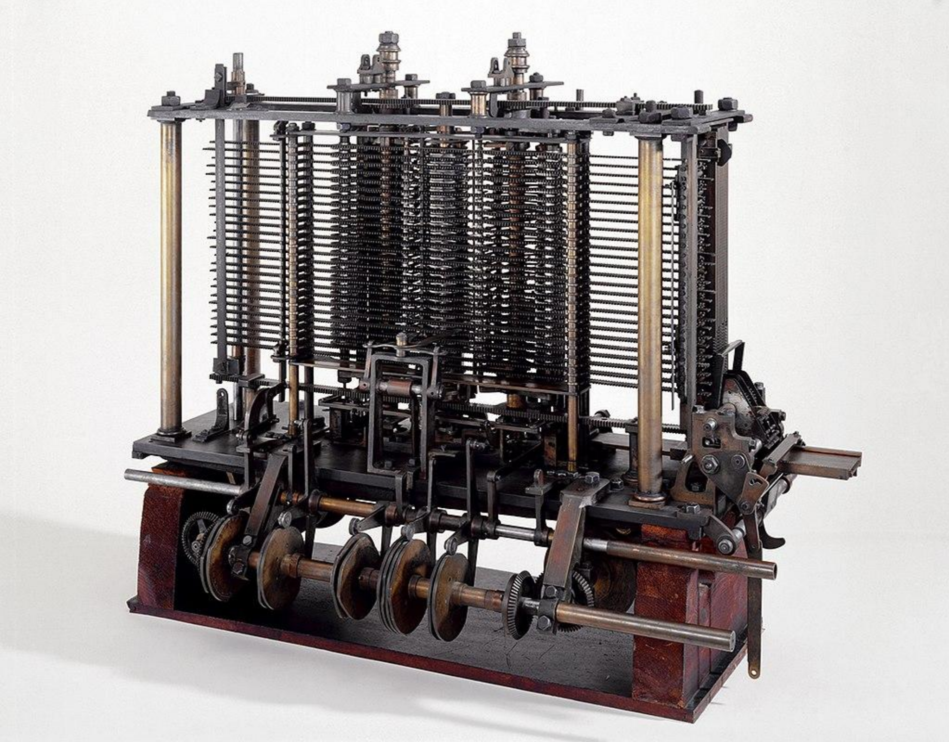


https://de.wikipedia.org/wiki/R.U.R.#/media/Datei:R.U.R._by_Karel_%C4%8Capek_1939.jpg

... it is beneath the dignity of excellent men to waste their time in calculation when any peasant could do the work just as accurately with the aid of a machine. —

Babbages Analytical Engine

- En mekanisk regnemaskine til *generelle anvendelser* fra matematikprofessoren Charles Babbage (1791-1871)
- Maskinen skulle drives af en dampmaskine, bestå af 55.000 dele og være 19 meter lang og tre meter høj.
- Input: hulkort, dengang brugt til at styre mekaniske vævestole.
- Output: Printer, en kurveplotter og en klokke.
- I 1878 anbefalede et udvalg, at den analytiske motor ikke skulle bygges.
- Maskinen, hvis designplaner anses for at være funktionelle, blev derefter glemt.



En del af regnemaskinen (udstillet på Science Museum, London)

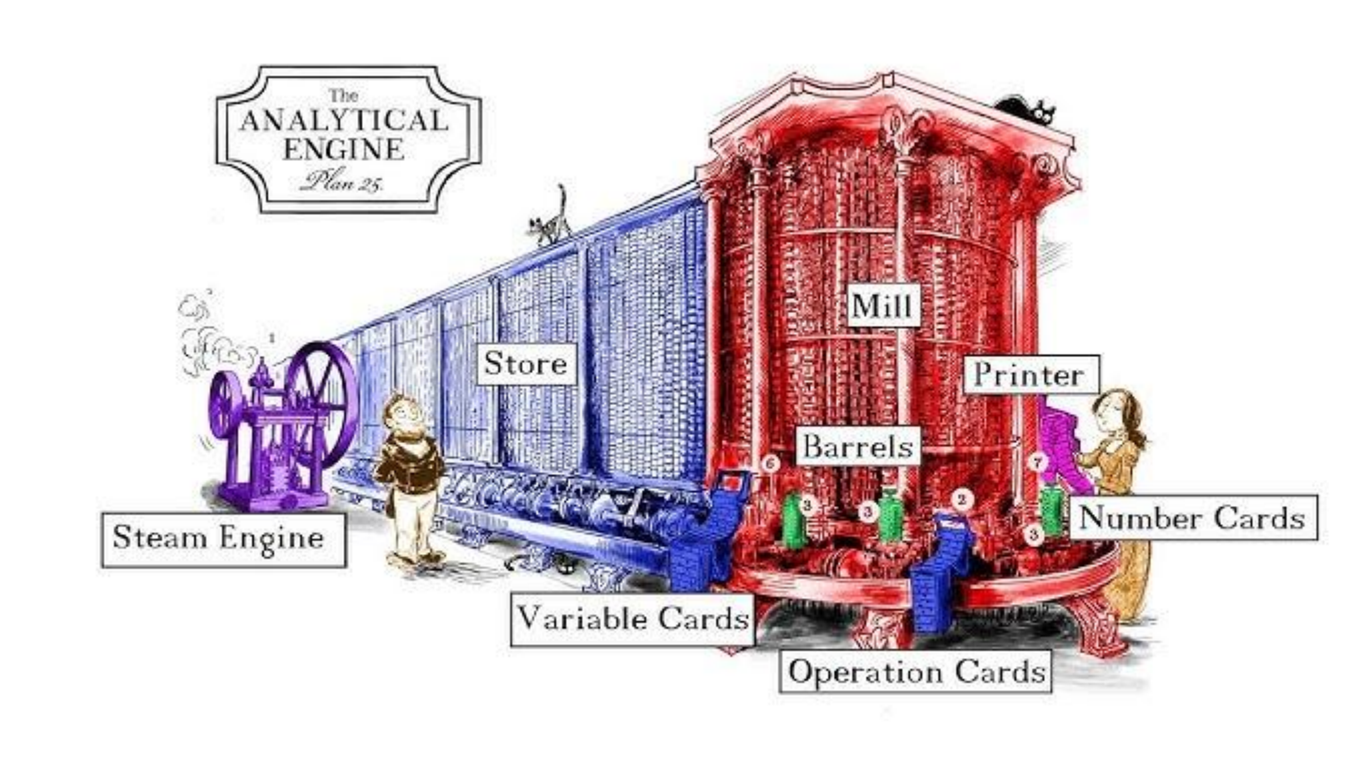
Plan 28 Blog

https://blog.plan28.org/

The mission of the project is to build a Babbage Analytical Engine for historical and educational purposes.

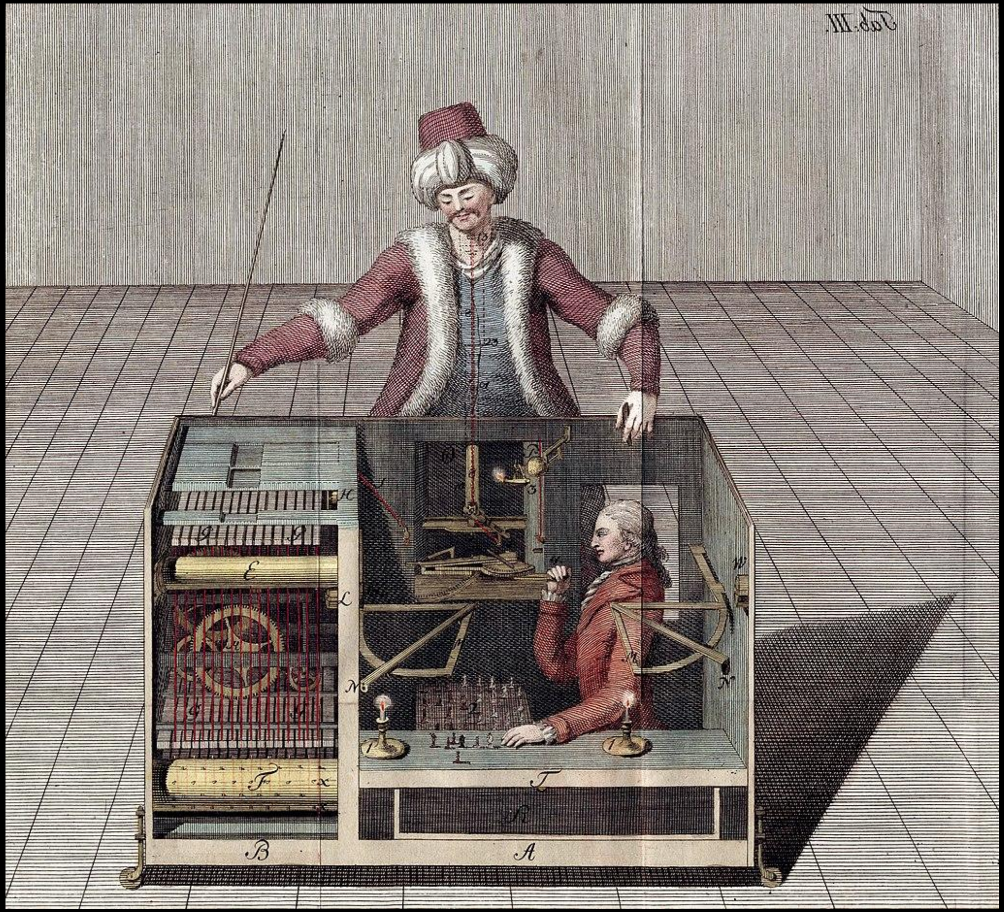
Film: 3:23-5:41

- https://www.youtube.com/watch?v=eMy4vSZ-J_I



”Skaktyrken”

- ”Skaktyrken” fra 1770 foregav at være en skakspillende maskine, men blev afsløret som et svindelnummer i 1820'erne.
- Skaktyrken blev fremvist i store dele af Europa samt Nord- og Sydamerika i over 80 år, før den blev ødelagt i en brand i 1854.
- Med en stærk skakspiller som operatør blev den til tider anset som uovervindelig.



Et tværsnit af ”skaktyrkeren” fra Racknitz, som dog tog fejl både med hensyn til operatørens position og automatens dimensioner.

Oversat fra: https://en.wikipedia.org/wiki/Mechanical_Turk

Be aware of fakes

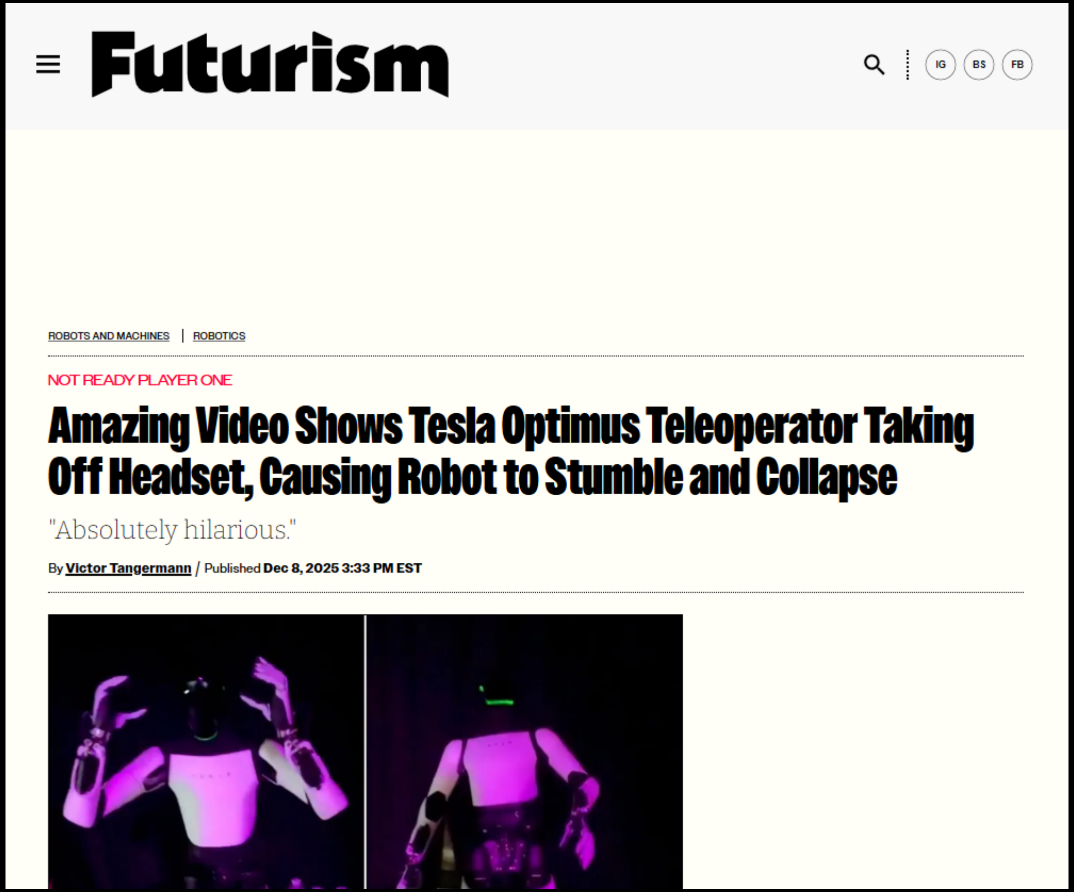
- https://www.youtube.com/watch?v=Eu5mYMavctM

Figure is the first-of-its-kind AI robotics company bringing a general-purpose humanoid to life. (https://www.figure.ai/company)

This story is part of TIME Best Inventions of 2025.
This was the scene inside the Silicon Valley headquarters an August morning this year. The three-year-old startup was in a sprint ahead of the October announcement of its next robot, the Figure 03, which was undergoing top-secret training when TIME visited. The robots folding towels were the company's previous model, the Figure 02, operating the same software that the Figure 03 will use.
Figure claims the 03 will be its first mass-producible humanoid, and that it will eventually even work on its own production line. The launch will be a critical moment for this startup of 360 people, which in September announced it had secured $1 billion in investment at a valuation of $39 billion, and which counts Nvidia, Jeff Bezos, OpenAI, and Microsoft among its investors.

https://time.com/7324233/figure-03-robot-humanoid-reveal/

Be aware of fakes



Futurism

IG BS FB

ROBOTS AND MACHINES | ROBOTICS

NOT READY PLAYER ONE

Amazing Video Shows Tesla Optimus Teleoperator Taking Off Headset, Causing Robot to Stumble and Collapse

"Absolutely hilarious."

By Victor Tangermann / Published Dec 8, 2025 3:33 PM EST

https://futurism.com/future-society/tesla-teleoperator-headset-optimus-fall

Will Lockett's Newsletter

Tesla's Optimus Is A Total Fiasco

The C-3PO wannabe is going nowhere.

WILL LOCKETT

FEB 09, 2026 · PAID

68 8 9 Share

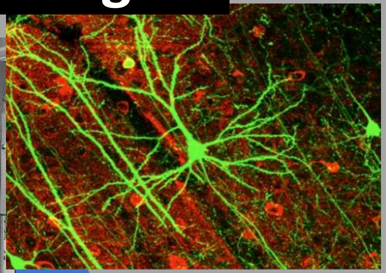
https://www.planetearthandbeyond.co/p/teslas-optimus-is-a-total-fiasco

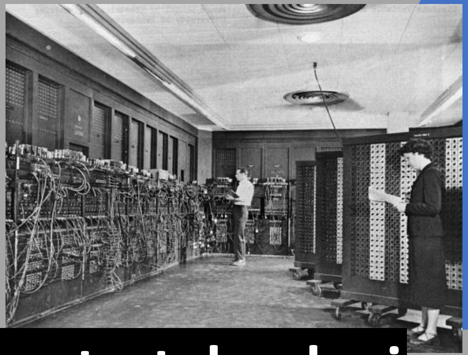
Overblik

- Hvad er intelligens? Hvad er kunstig intelligens?
- **Trekant: KI, Robot- og Computerteknologi**
  - Indtil 2. Verdenskrig
  - **1940-1956: Begyndelsen**
  - 1957-1984: Kampen mellem de symbolske og subsymbolske metoder
  - 1985-2005: Evolutionære fremskridt og Bias/Variance dilemmaet
  - 2005-: Deep Neural Networks tager over
- ChatGPT og embodiment

Trekant 1940-56: KI, Computer- og Robotteknologi

Kunstig intelligens





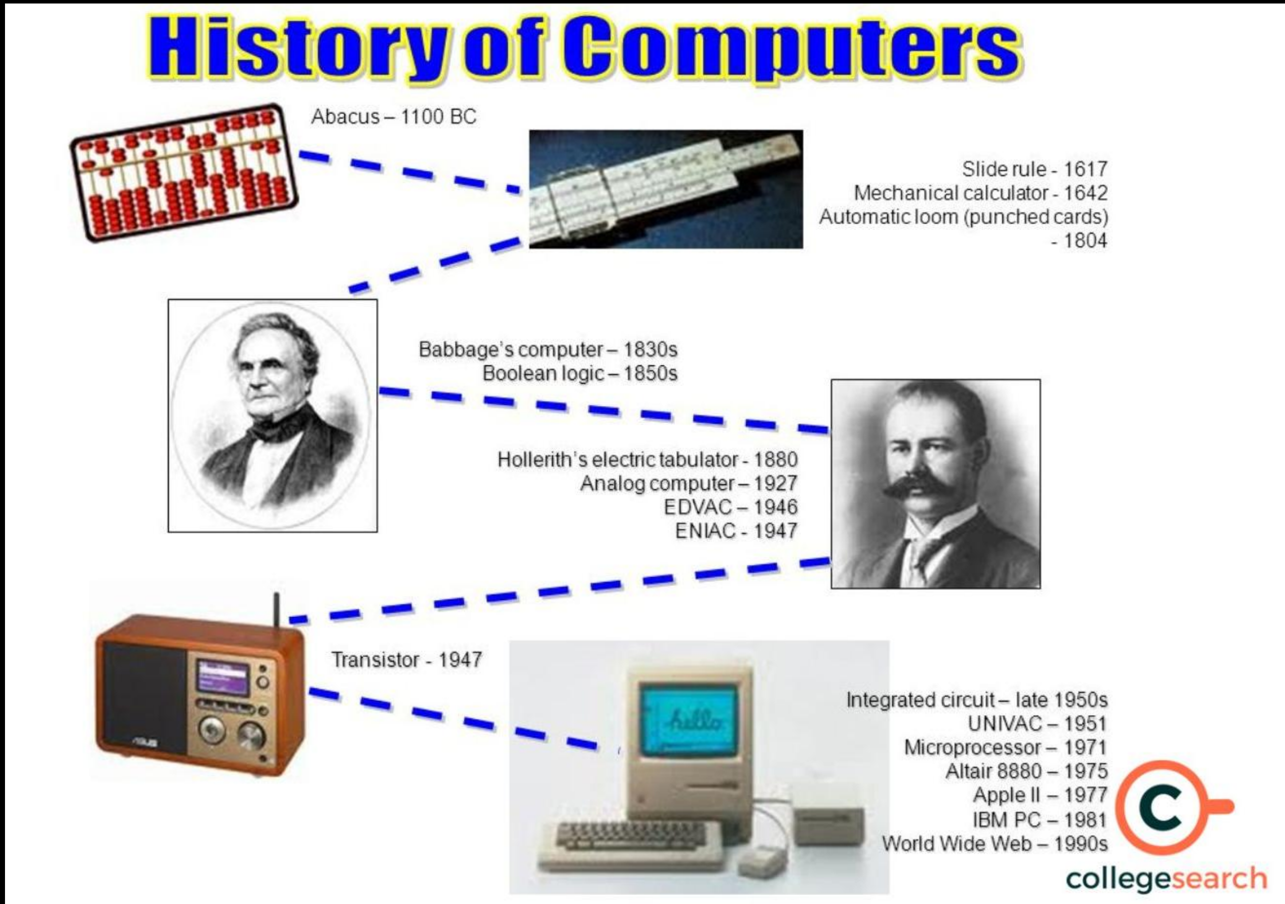
Computerteknologi

Robotteknologi

Computer: Definition

En computer er en enhed, der behandler data ved hjælp af programmerbare beregningsregler.

- Den består grundlæggende af:
  - *input-enhed(er)* (f.eks. tastatur),
  - *en bearbejdningsenhed (*processor(er) med hukommelse (cache/ram)) til at gemme oplysningerne midlertidigt under behandlingen,
  - *et baggrundslager* til opbevaring af styresystem, computerprogrammer og data
  - output-enhed(er) til visning af resultatet (f.eks. skærm eller printer).



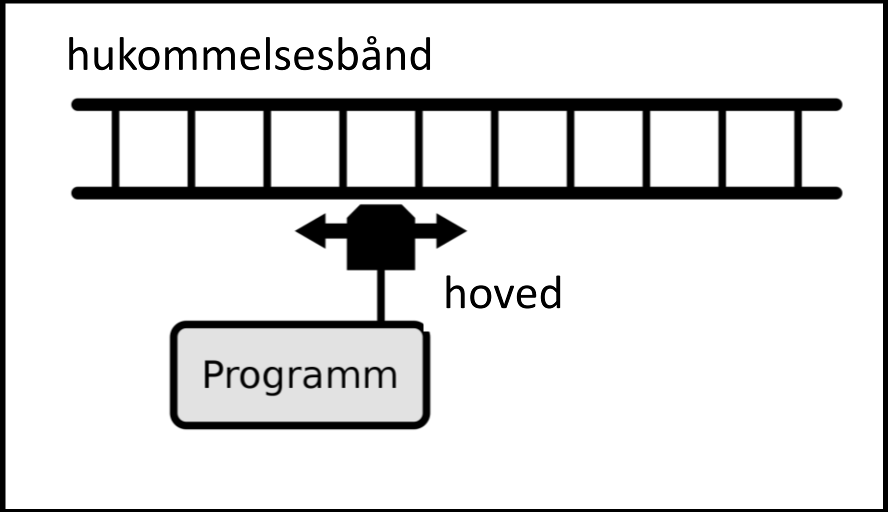
Kilde: https://www.collegesearch.in/articles/history-of-computer

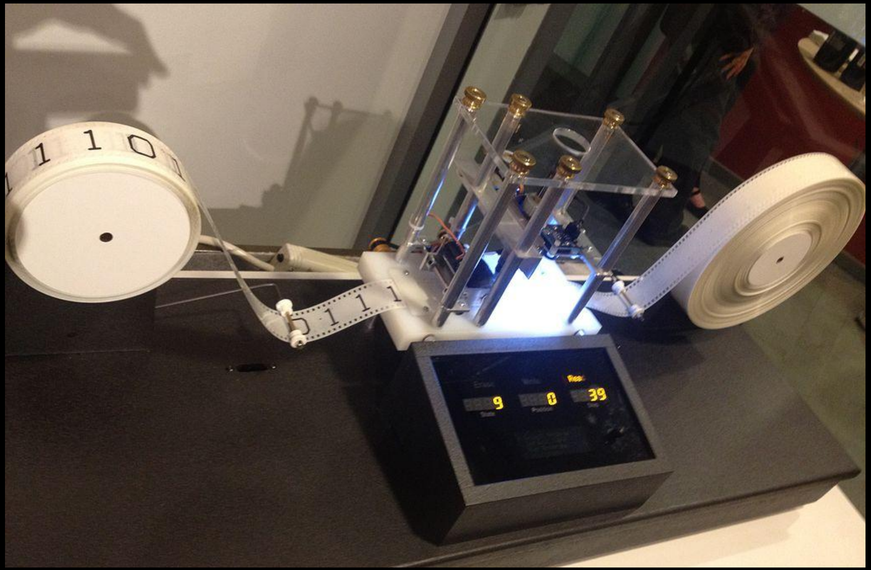
Computer: Turing-maskine

En *Turing-maskine* er en matematisk beregningsmodel, der beskriver en abstrakt maskine, som *manipulerer symboler* på en båndstrimmel i henhold til en *tabel med regler*.

- Maskinen opererer på et *uendeligt hukommelsesbånd*, som hver kan indeholde et enkelt symbol fra et begrænset sæt af symboler, kaldet *maskinens alfabet*.
- På trods af modellens enkelhed er den i stand til at *implementere enhver computeralgoritme*.

Oversæt fra https://en.wikipedia.org/wiki/Turing_machine

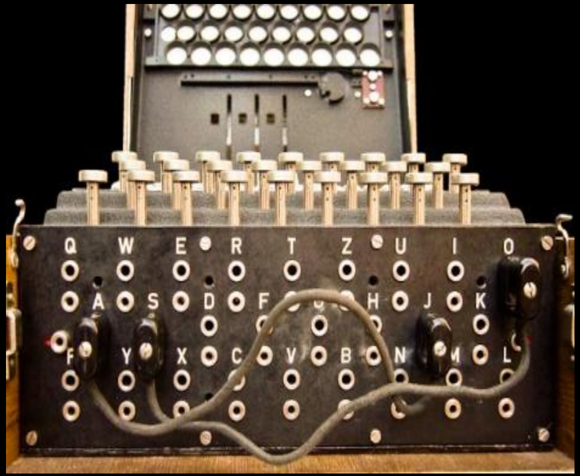


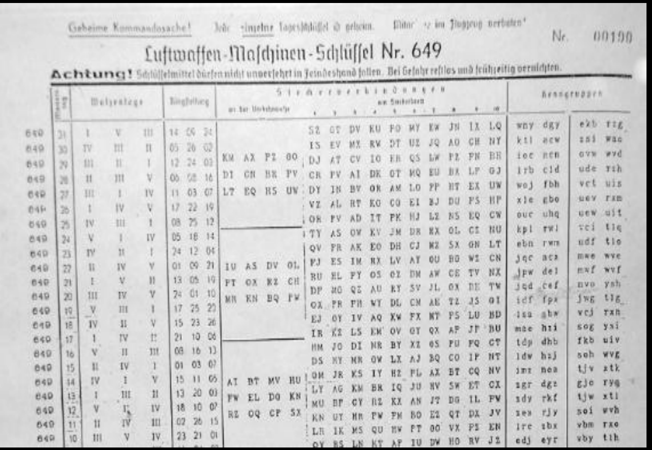


Kilde: https://de.wikipedia.org/wiki/Turingmaschine

Forløber for computere under 2. Verdenskrig: Enigma

- Enigma
  - var en elektromekanisk chiffermaskine,
  - blev brugt af det tyske militær til både *kryptering* og *dekryptering* af meddelelser.
  - var let anvendelig
  - (eller bedre deres kode) blev anset for at være umulig at knække.
- Tyskerne
  - troede koden var umuligt at knække
  - tog fejl



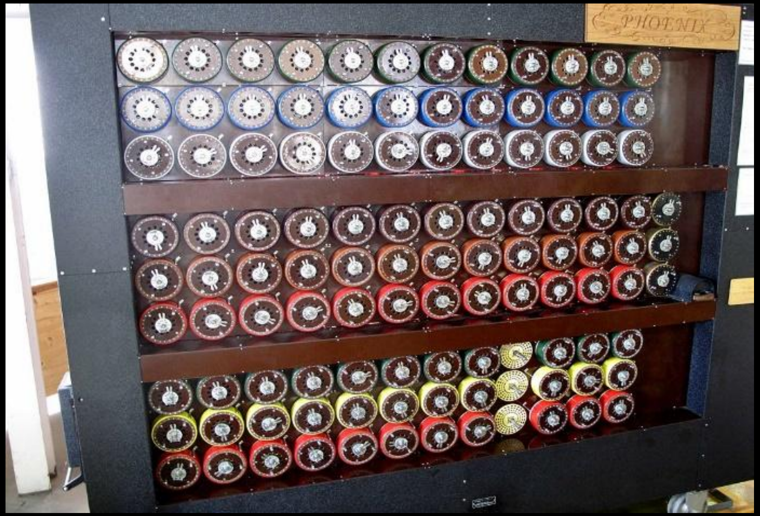


The Imitation Game

Forløber for computere under 2. Verdenskrig: Turing-bomben

- Turing-bomben blev brugt af britiske kryptologer til at dechifrere tyske hemmelige beskeder under WW2.
- En vigtig svaghed var, at et bogstav kunne ikke blive krypteret som sig selv.
- Bomben var en elektromekanisk simulation af flere Enigma-maskiner

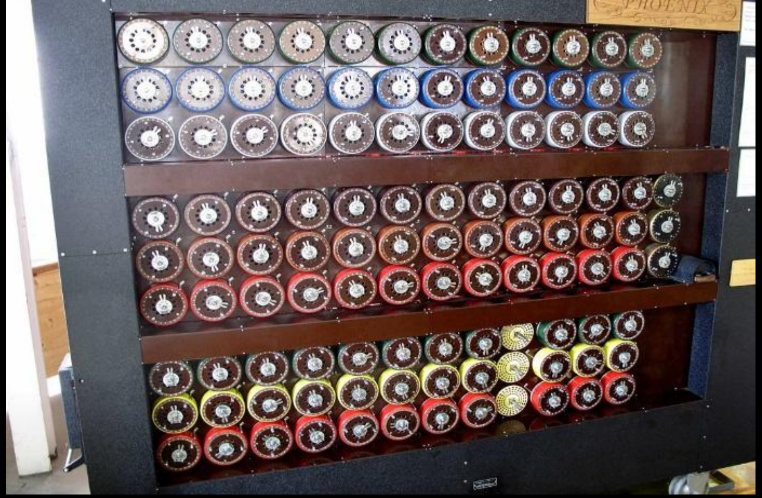
Oversæt fra: https://en.wikipedia.org/wiki/Bombe



Turing Bombe

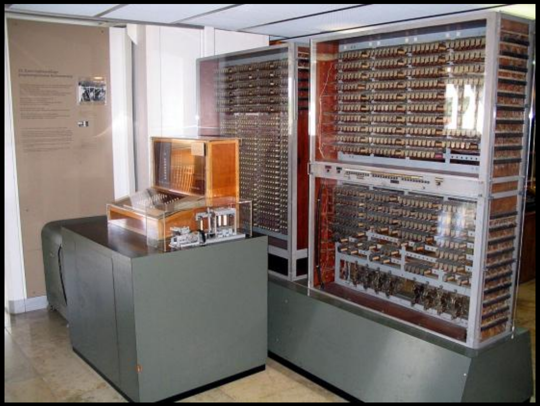
Enigmas svaghed

- Et bogstav kunne ikke repræsentere sig selv!
- Figuren viser, hvordan antallet af muligheder for, hvor ordet *Oberkommandowehrmacht* kunne optræde, kan reduceres
  - Kun 8 ud af 27 mulige positioner kan indtages
- Problemets forenkling, gjorde at Enigma kunne knækkes ved hjælp af den daværende ”computerteknologi”
- Den skulle dog presses til det yderste



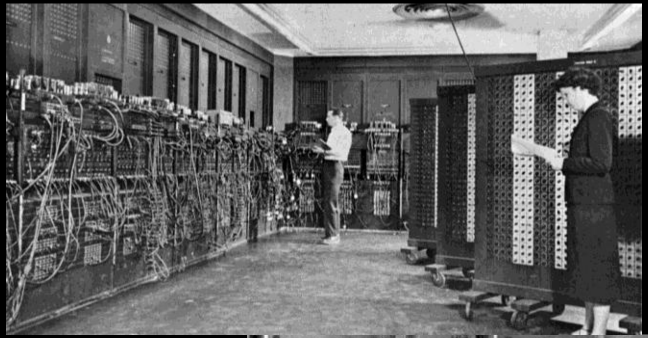
Hvorfor var hverken Enigma eller Turing-bomben en computer?

De første computere: Z3 (1941) og ENIAC (1945)

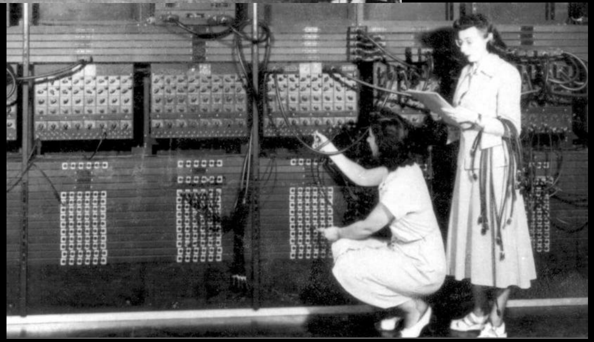


Z3 (udviklet af Konrad Zuse) var den første fungerende, programmerbare, fuldt automatiske maskine, med hovedparten af de egenskaber som normalt bruges som definition af en computer.

https://da.wikipedia.org/wiki/Z3

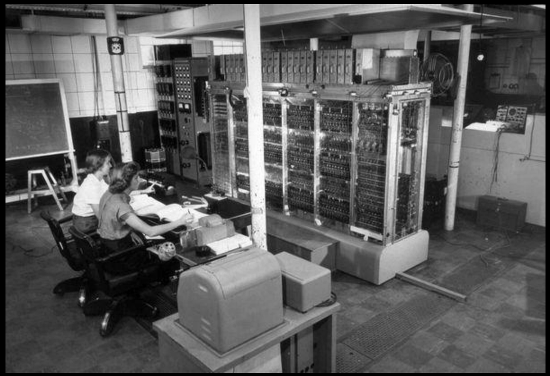


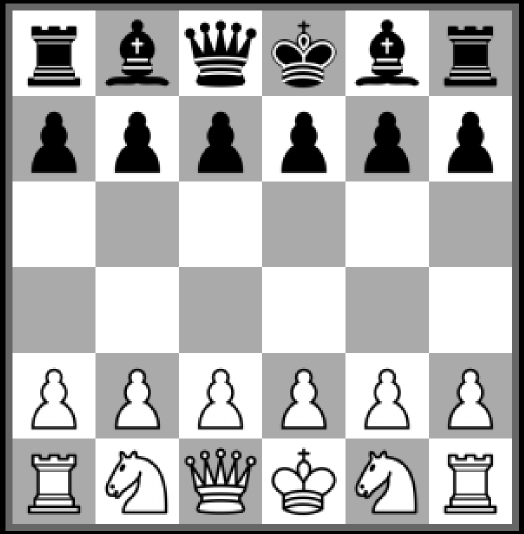
- ENIAC (Electronic Numerical Integrator and Calculator) var den første amerikanske computer.
- Fyldte ca. *150 kvadratmeter* og vejede *30 tons*.
- ENIAC programmer var fastlagt ved hjælp af kabler.



En KI succeshistorie: skak-computeren

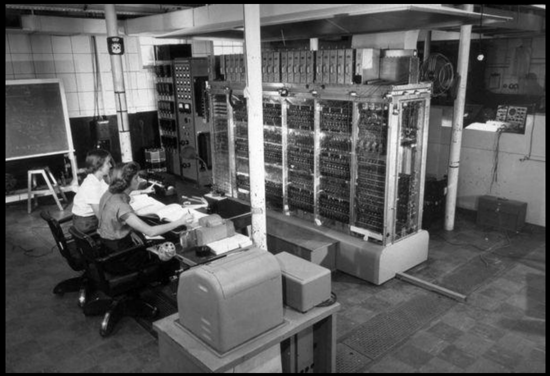
MANIAC Computer (1951)

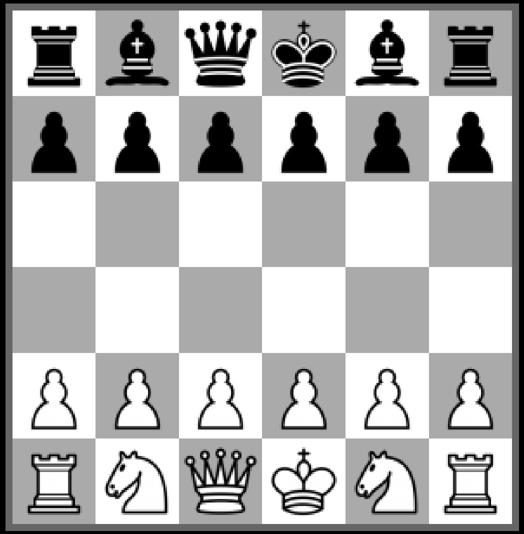




En KI succeshistorie: skak-computeren

MANIAC Computer (1951)

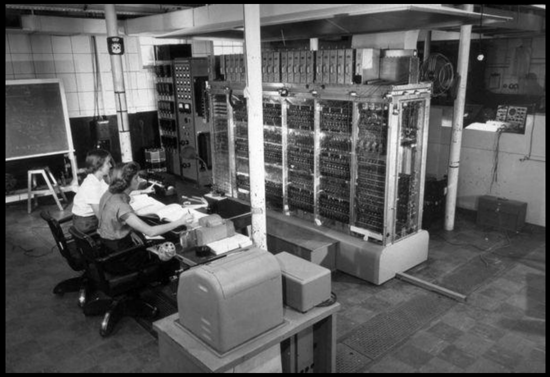


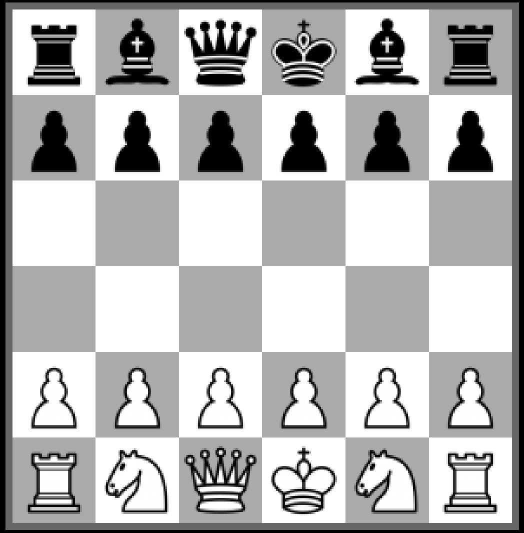


IBM's Deep Blue vinder mod Kasparov (1997)

En KI succeshistorie: skak-computeren

MANIAC Computer (1951)





IBM’s Deep Blue vinder mod Kasparov (1997)

Det har taget mindre end 50 år og var meget forudsigeligt!

Vil det samme ske med andre kognitive menneskelige færdigheder?

Transhumanisme og Posthumanisme

Transhumanismen er en international kulturel bevægelse, der… går ind for, at vi gennem teknologien og videnskaben overskrider (transcenderer) en lang række af de grænser og betingelser, vi i dag forstår ved det menneskelige.

Selve begrebet "transhumanisme" stammer fra 1957, hvor biologen Julian Huxley … brugte det i en artikel om biologiens nye muligheder for mennesket.

Men ideen om at udnytte teknologiens muligheder til at ændre grundlæggende på mennesket fik først fart i begyndelsen af 1980'erne … (www.transhumanism.org).

Kilde: Transhumanisme - det nye menneske

https://auhist.au.dk/showroom/praesentationer/aviser-blade-og-magasiner-fra-au/augustus/2007/transhumanisme/indledning

Transhumanisme og **Posthumanisme**

An AI takeover [en særlig form af posthumanisme] is an imagined scenario in which artificial intelligence (AI) emerges as the dominant form of intelligence on Earth and computer programs or robots effectively take control of the planet away from the human species …

Possible scenarios include replacement of the entire human workforce due to automation, takeover by an artificial superintelligence (ASI), and the notion of a robot uprising.

Stories of AI takeovers have been popular throughout science fiction, but recent advancements have made the threat more real. Some public figures, such as Stephen Hawking and Elon Musk, have advocated research into precautionary measures to ensure future superintelligent machines remain under human control.

Kilde: https://en.wikipedia.org/wiki/AI_takeover

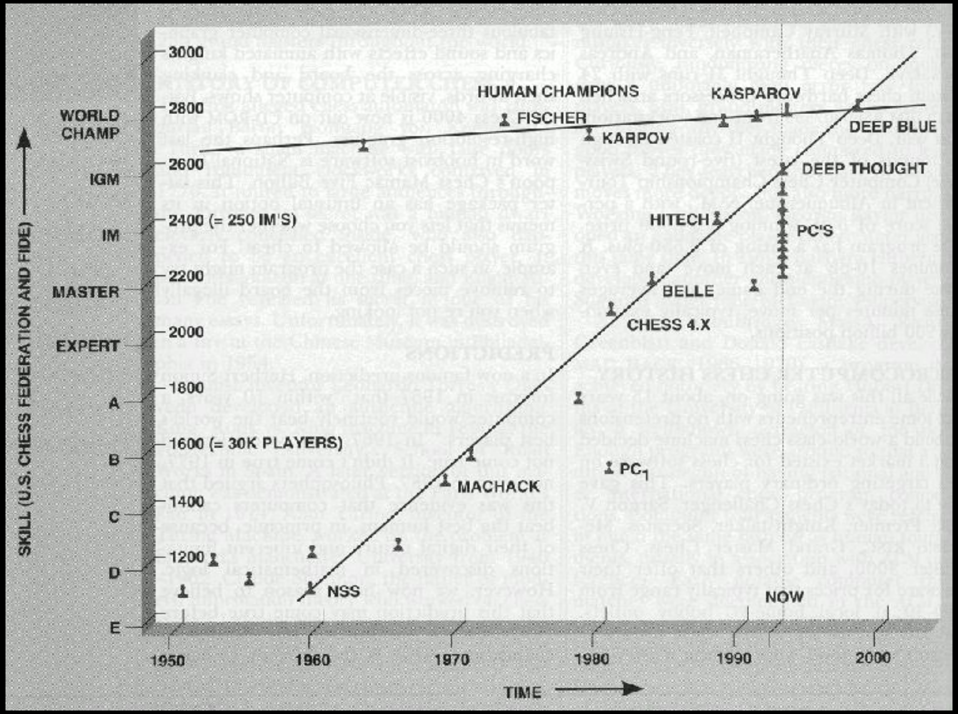
Historisk Analogi? Den første landing af Cristopher Columbus I Amerika

De indfødte amerikaner vare måske lidt for naive

Puebla y Tolín, Dióscoro Teófilo (1862)

Tidsforløbet af computernes og menneskernes skak evner

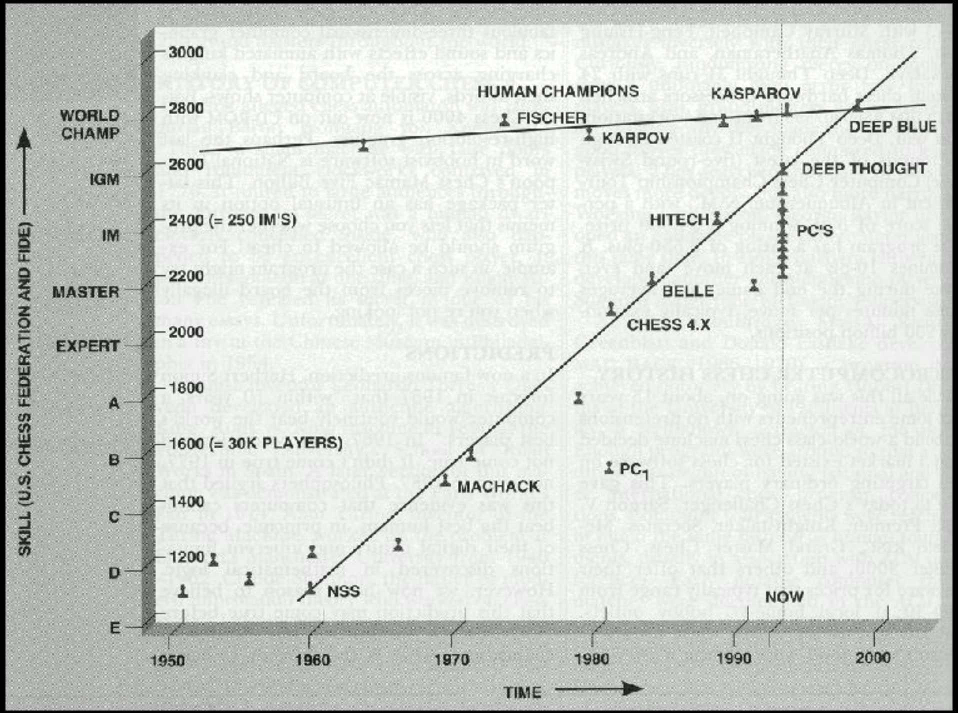
Elo-rating er en statistisk metode til at afgøre styrkeforskellen på individuelle spillere i to-personers spil; jo højere tallet er, jo større styrke. https://da.wikipedia.org/wiki/Elo-rating



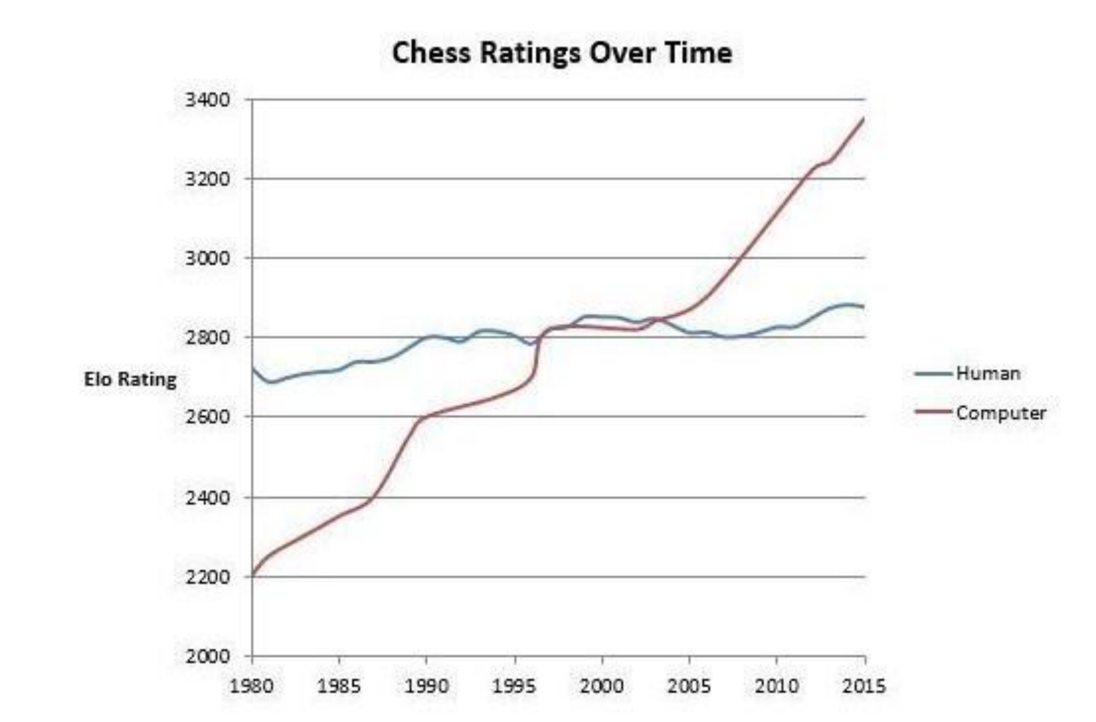
Coles 2002.

Tidsforløbet af computernes og menneskernes skak evner

Elo-rating er en statistisk metode til at afgøre styrkeforskellen på individuelle spillere i to-personers spil; jo højere tallet er, jo større styrke. https://da.wikipedia.org/wiki/Elo-rating



Coles 2002.



Erik Brynjolfsson, https://x.com/erikbryn/status/721378122927620097?mx=2

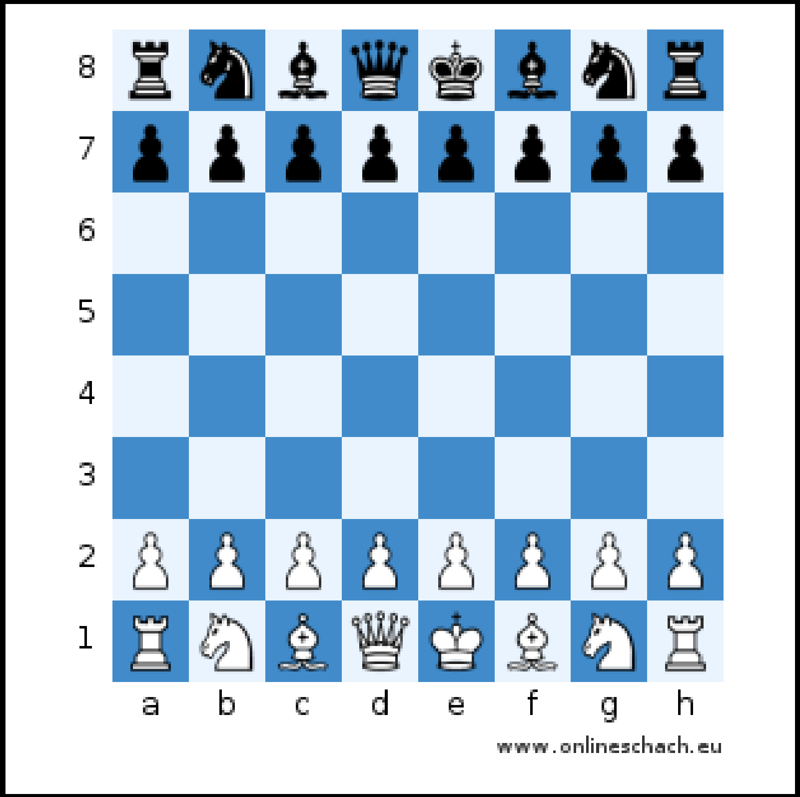
Elo-rating Magnus Carlsen (2025): 2831

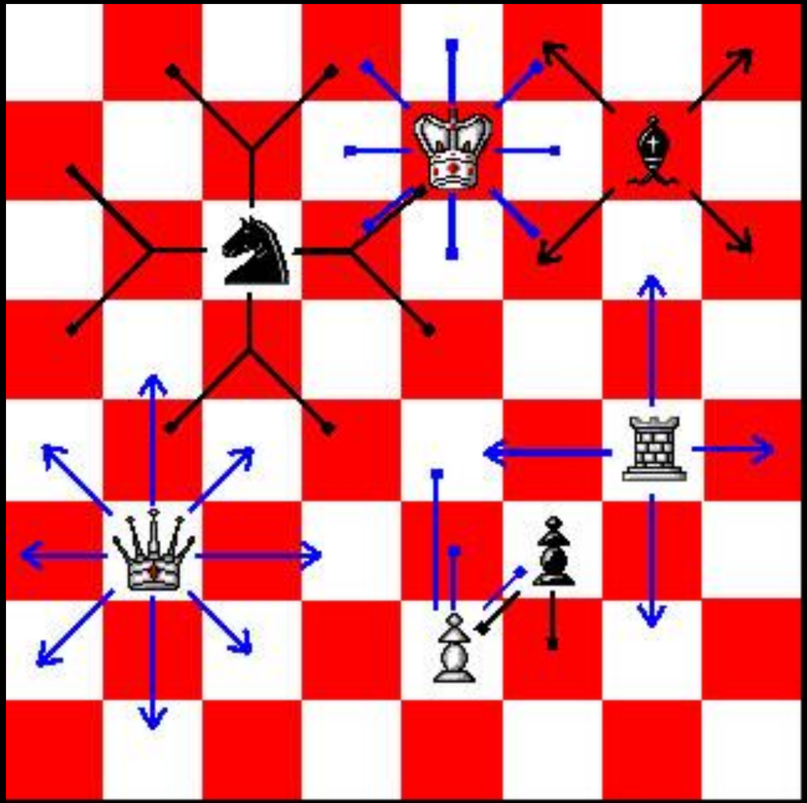
Hvordan fungerer skak-computere?

- Tre skridt
  - Trækgenerator
  - Evalueringsfunktion
  - Min-Max search

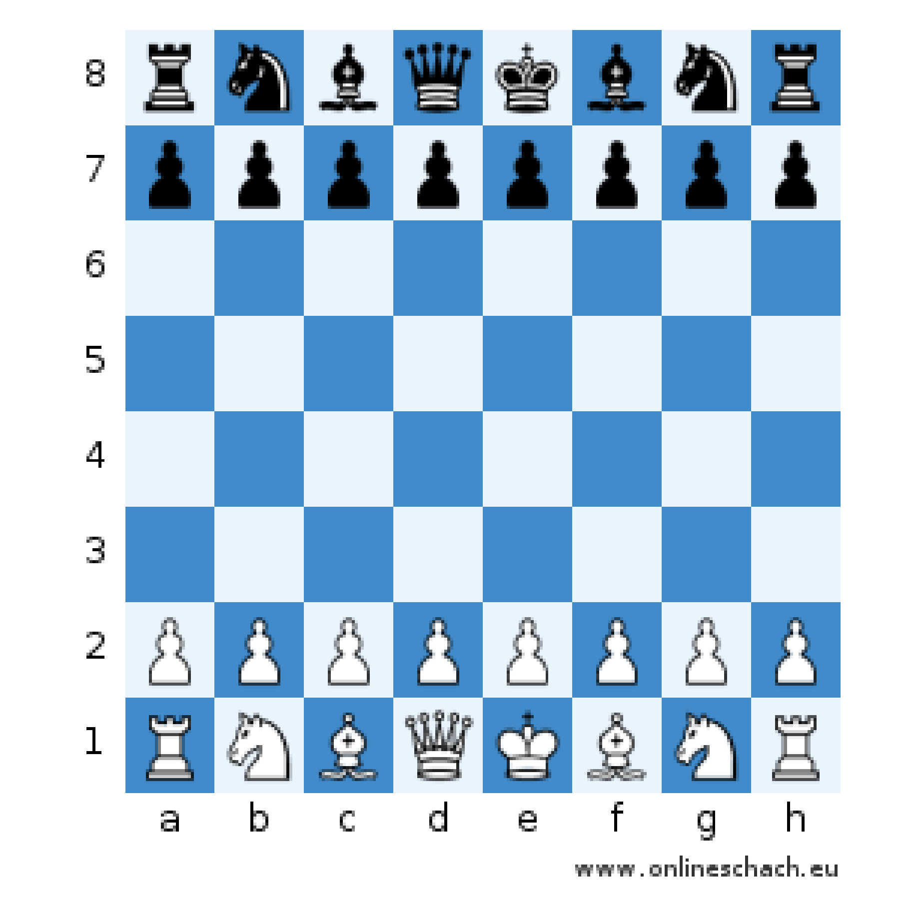
https://da.wikipedia.org/wiki/Computerskak#/media/Fil:Fidelitycc_19.jpg

Trækgenerator



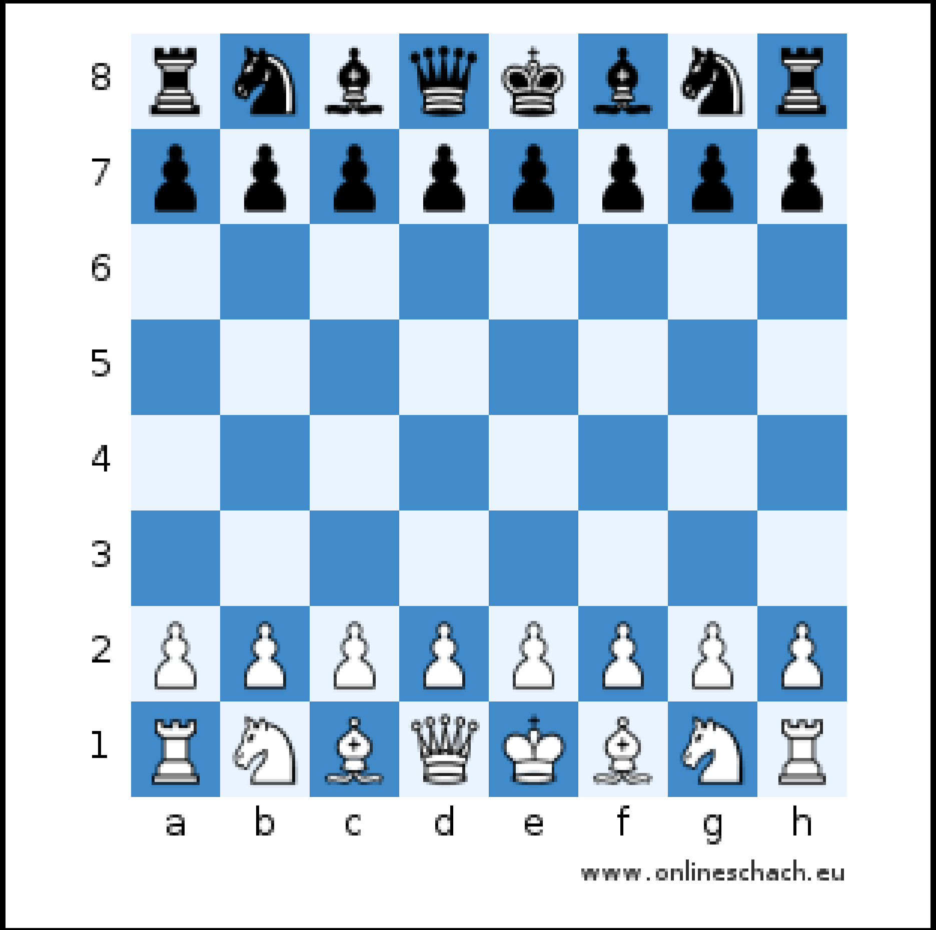


Kompleksitet: Antal trækmuligheder



- Muligheder ved første træk?

Kompleksitet: Antal trækmuligheder

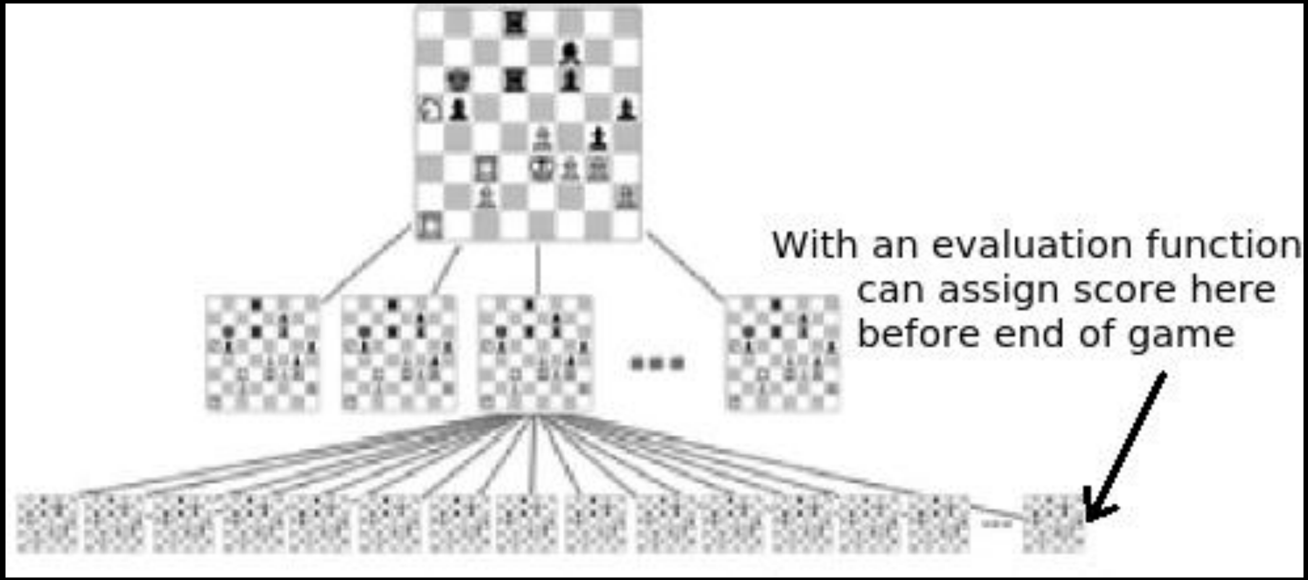


- Muligheder ved første træk?
  - 20
- Gennemsnitligt antal trækmuligheder
  - 40
- Gennemsnitligt antal træk
  - 80
- Kompleksitet
  - ca. 40^{80}
- Antal atomer i universet
  - 10^{78} - 10^{82}

Evalueringsfunktion og søgning

https://thechessworld.com/articles/training-techniques/7-most-important-factors-in-chess-position-analysis/

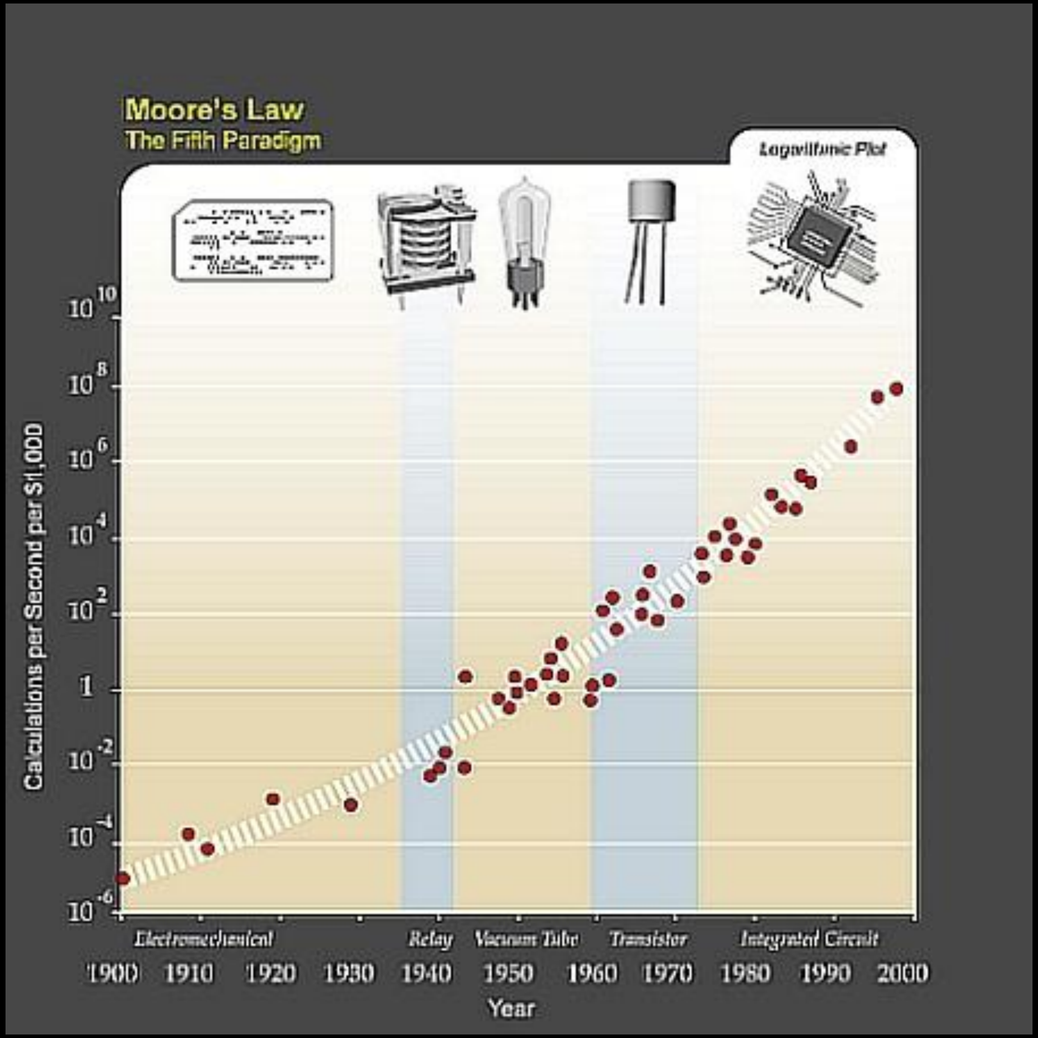
Min-Max search



Moores Lov

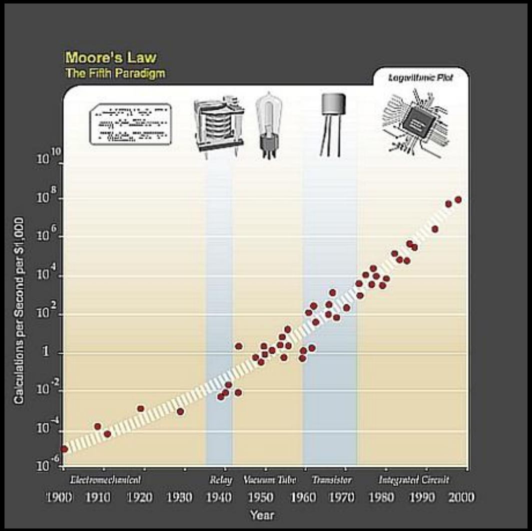
Moores Lov siger – let forenklet – at hastigheden af regneprocessere på computere fordobles hver 18. måned.

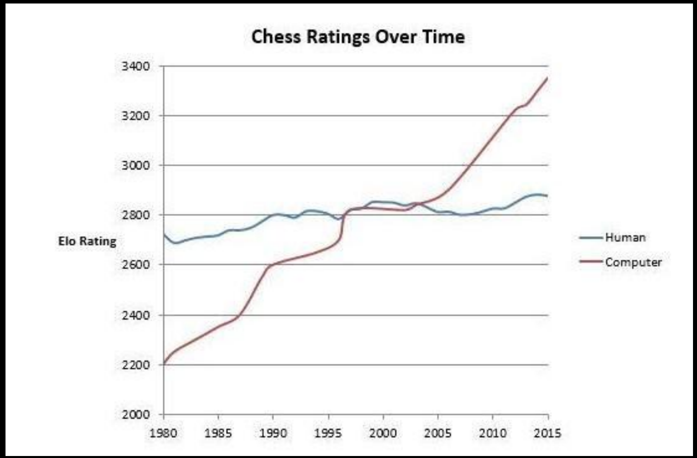
Gordon Earle Moore (født 1929) medstifter af virksomheden Intel.



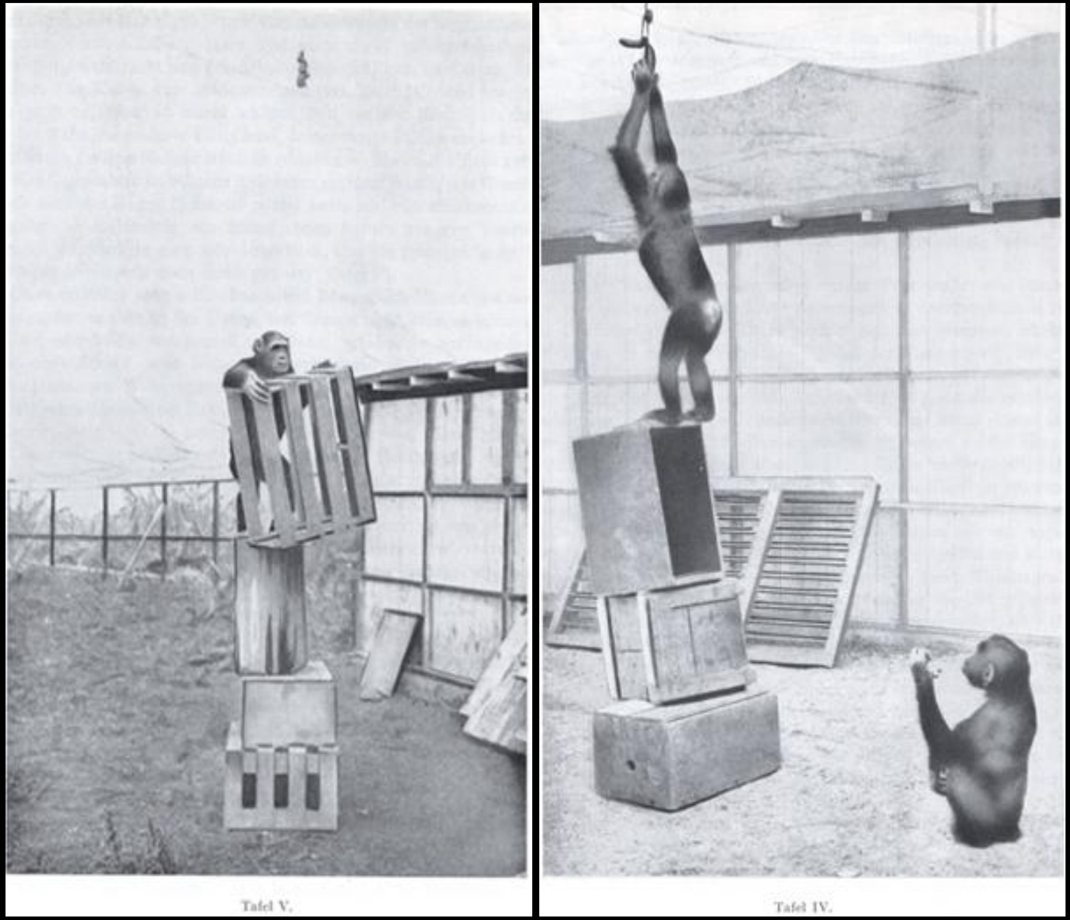
Menneske- og maskinskak og Moores lov

- En god spiller kan forudsige de næste 10 træk.
  - Maks. 60 positioner pr. minut
- Deep Blue kunne evaluere 200.000.000 positioner på et sekund
- I 2009 nåede et skakprogram, der kørte på en mobiltelefon, stormesterniveauet.





Abeeksperimenter

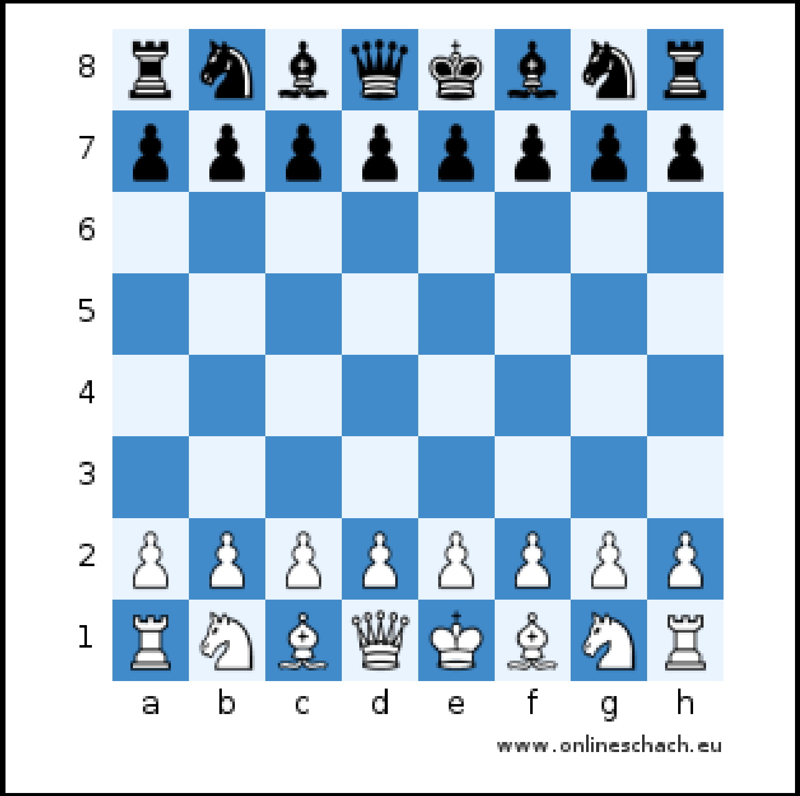


Köhler, Wolfgang (1963). Intelligenzprüfungen am Menschenaffen. Berlin, Göttingen, Heidelberg: Springer.

Der findes ingen robotter, der løser denne type opgave (uden en masse speciel programmering).

Skak er et perfekt eksempel på et symbolsk system

- Diskret tilstandsrum (skakbræt med 64 felter og 24 figurer)
- Figurernes bevægelsesmuligheder er entydig defineret
- Det er klart defineret, hvornår man har vundet, tabt, eller spillet remis
- Symbolske metoder definerer et lignende symbolsk system



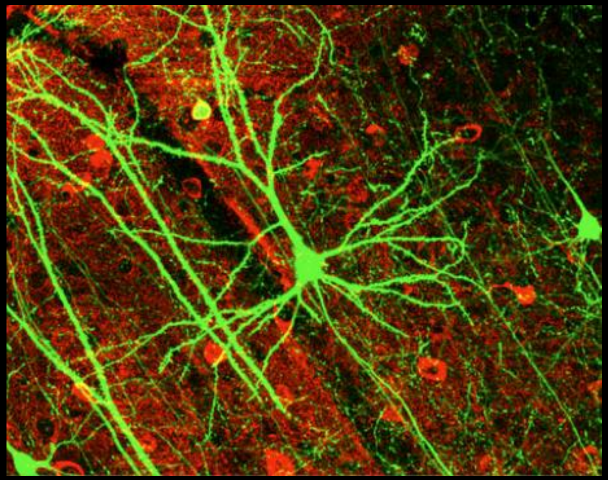
Hjerne og neuroner

- Neuroner er hjernens grundlæggende byggesten
- Antallet neuroner omkring 10^{12}
- Antallet af synapser 10^{15}

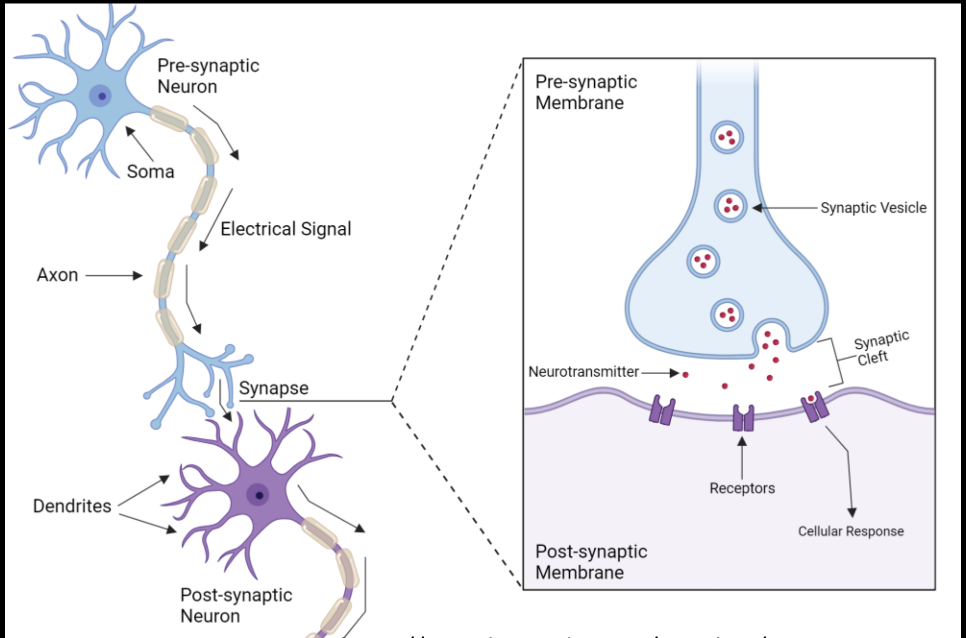
Synapsen er det ganske lille mellemrum mellem to nerve-ender. Synapsen er overgangsstedet mellem to neuroner, hvorigennem et neuron kommunikerer med det næste neuron.

NATIONALT VIDENSCENTER FOR DEMENS

Neuroner under mikroskopet

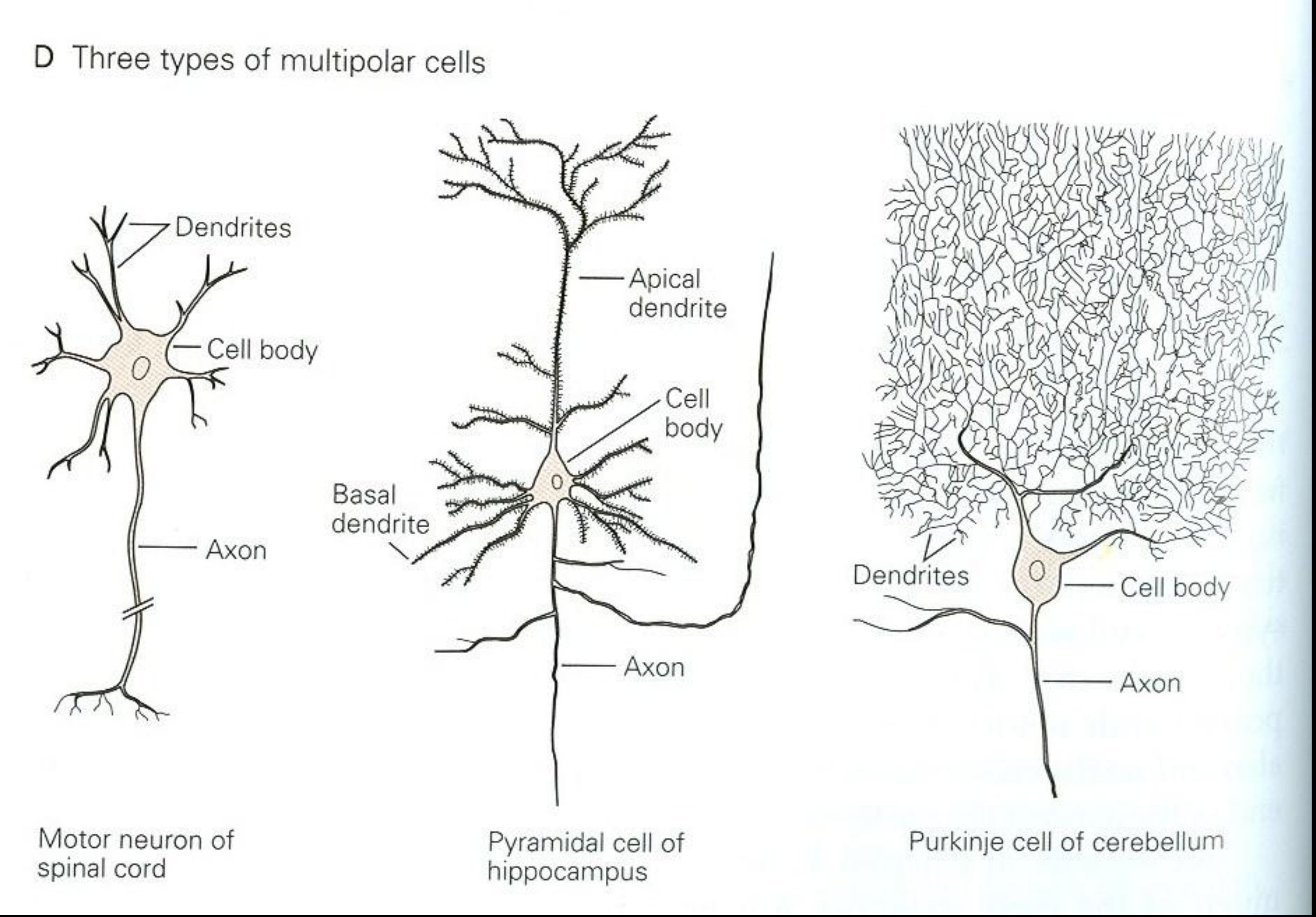


Neuroner og synapser (skematisk)



https://www.biorender.com/template/neuron-synapse

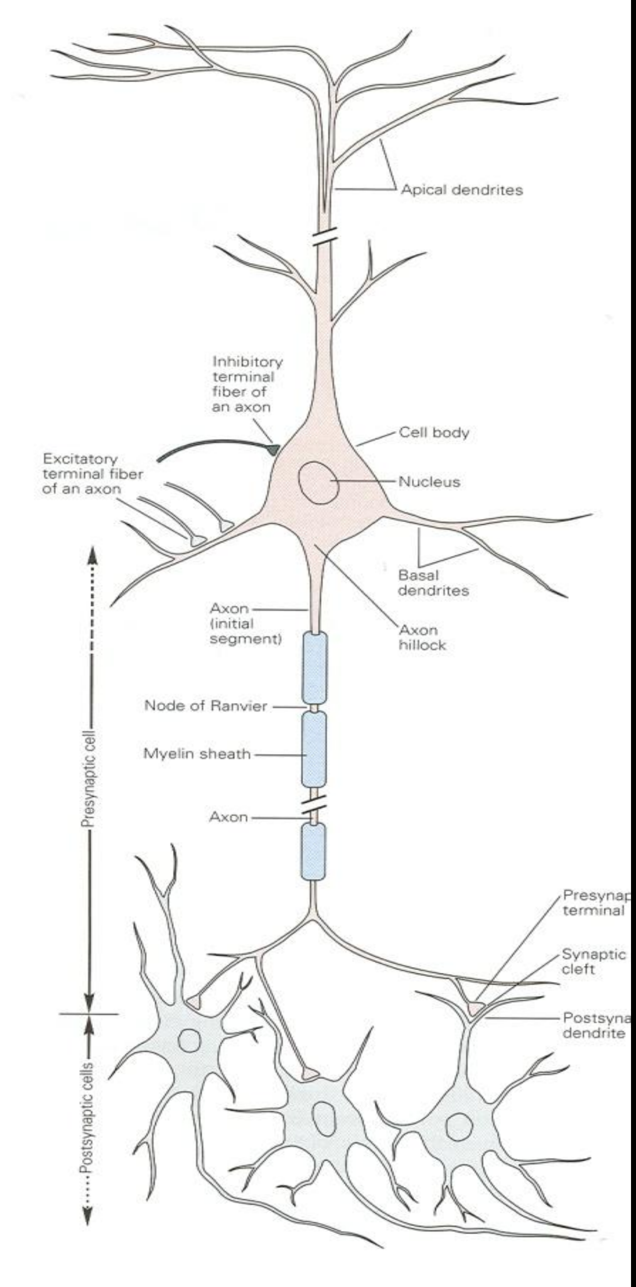
Forskellige slags neuroner



Neuron (skematisk)

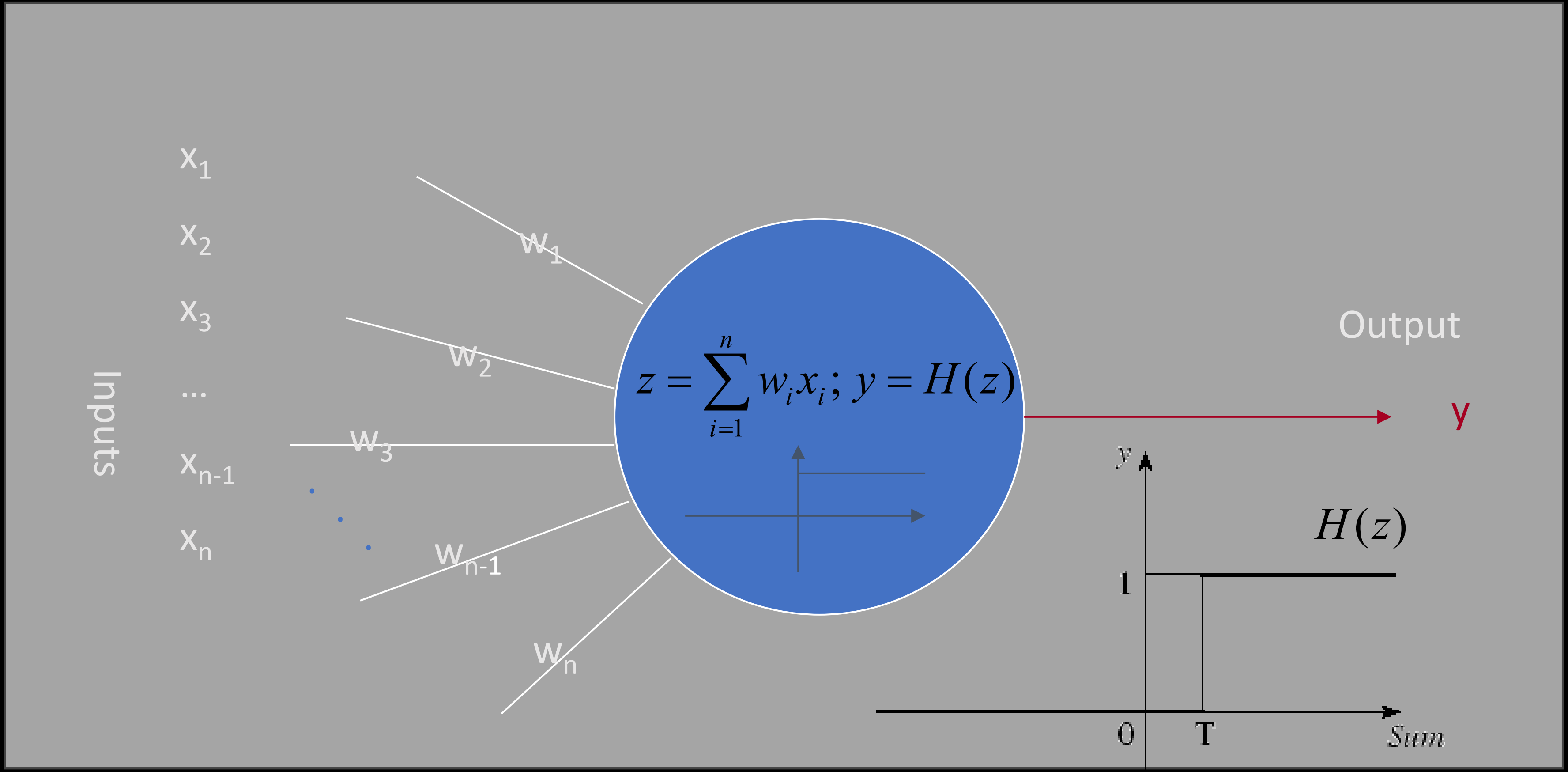
- Struktur
  - Celles krop
  - Dendritter
  - Akson
  - Synapser
- Længde af neurale forbindelser:
  - Mere end 100.000 km

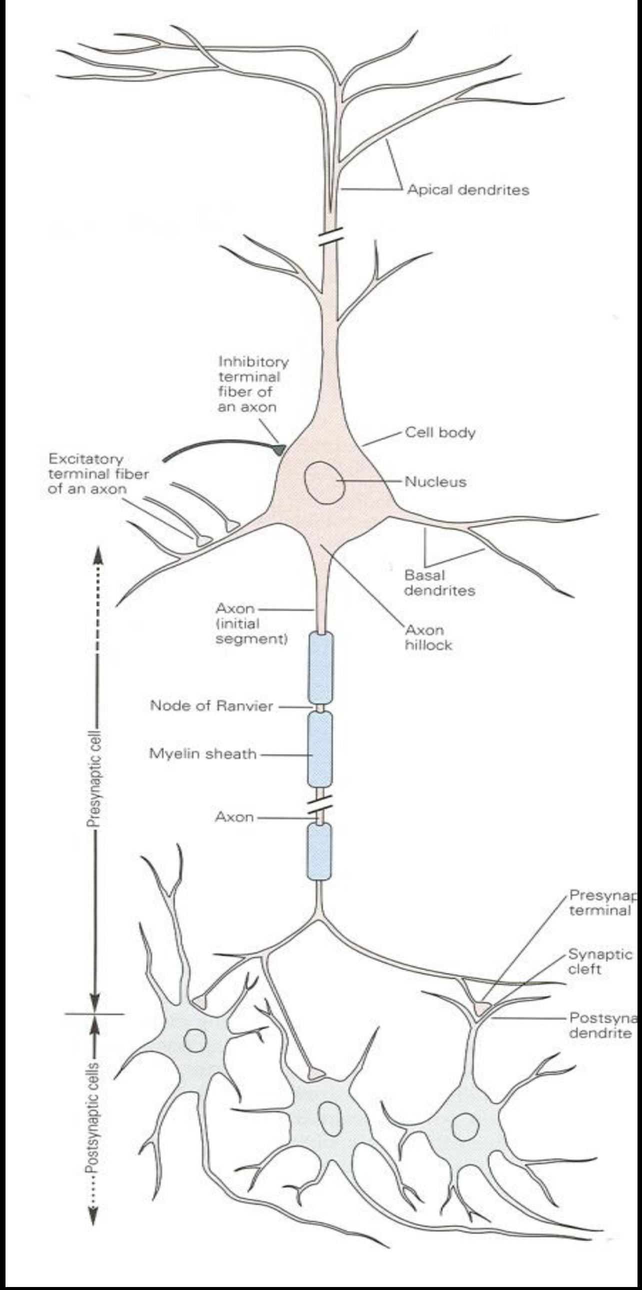
Hvis der i alt er ca. 10^{12} neuroner og 10^{15} synaptiske forbindelser, hvor mange synapser har en neuron i gennemsnit?



Kunstige neuroner

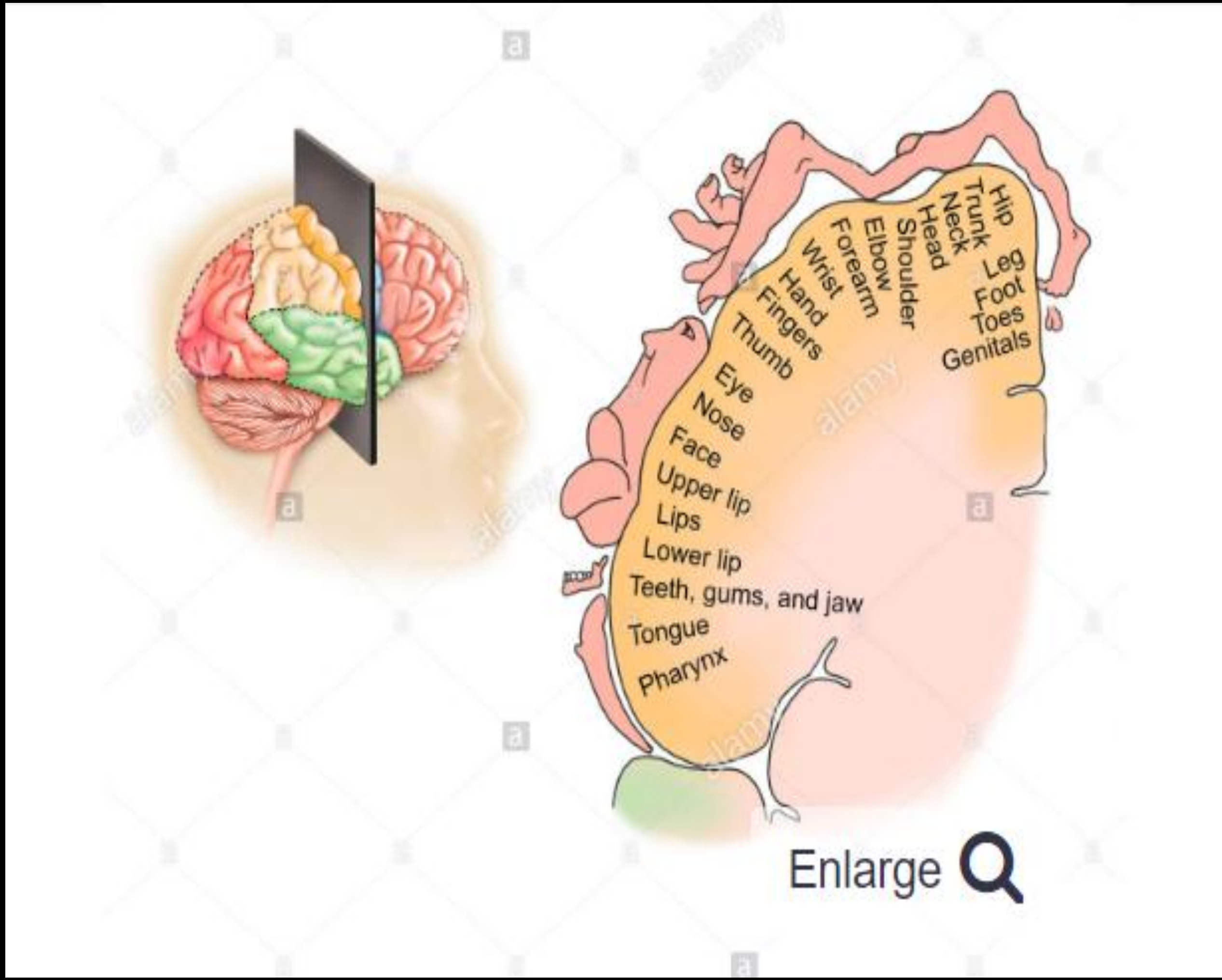
En simpel model af en neuron, som stadigvæk bruges er McCullogh-Pitts model (1943)





Hjernen og neuroner: Kortikal organisation og sansernes resolution

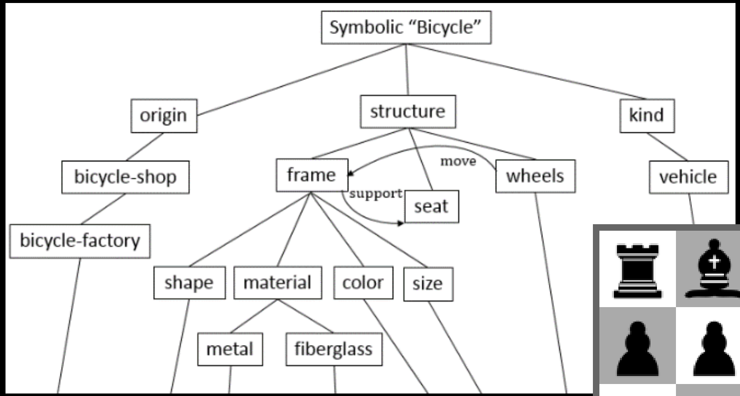
- Hjernes organisation, især for de sensoriske systemer, beskrives ofte i form af kort.
- Sensorisk information fra foden projiceres til et kortikalt sted, og projektionerne fra hånden målrettes mod et andet sted.
- Som et resultat ligner den kortikale repræsentation af kroppen et kort (eller homunculus).
- Kortets resolution er *ikke* homogen

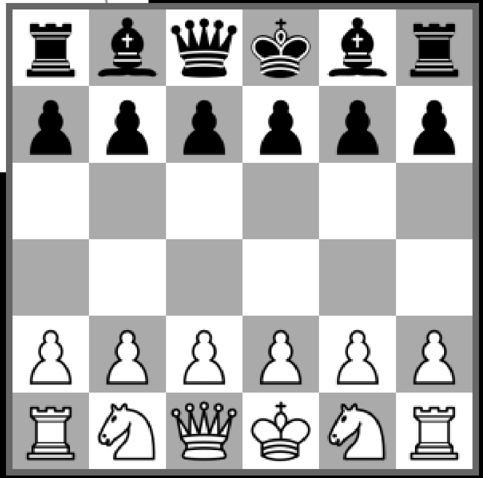


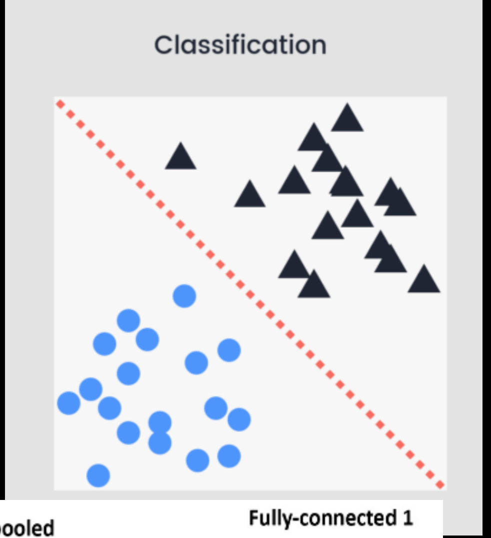
Motor Homunculus Model

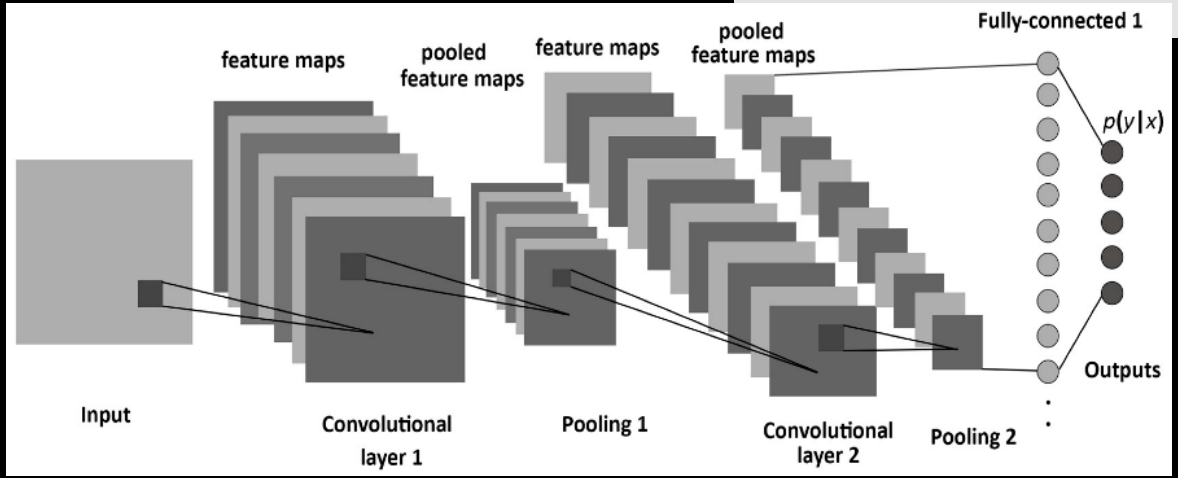
https://fineartamerica.com/featured/motor-homunculus-model-natural-history-museum-londonscience-photo-library.html

Symbolsk vs. subsymbolsk KI





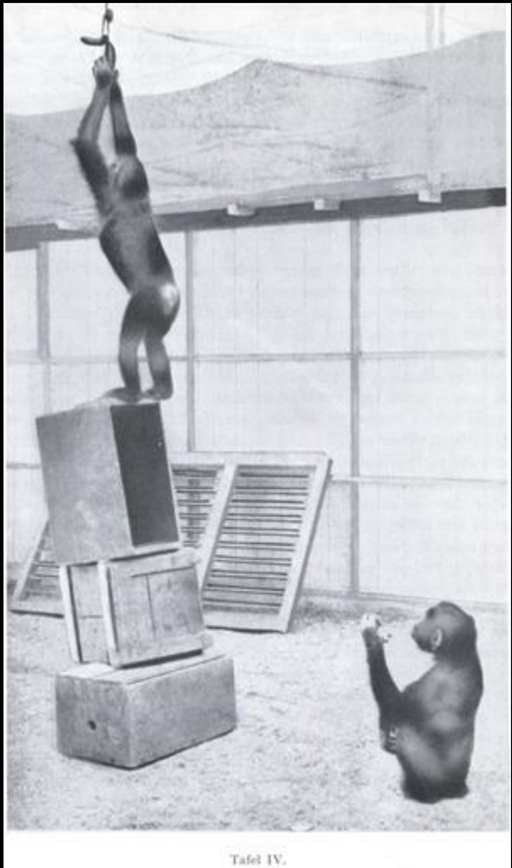




- Symbolsk tilgang
  - Intelligens er baseret på diskrete symboler i formelle systemer.
  - Skakcomputere og andre ekspertsystemer.

- Sub-symbolsk tilgang
  - Behandlingen skal starte med den sensoriske information.
  - Kunstige Neurale Netværk (KNN)

Abe-Eksperimenter



Köhler, Wolfgang (1963). Intelligenzprüfungen am Menschenaffen. Berlin, Göttingen, Heidelberg: Springer.

Der findes ingen robotter, der løser denne type opgave (uden en masse speciel programmering).

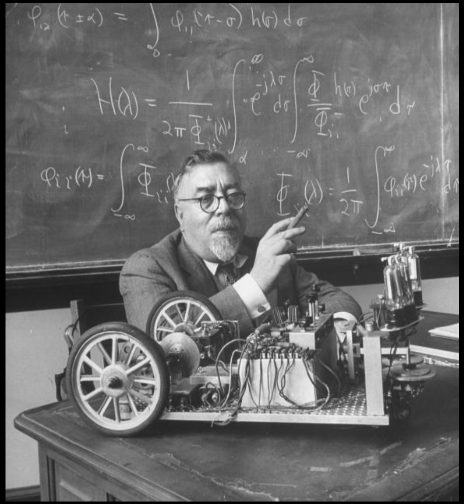
Hvad sker på det symbolske og det sub-symbolske niveau?

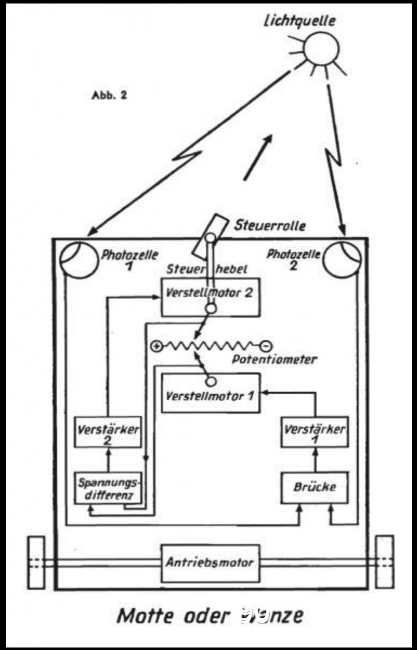
- Det symbolske niveau (planlægning)
    - Man skal lave en plan for, hvordan man organiserer kasserne, og i hvilken rækkefølge man kan stable dem oven på hinanden
- Det subsymbolske niveau (sensorik)
    - Hvor er kasserne?
    - Hvordan griber man kasserne?
    - Hvordan og hvor skal man placere en kasse, således at der opstår en stabil struktur?

Norbert Wiener: Palomilla, en af de første robotter (1949)

- Robotten kunne bevæge sig mod lyset ved hjælp af et internt kontrolsystem, som møl, eller kravle væk fra lyset, som et insekt.
- Når enheden bevægede sig, ændrede dens sensoriske elementer retning og styrede dermed den motoriserede bevægelse.
- På denne måde var Palomilla et af de første kunstige væsener, der simulerede adfærd i et integreret kredsløb.

Oversat fra: Ute Holl, Emanuel Welinder. Die anthropologische Differenz der Medien. Wissenschaft und Phantasma





Robot (Definition)

En robot er en programmerbar maskine, der ved interaktion (vekselvirkning) med sine omgivelser autonomt (selvstyrende) kan udføre en mangfoldighed af opgaver.

Danmarks Nationalleksikon

A robot is an autonomous machine capable of sensing its environment, carrying out computations to make decisions, and performing actions in the real world.

IEEE

Er Palomilla en robot?

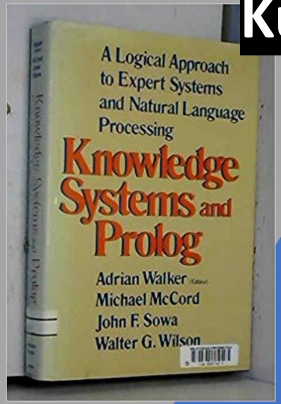
Er en moderne vaskemaskine roboter?

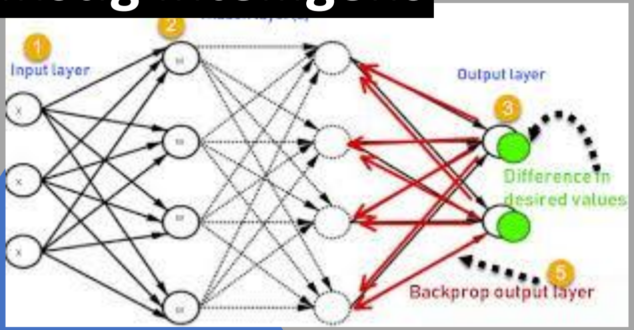
Overblik

- Hvad er intelligens? Hvad er kunstig intelligens?
- **Trekant: KI, Robot- og Computerteknologi**
  - Indtil 2. Verdenskrig
  - 1940-1956: Begyndelsen
  - **1957-1984: Kampen mellem de symbolske og subsymbolske metoder**
  - 1985-2005: Evolutionære fremskridt og Bias/Variance dilemmaet
  - 2005-: Deep Neural Networks tager over
- ChatGPT og embodiment

Trekant 1957-84: KI, Computer- og Robotteknologi

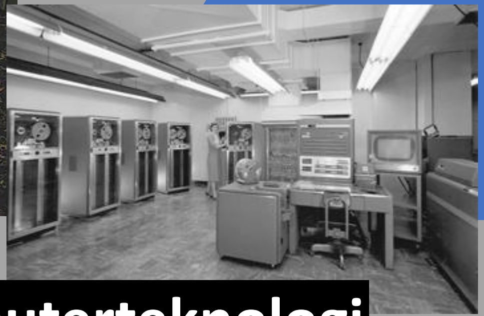
Kunstig intelligens





Lisp maskine
Symbolics 3640
1984

IBM-704 main-frame
computer
(sent 50er)

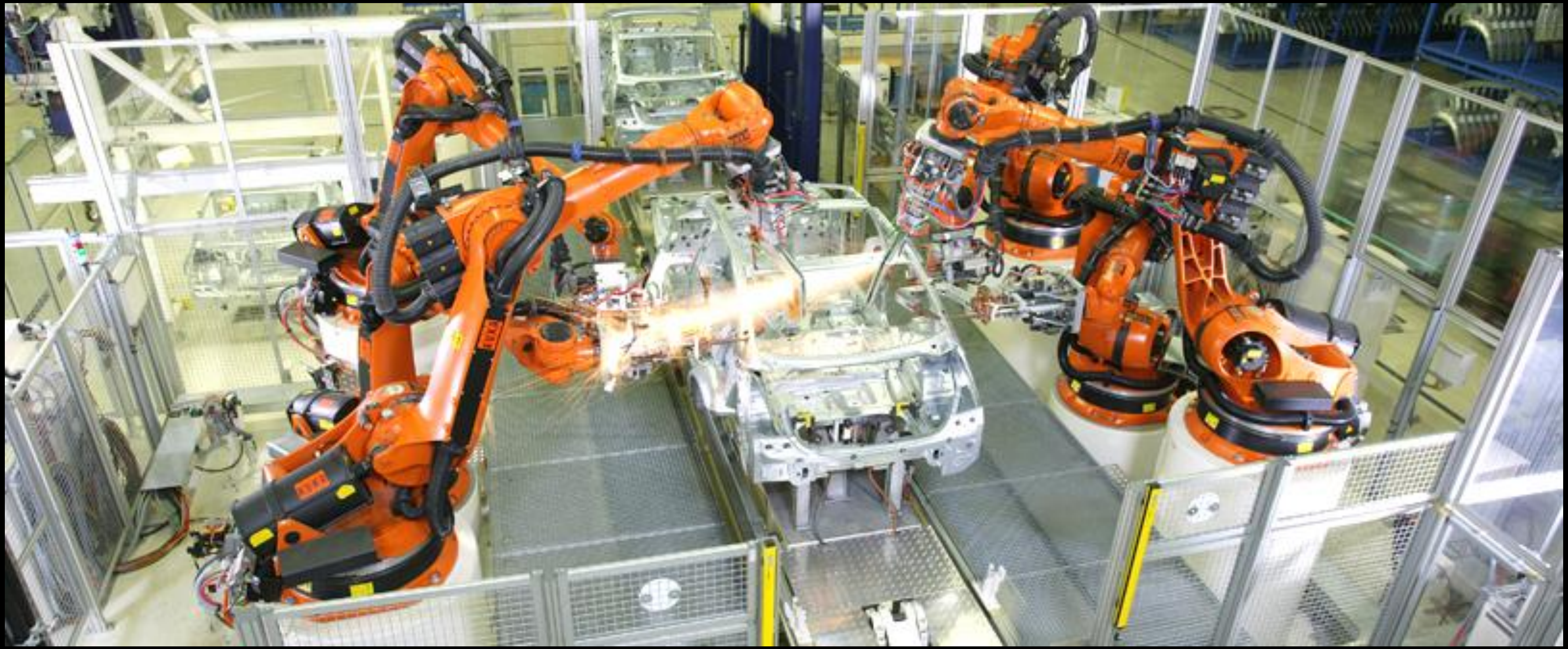


Computerteknologi

Robotteknologi

Industrial Robotics startede i 1960'erne

- Første industrirobot (1961)
  - Unimate-robotten arbejdede på et General Motors-samlebånd
- Første industrirobot med seks frihedsgrader (1973)
  - Disse robotter har erstattet gentagende bevægelser i masseproduktion, hvilket førte til en høj grad af velstand i mange lande



Ekspertsystem

Et computerprogram, der kan støtte eller udføre beslutnings- eller diagnoseprocesser på samme måde, som en menneskelig ekspert ville gøre det.

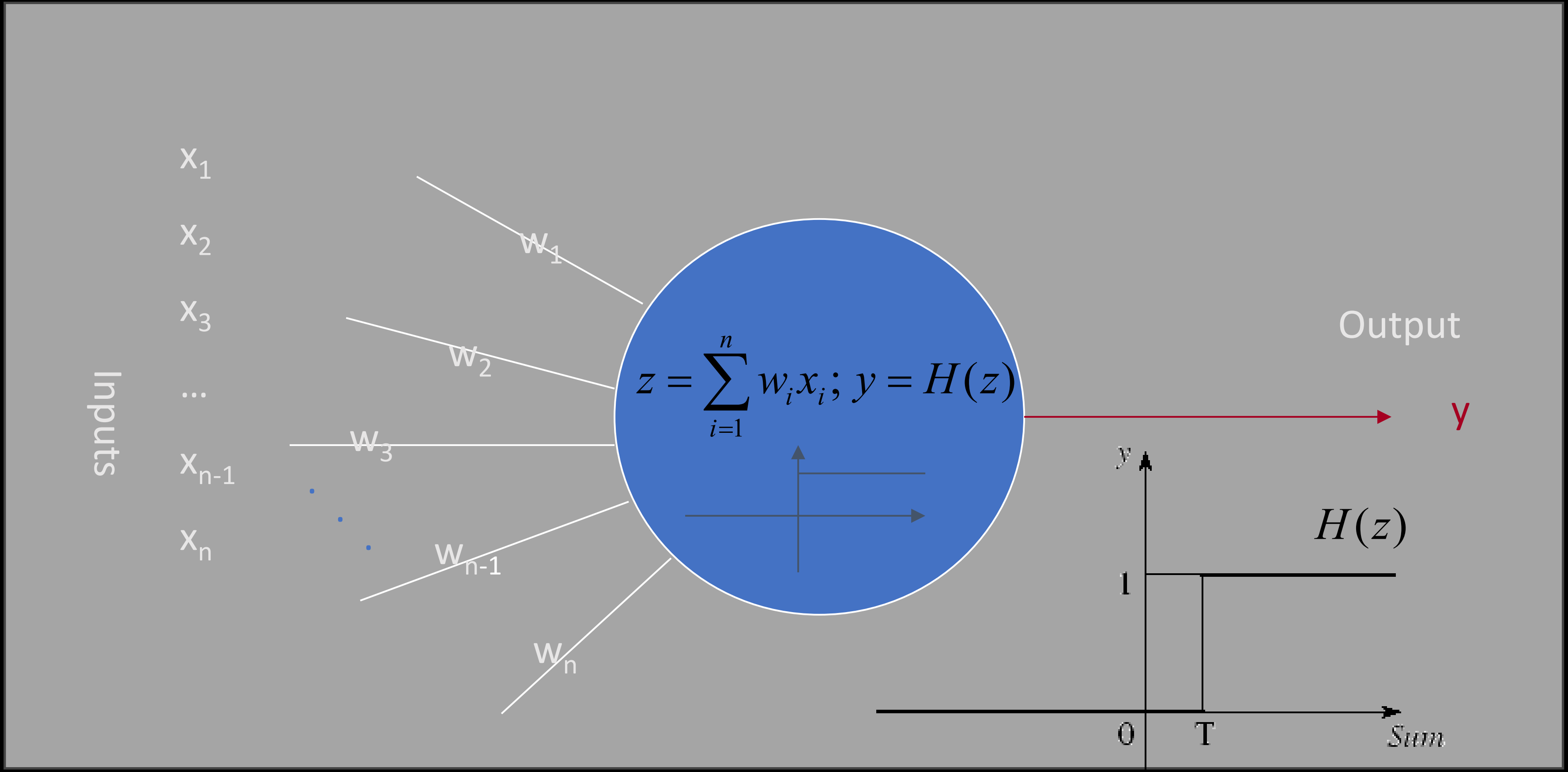
Gyldendal Den Store Danske

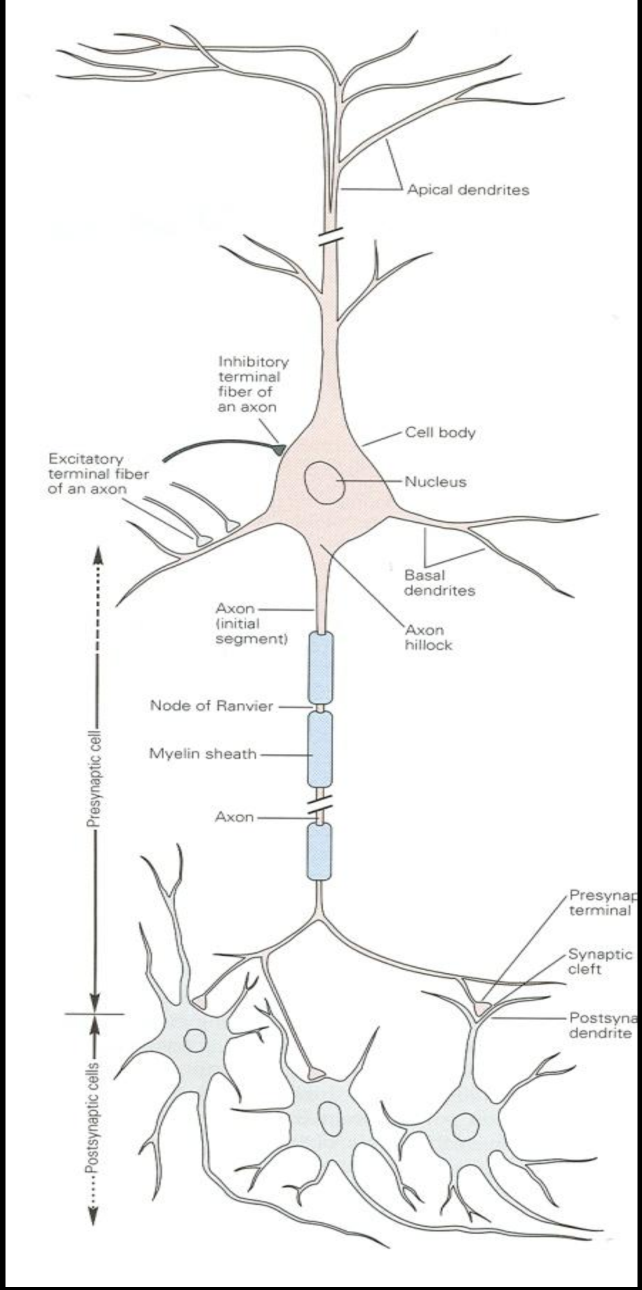
- Ekspertsystemer omfatter en bred vifte af systemer, som løser komplekse problemer i en snævert domæne (design af elektriske kredsløb, klassifikation af bakterier (MYCIN, 1970)).
- De første ekspertsystemer kom allerede i 1950'erne
- begrænsede af den ringe computerkraft og helt urealistiske forventninger.
- Algoritmer baserede på *symbolske regler*

Lisp maskine Symbolics 3640 (1984)

Kunstige neuroner

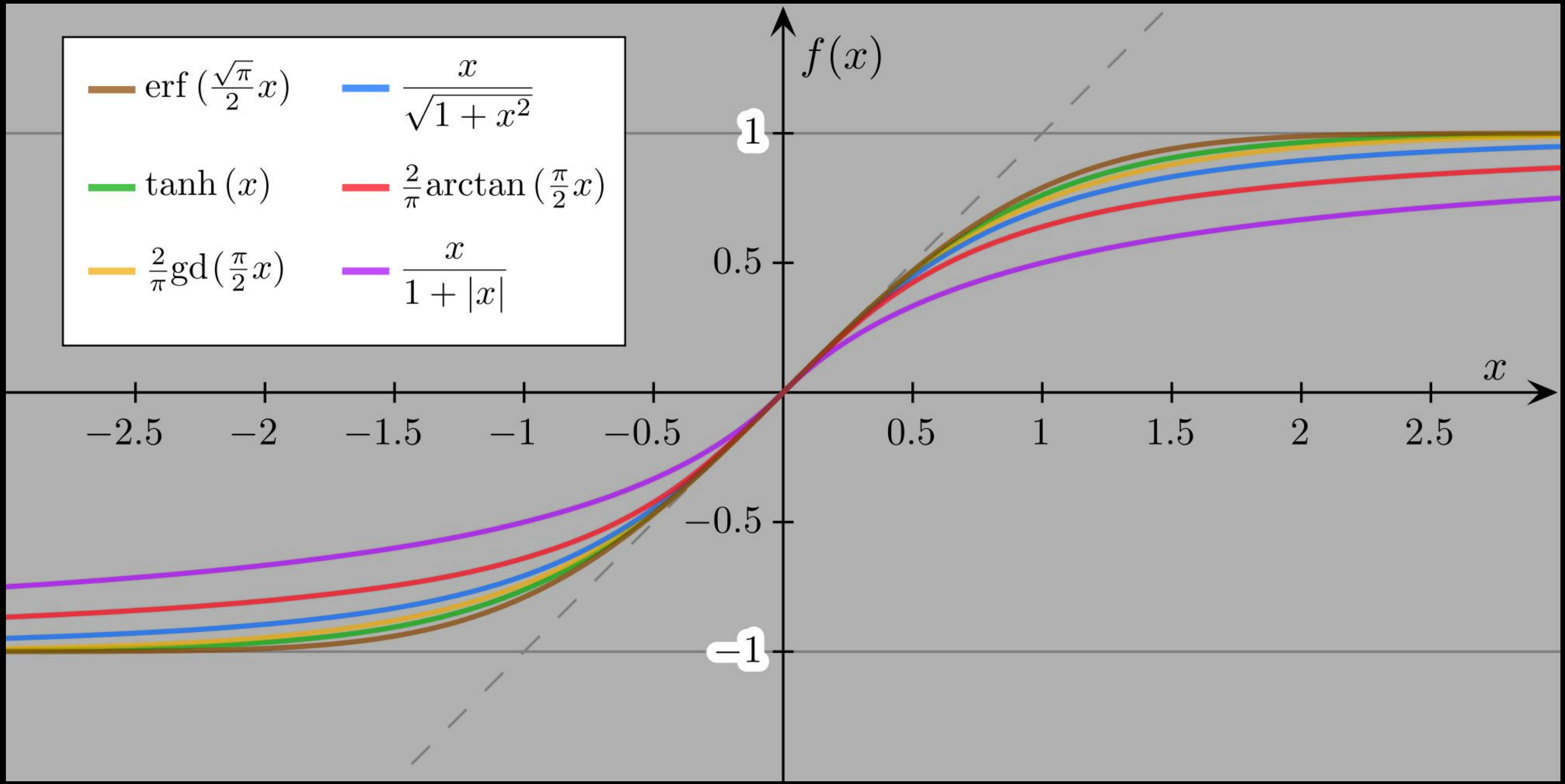
En simpel model af en neuron, som stadigvæk bruges er McCullogh-Pitts model (1943)



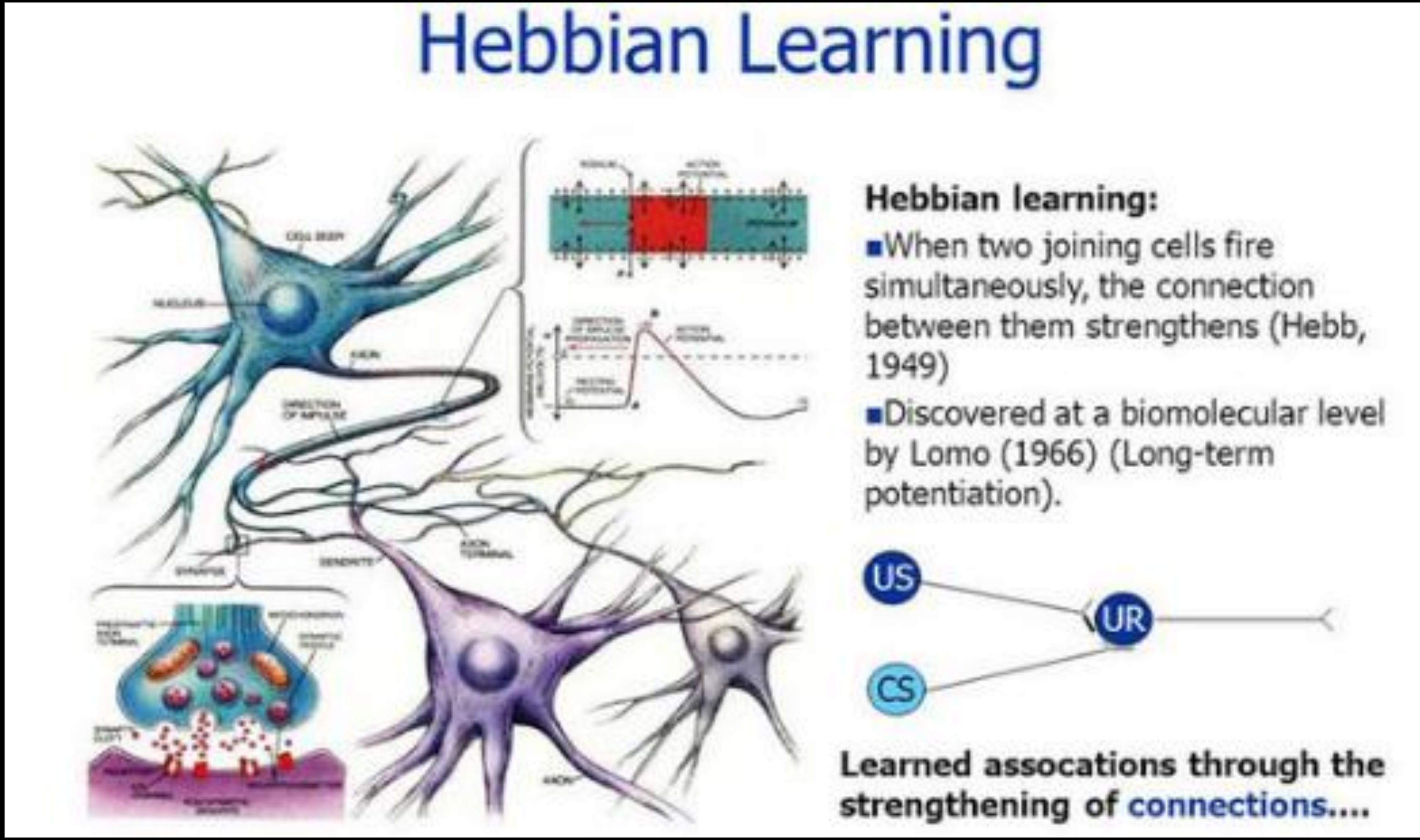


Fra tærskelfunktion til sigmoide funktioner

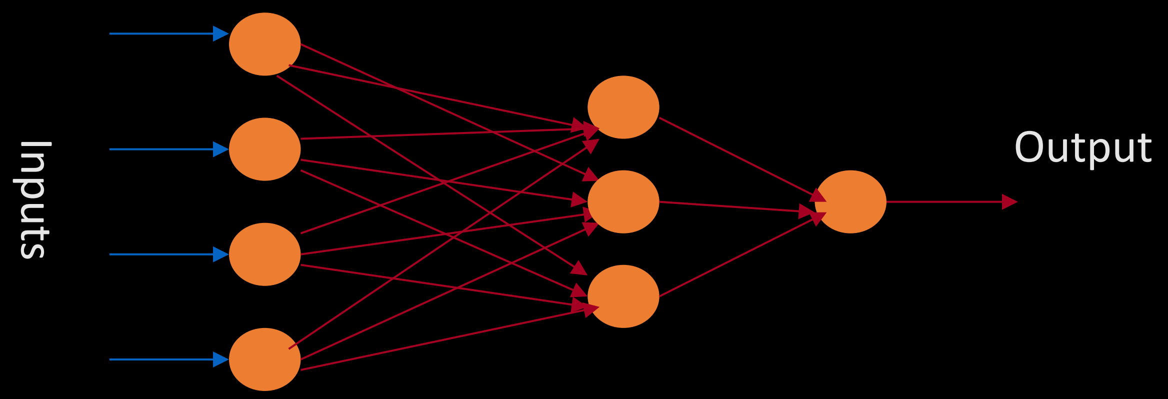
Forskellige sigmoide funktioner



Hebbs regel


Hebbian Learning
Hebbian learning:
When two joining cells fire simultaneously, the connection between them strengthens (Hebb, 1949)
Discovered at a biomolecular level by Lomo (1966) (Long-term potentiation).
US
UR
CS
Learned assocations through the strengthening of connections....

Artificial Neural Networks (Cybernetics, Connectionism)

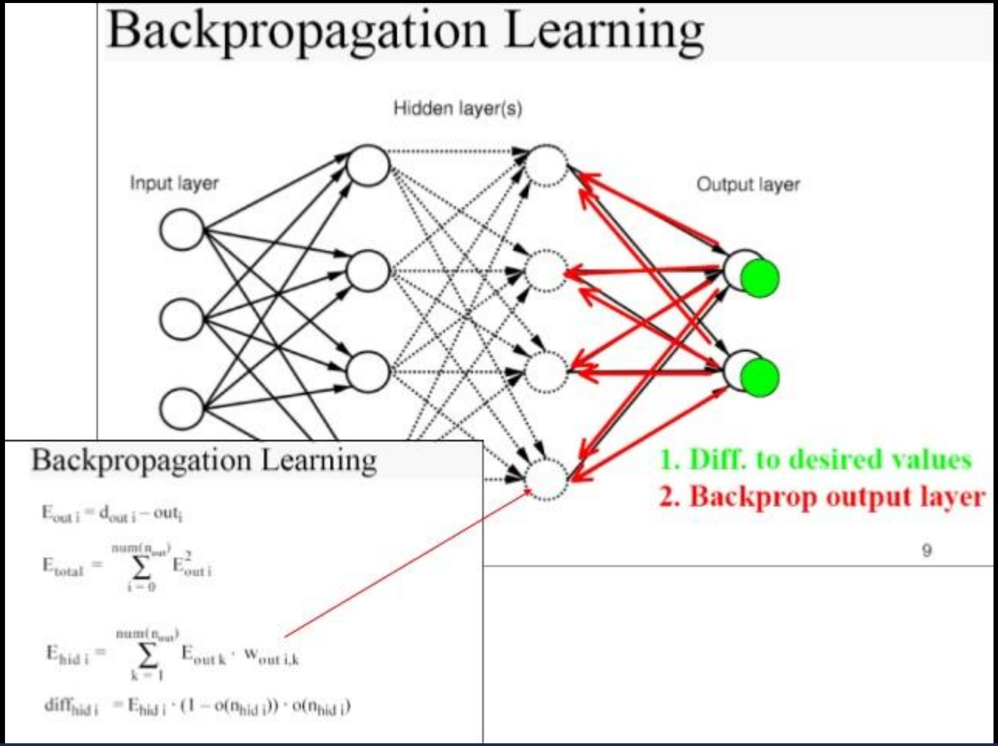


Et typisk neuralt netværk består af en række simple beregningsenheder, kaldet (kunstige) neuroner. En neuron har i lighed med biologiske nerveceller et antal input-kanaler ... og en enkelt output-kanal ... Disse kanaler bærer signaler mellem neuronerne og forbindes med hinanden i kontaktpunkter, der kaldes (kunstige) synapser.

Danmarks Nationalleksikon

- Formålet med det neurale netværk er at omdanne input til meningsfulde output.
- Perceptron (1957) fra Frank Rosenblatt

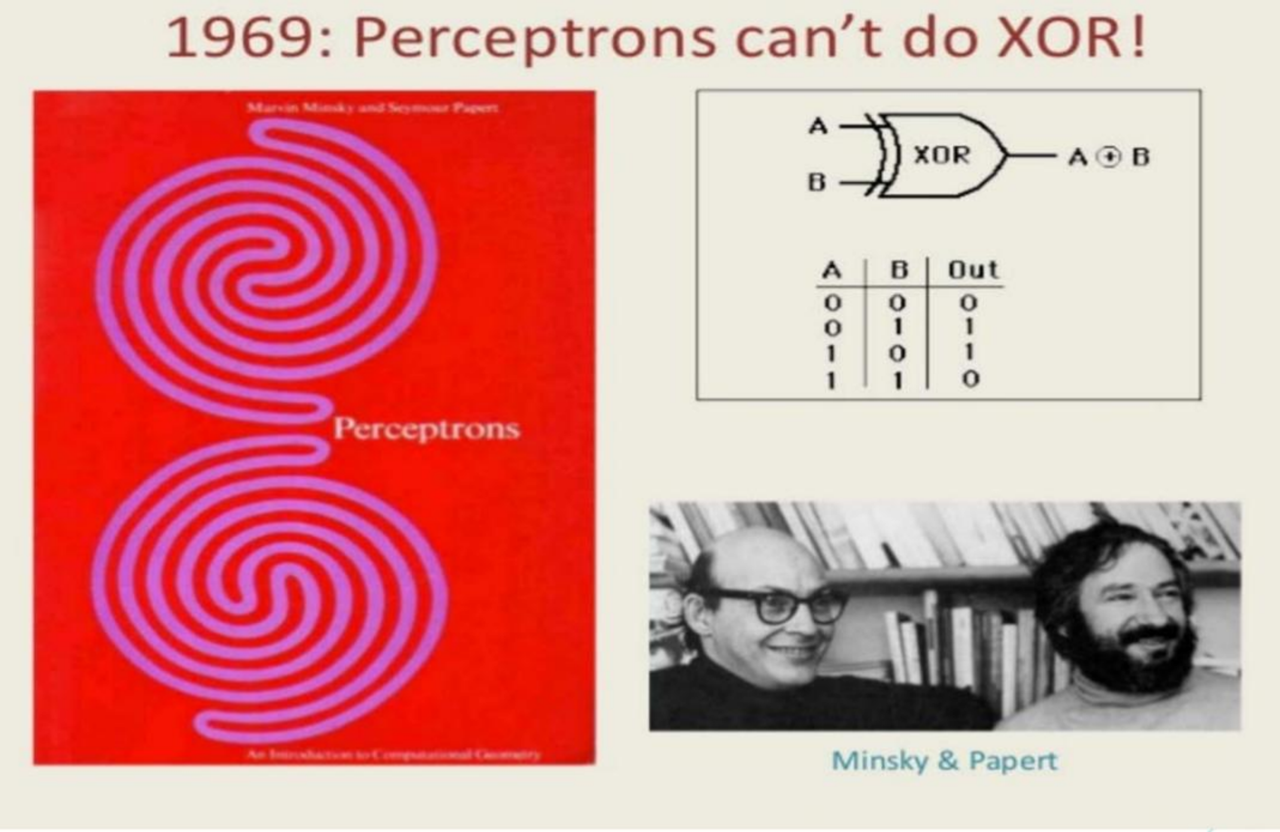
Backpropagation (1970-1982)



Backpropagation of error (henceforth BP) is a method for training feed-forward neural networks (Artificial Neural Networks). A specific implementation of BP is an iterative procedure that adjusts network weight parameters according to the gradient of an error measure.

SPRINGER NATURE Link

Første store angreb: Minsky og Papert (1969)



MINSKY M., PAPERT S. (1969), *Perceptrons: An Introduction to Computational Geometry,* Cambridge MA, The MIT Press.

Coining the term AI

At the "Dartmouth founding meeting, John McCarthy and Marvin Minsky, coined the term "artificial intelligence" (AI) in 1956.

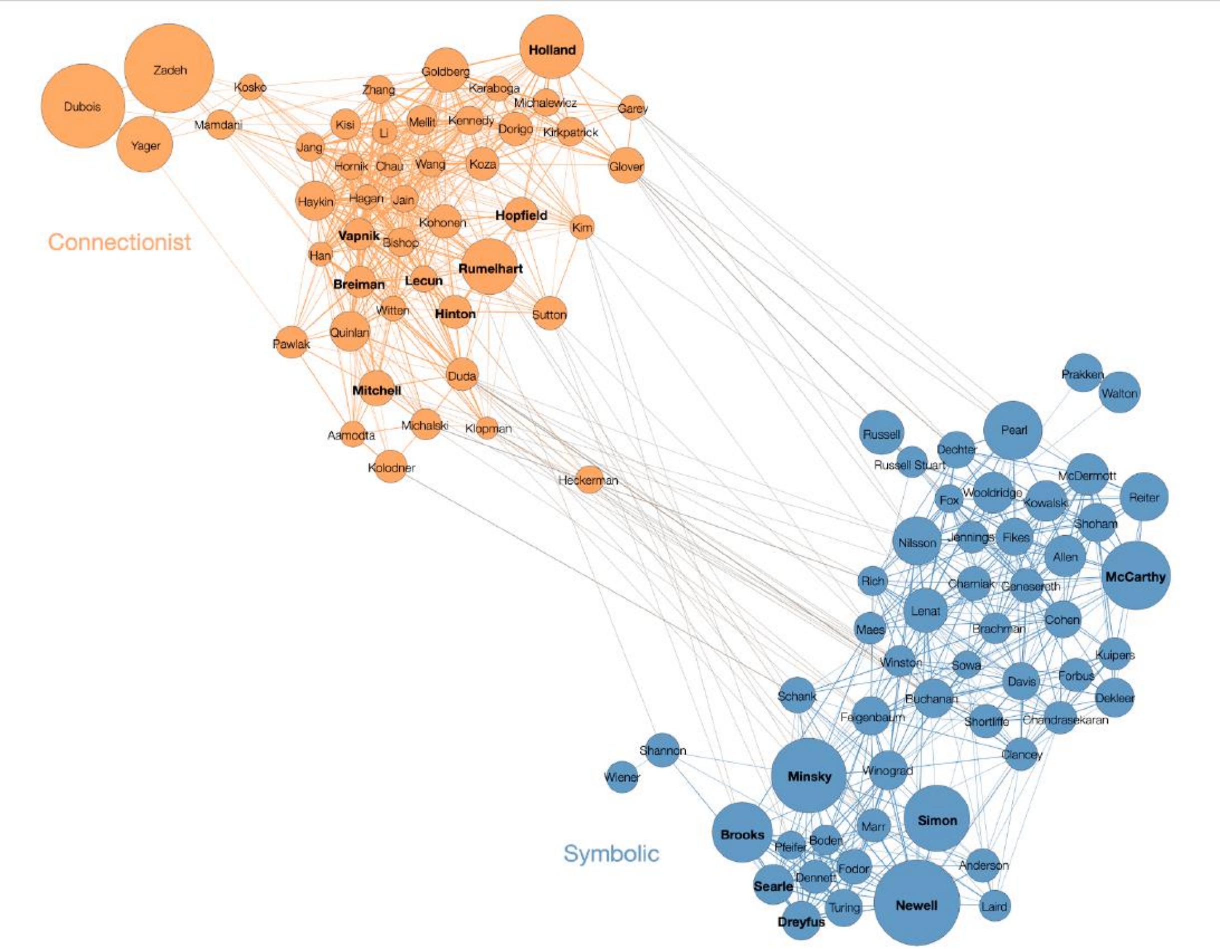
Their intention was to oppose the connectionism of early cybernetics.

The purpose of "symbolic" AI was to implement rules in computers via programs, so that high-level representations could be manipulated."

Cardon et al (2018)

- Klog populisme: Lav en korrekt udtalelse, som offentligheden dog vil overse, og dermed sikre finansiering i de næste 10 år.

Kamp mellem den symbolske og subsymbolske tilgang

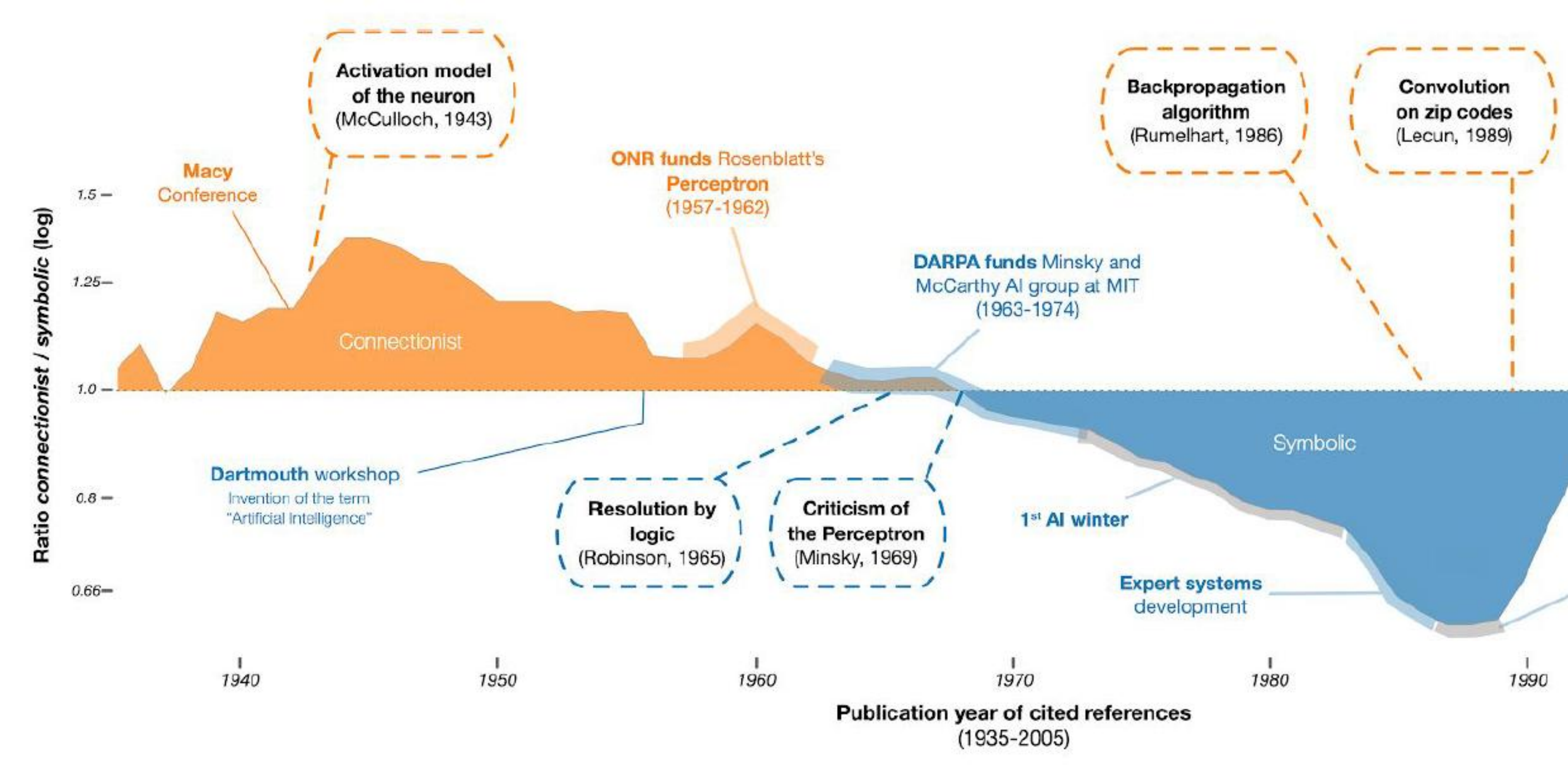


Cardon et al (2018) NEURONS SPIKE BACK
The Invention of Inductive Machines and the Artificial Intelligence Controversy

Co-citation netværk af de 100 mest citerede forfattere i videnskabelige publikationer, der nævner “Kunstig intelligens” (februar 2018)

- Symbolsk tilgang
  - Intelligens er baseret på regler for at manipulere diskrete symboler i formelle systemer.
    - Skakcomputere og andre ekspertsystemer.
- Subsymbolsk tilgang
  - Behandling bør starte med den sensoriske information.
  - Systemet lærer fra erfaring
    - Kunstige neurale netværk

Udvikling af den akademiske indflydelse på den sub-symbolske og symbolske tilgang



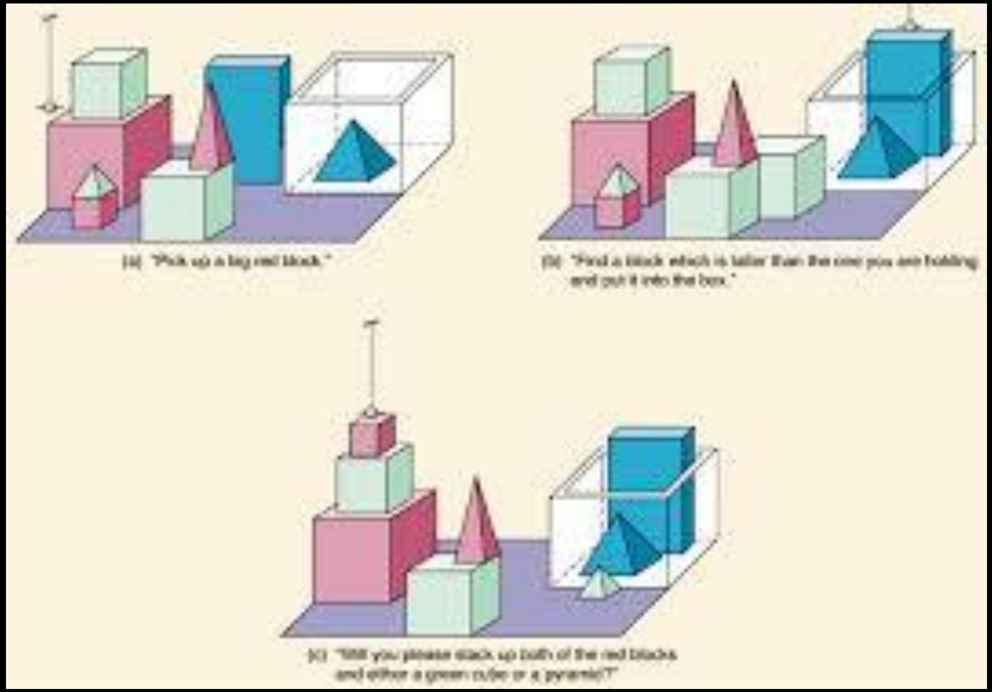
Cardon et al (2018)

Grafen viser ændringerne i forholdet mellem antallet af publikationer citeret i det subsymbolske korpus (orange) og det symbolske korpus (blå), begge justeret i forhold til det samlede antal publikationer i WoS.

Symbolsk tilgang: GOFAI

- GOFAI: Good Old-Fashioned Artificial Intelligence
  - John Haugeland, *Artificial Intelligence: The Very Idea* (1985)

Goofy
(Fedtmule)



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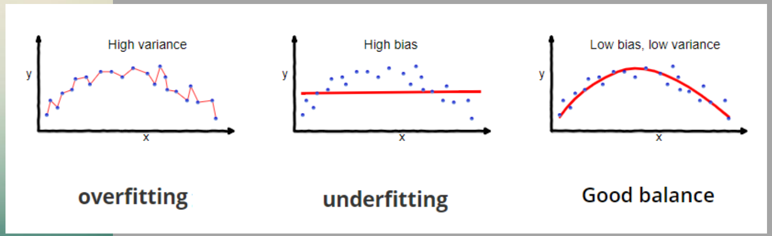
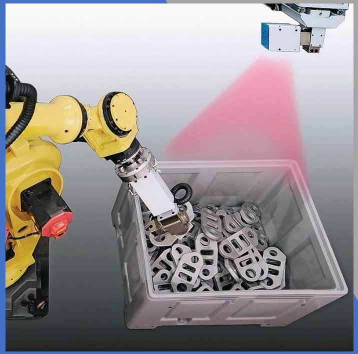
Trekant 1985-2005: KI, Computer- og Robotteknologi

Kunstig intelligens



IBM PC (model 5150)
1981

Computerteknologi



Robotteknologi

Personal Computer og WWW

IBM PC (model 5150) - 1981

Offentligt tilgængeligt fra 1990

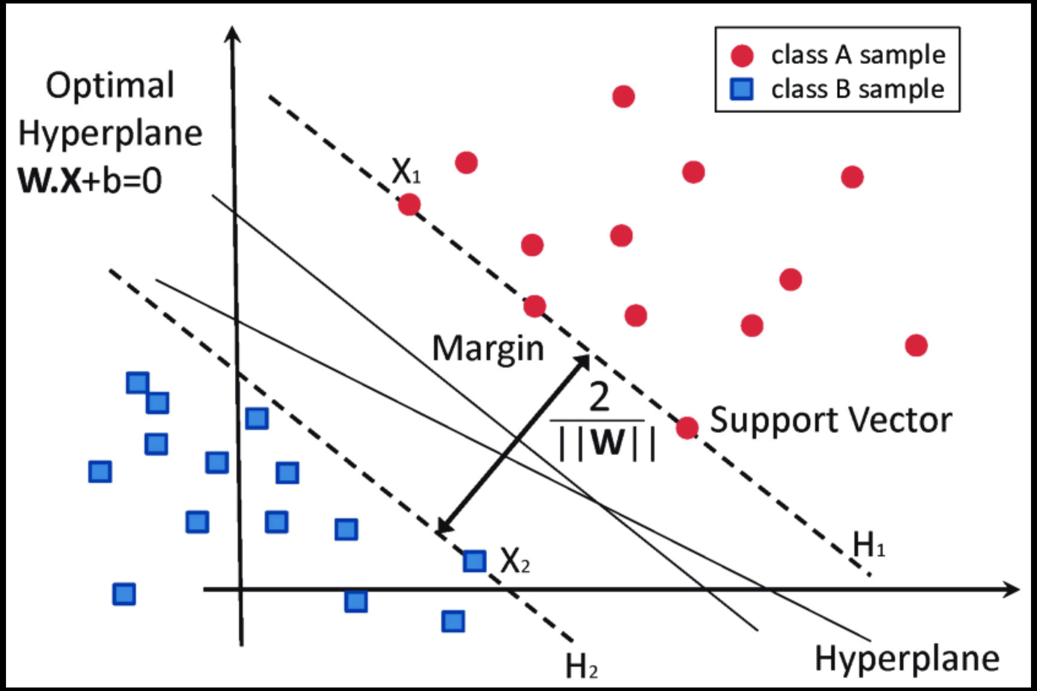
Udviklingen inden for robotteknologi

Kameraer kommer ind i robotløsninger

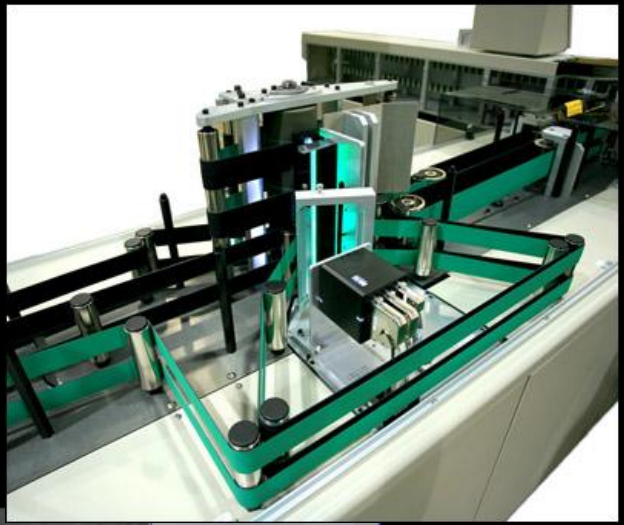
Asimo (2000-2022)
Humanoid Robotics hos Honda

Status inden for KI

- Mange forskellige metoder var i brug
  - Support Vector Machines (SVMs)
  - Neurale Netværk
  - Statistiske metoder
  - K-Nearest Neighbor (KNN)
  - …
- Meget tid gik til data pre-processing

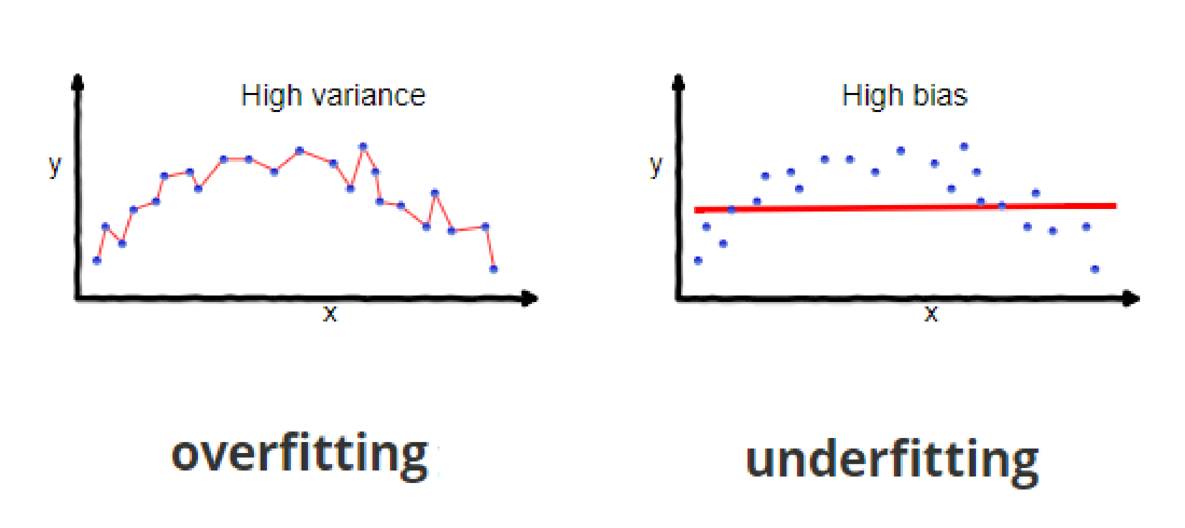


Kunstig intelligens



Det store gennembrud er ikke kommet!

Andet store angreb på neurale netværk: Dilemmaet mellem bias og varians (1992)



Geman, Stuart; E. Bienenstock; R. Doursat (1992). "Neural networks and the bias/variance dilemma". Neural Computation. 4: 1–58

Inferring this complexity from examples, that is, learning it, although theoretically achievable, is, for all practical matters, not feasible: ***too many examples would be needed.***

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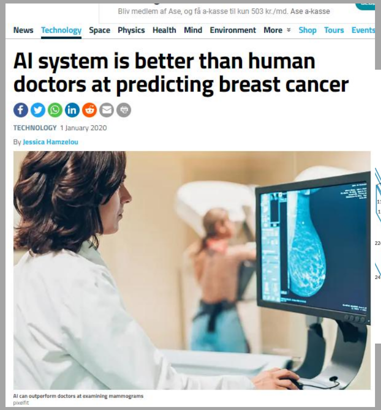
- ChatGPT og embodiment

Trekant 2006 - nu:
KI, Computer- og Robotteknologi

Kunstig intelligens

Deep Learning


Bliv medlem af Ase, og få a-kasse til kun 503 kr./md. Ase a-kasse

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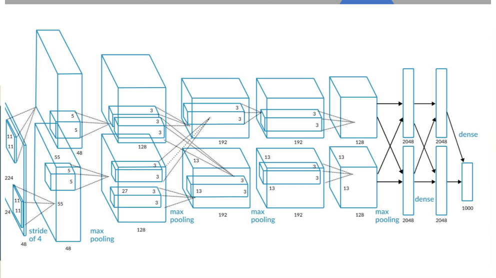
AI system is better than human doctors at predicting breast cancer

TECHNOLOGY 1 January 2020

By Jessica Hamzelou

AI can outperform doctors at examining mammograms




Chinese man caught by facial recognition at pop concert

13 April 2018

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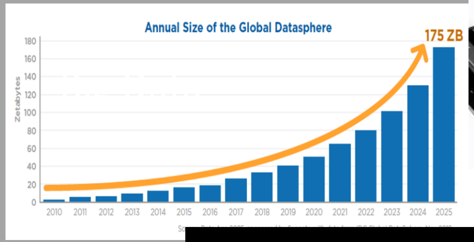
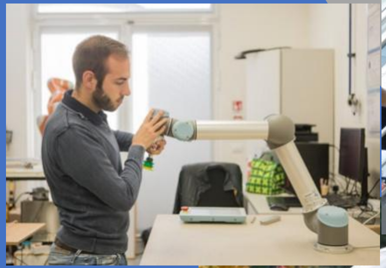
(File Photo) There are an estimated 170 million CCTV cameras already in place in China

GPUs





Computerteknologi

Robotteknologi

10-02-2026

Deep Learning beats it all


Bliv medlem af Ase, og få a-kasse til kun 503 kr./md. Ase a-kasse

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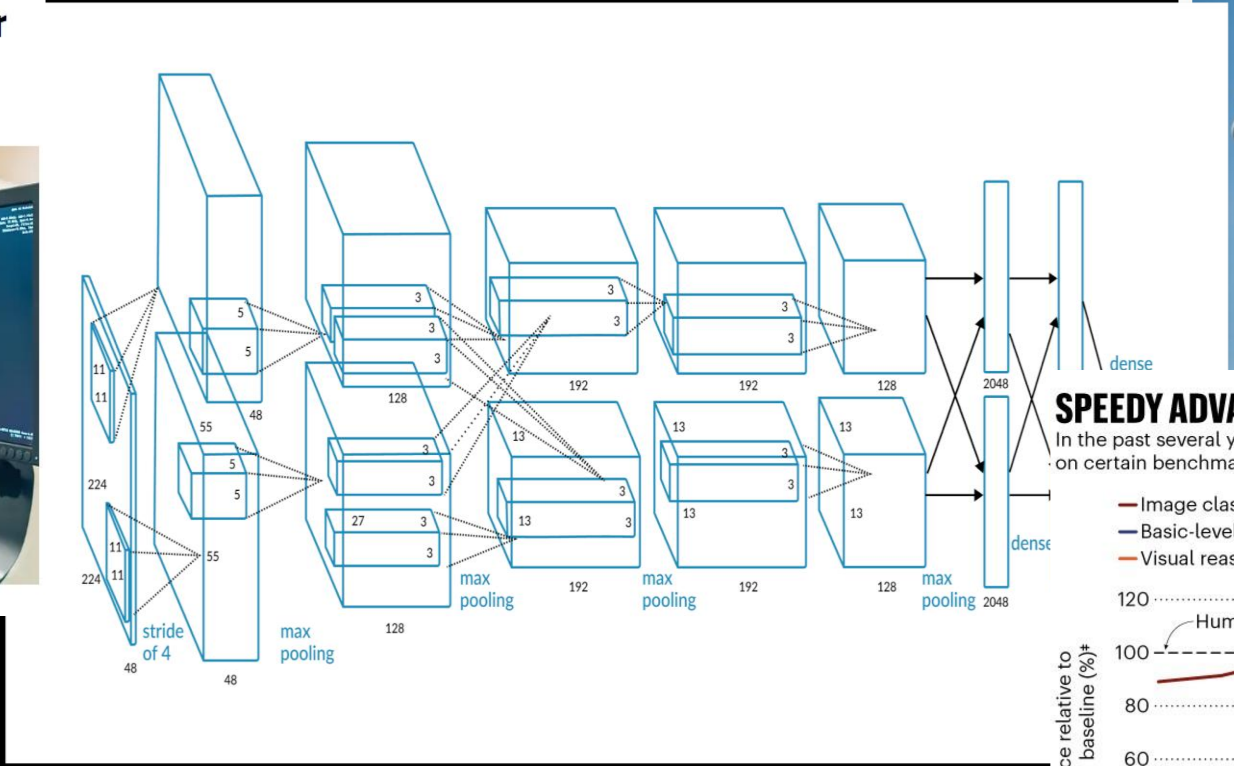
pixelfit

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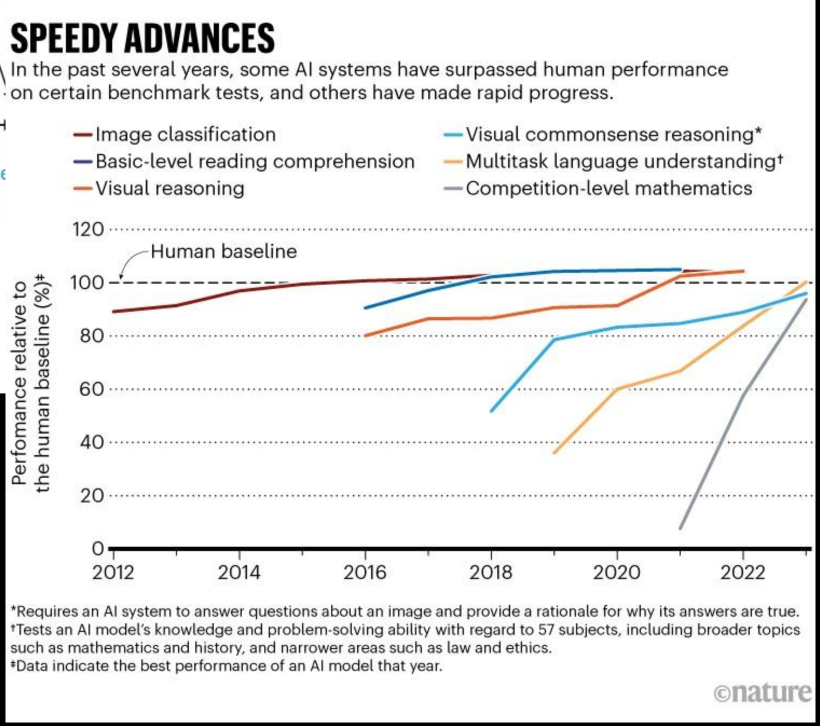
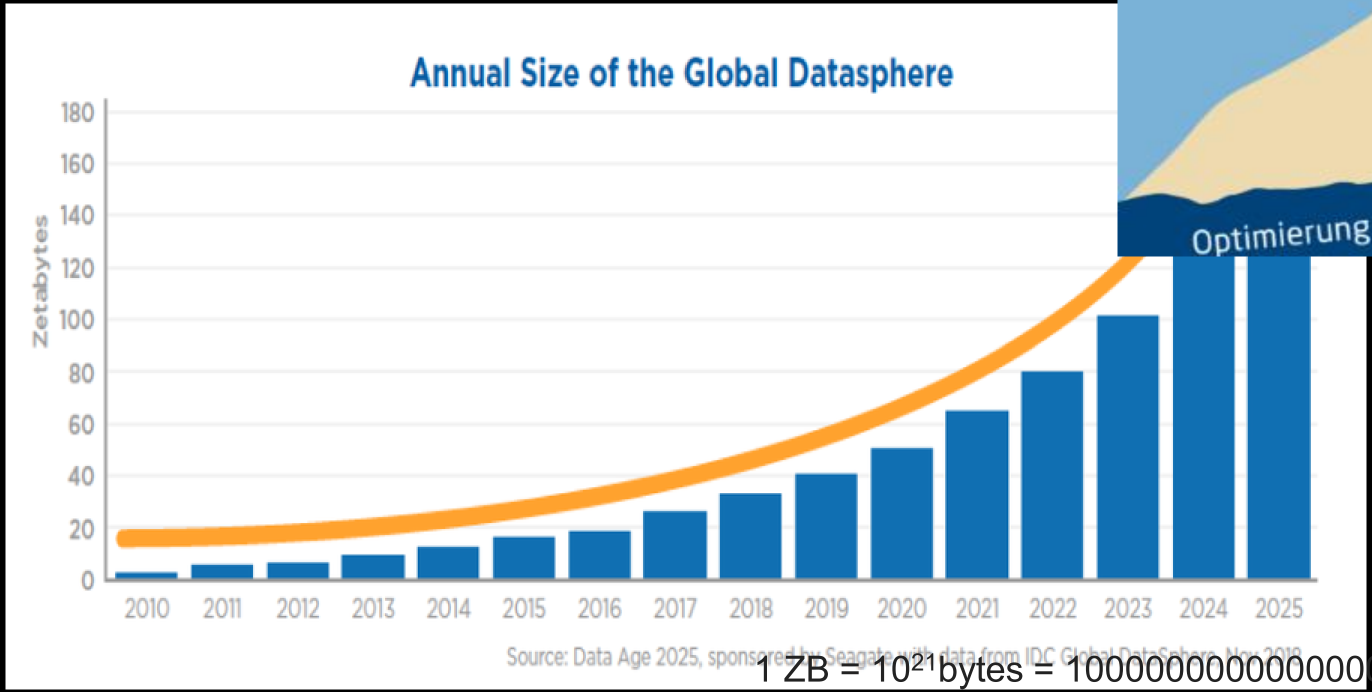
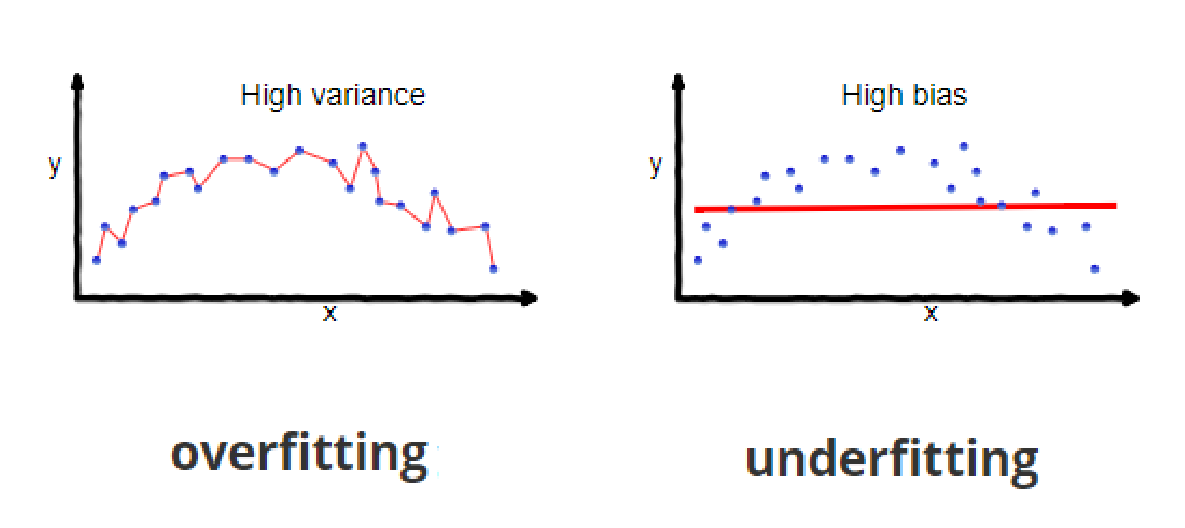


Big Data



1 ZB = 10^{21} bytes = 1000000000000000000000bytes

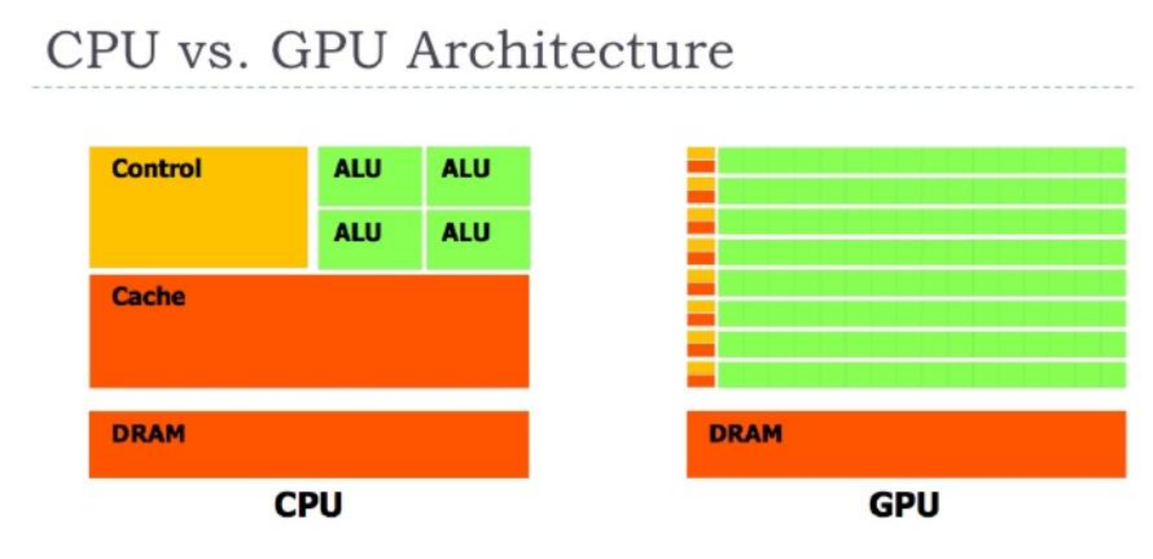
Andet store angreb på neurale netværk: Dilemmaet mellem bias og varians (1992)

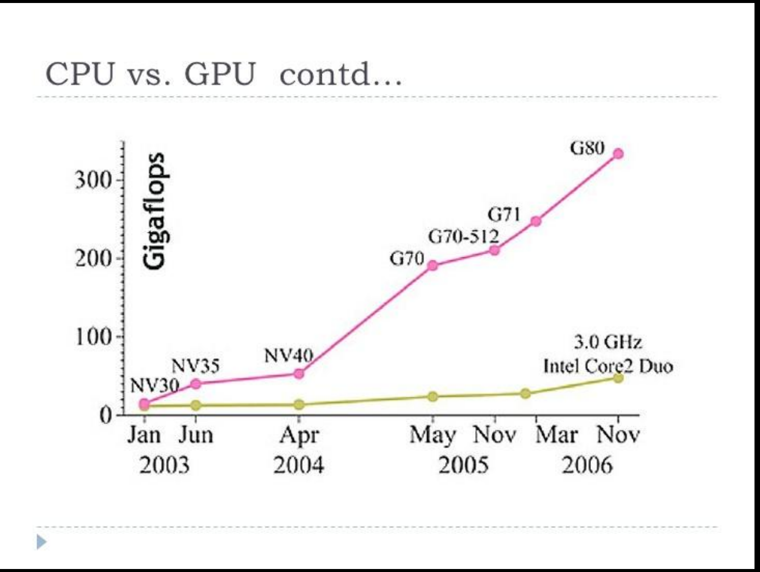


Geman, Stuart; E. Bienenstock; R. Doursat (1992). "Neural networks and the bias/variance dilemma". Neural Computation. 4: 1–58

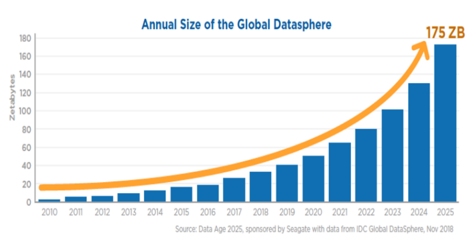
*Inferring this complexity from examples, that is, learning it, although theoretically achievable, is, for all practical matters, not feasible: **too many examples would be needed.***

Parallel Processing
Graphical Processing Units (GPUs)





Big Data er nøglen til at løse Bias/Variance dilemmaet



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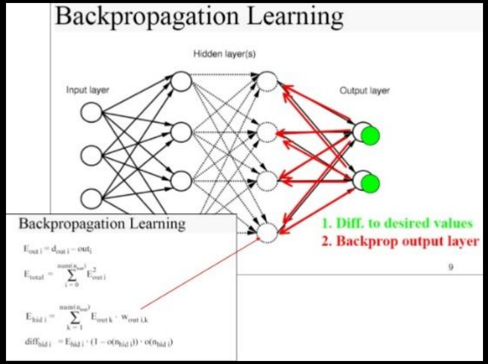

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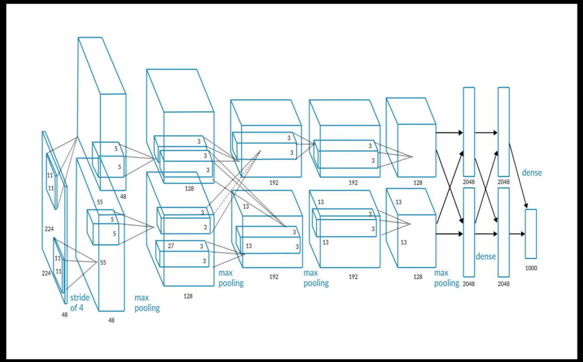
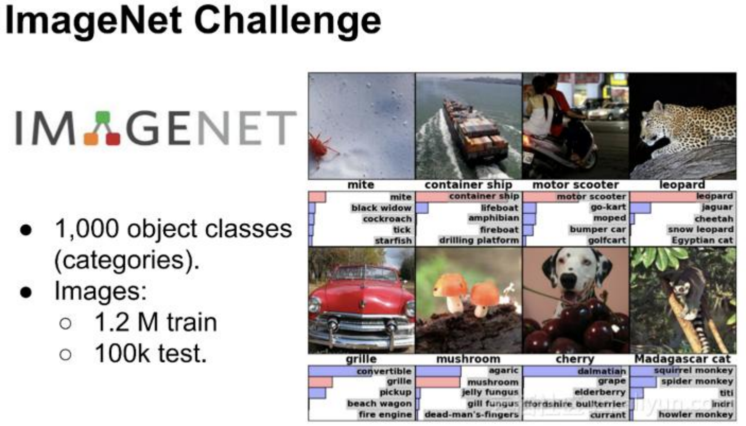


Geman et al (1992): Inferring this complexity from examples, that is, learning it, although theoretically achievable, is, for all practical matters, not feasible: ***too many examples would be needed.***

Mange succesrige applikationer med en dominerede metode: Deep Neural Networks



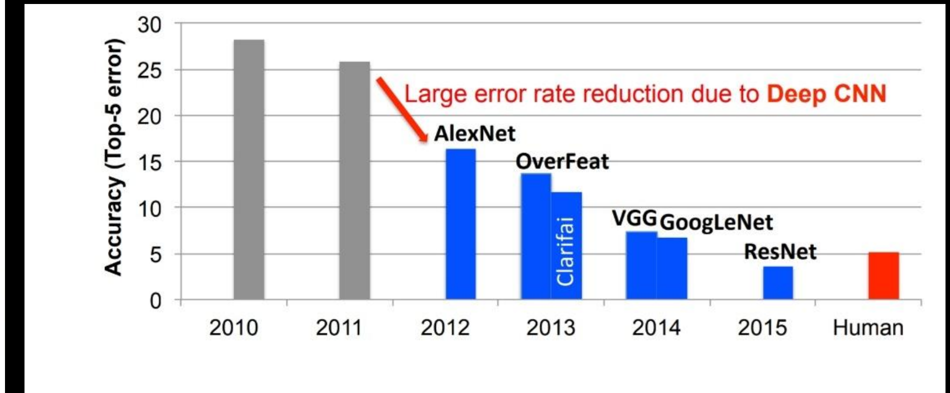


Fig. 7. Results from the ImageNet Challenge [14].

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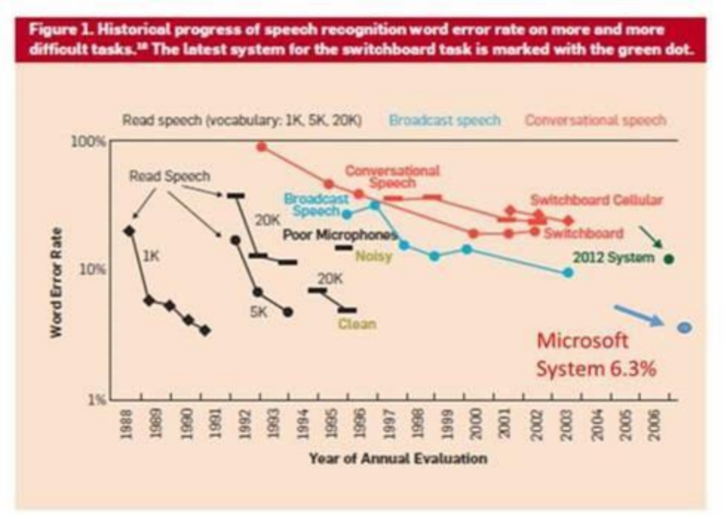

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13 April 2018

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(File Photo) There are an estimated 170 million CCTV cameras already in place in China

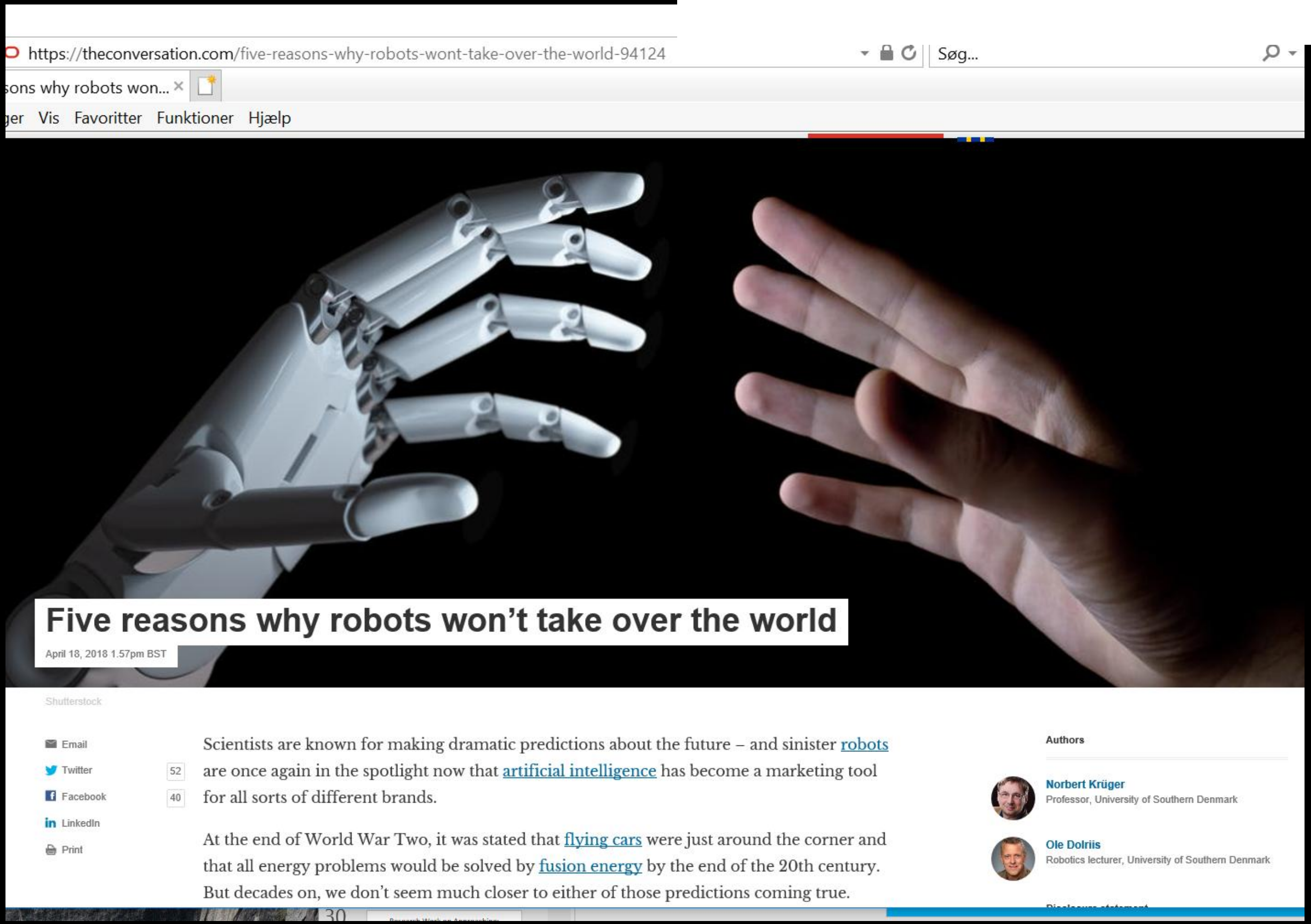
Five reasons why robots won't take over the world

theconversation.com (18.4.2018)

Norbert Krüger
Professor, University of Southern Denmark

Ole Dolriis
Robotics lecturer, University of Southern Denmark

- Missing human-like hands
- No reliable tactile perception
- Complexity of control in manipulation
- Complexities of human-robot interaction
- Human reason: Humans can always decide against using a certain technology



https://theconversation.com/five-reasons-why-robots-wont-take-over-the-world-94124

sons why robots won...

ger Vis Favoritter Funktioner Hjælp

Five reasons why robots won't take over the world

April 18, 2018 1.57pm BST

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Scientists are known for making dramatic predictions about the future – and sinister robots are once again in the spotlight now that artificial intelligence has become a marketing tool for all sorts of different brands.

At the end of World War Two, it was stated that flying cars were just around the corner and that all energy problems would be solved by fusion energy by the end of the 20th century. But decades on, we don't seem much closer to either of those predictions coming true.

Authors

Norbert Krüger
Professor, University of Southern Denmark

Ole Dolriis
Robotics lecturer, University of Southern Denmark

https://theconversation.com/five-reasons-why-robots-wont-take-over-the-world-94124

Robotteknologi

Autonomous cars

https://innovationatwork.ieee.org/autonomous-vehicles-for-today-and-for-the-future/

Cobots

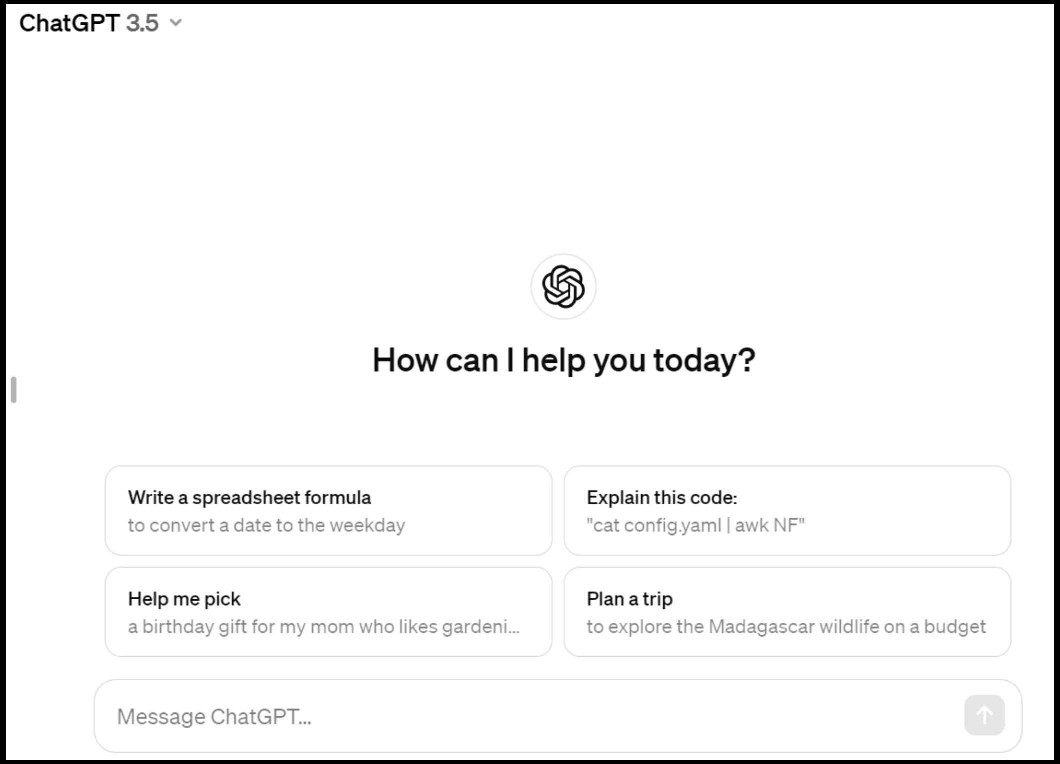
https://www.metal-supply.dk/article/view/202094/universal_robots_vokser_eksplosivt?ref=rss

Overblik

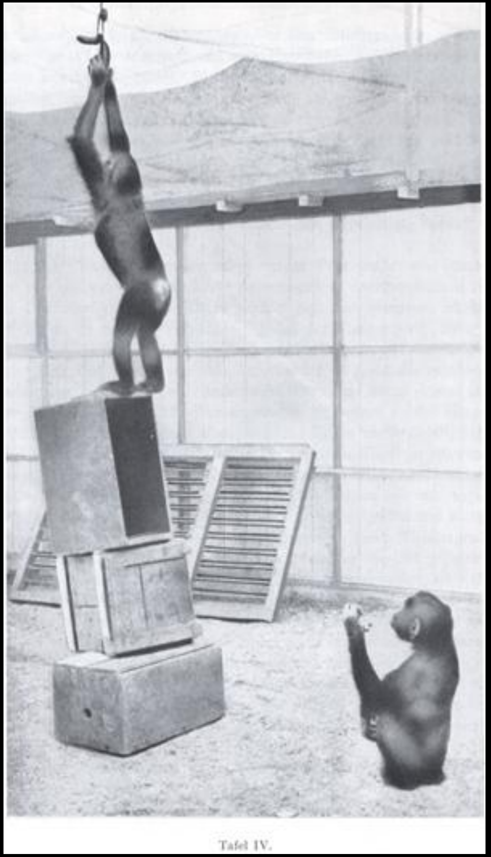
- Hvad er intelligens? Hvad er kunstig intelligens?
- Trekant: KI, Robot- og Computerteknologi
  - Indtil 2. Verdenskrig
  - 1940-1956: Begyndelsen
  - 1957-1984: Kampen mellem de symbolske og subsymbolske metoder
  - 1985-2005: Evolutionære fremskridt og Bias/Variance dilemmaet
  - 2005-: Deep Neural Networks tager over
- ChatGPT og embodiment

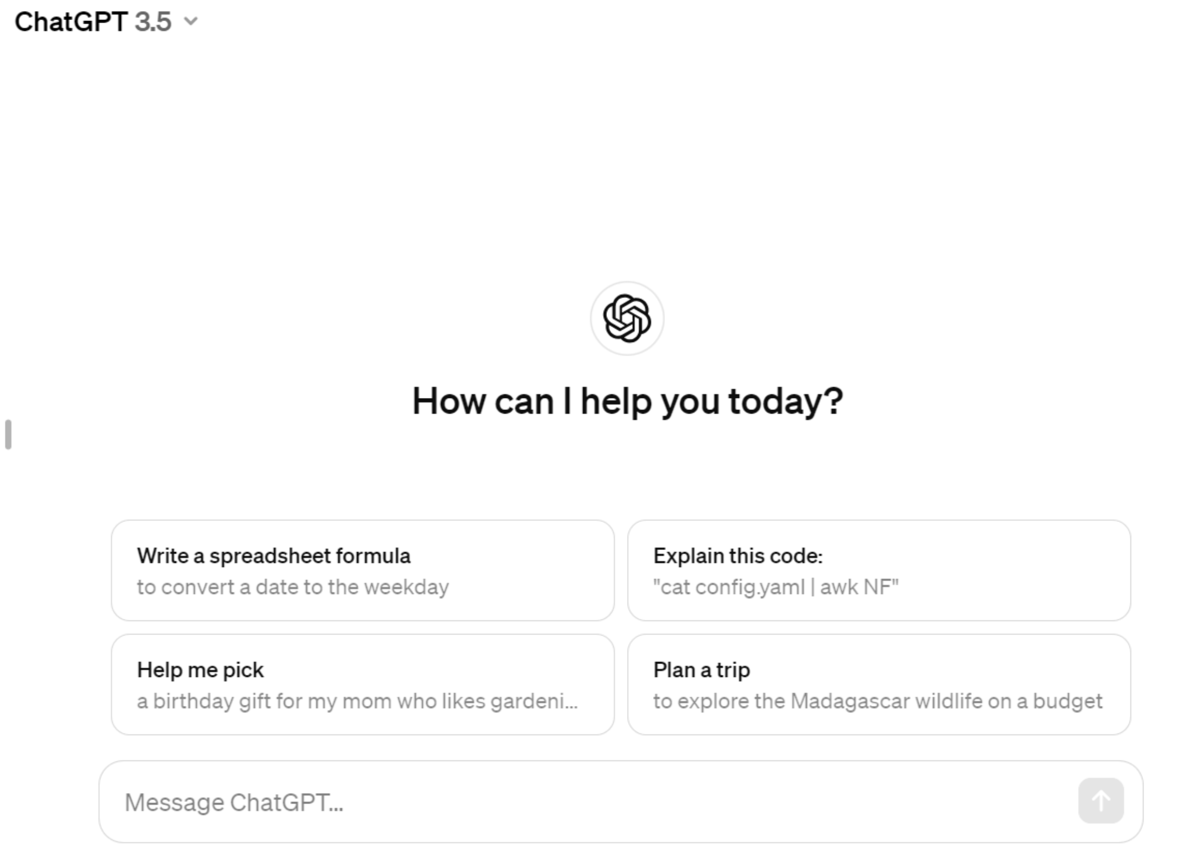
ChatGPT

- Den 30. november 2022 lancerede OpenAI ChatGPT (en forkortelse for "Chat Generative Pre-trained Transformer").
- ChatGPT
  - er i stand til at
    - besvare vanskelige akademiske spørgsmål (selvom det nogle gange også kan være helt forkert),
    - løse programmeringsopgaver
    - og skrive digte.
  - er baseret på Machine Learning.
  - har åbnet døren til applikationer, der hidtil ikke var set.



ChatGPT og Abe-Eksperimenter





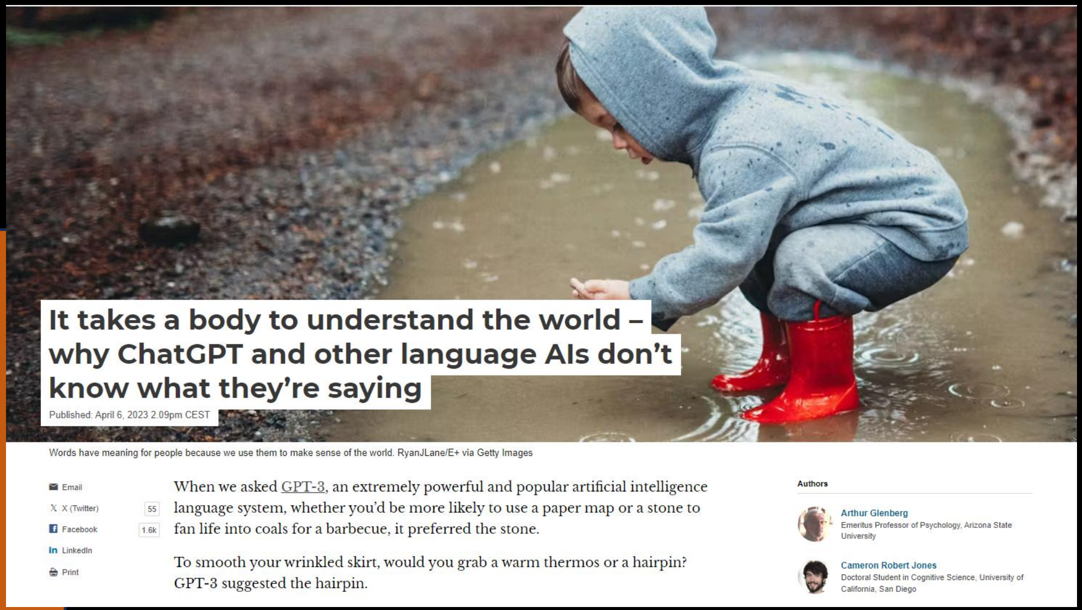
Köhler, Wolfgang (1963). Intelligenzprüfungen am Menschenaffen. Berlin, Göttingen, Heidelberg: Springer.

Limits to ChatGPT

Although GPT-3 is extremely good at learning the rules of what follows what in human language, it doesn’t have the foggiest idea what any of those words mean to a human being. And how could it?

Humans are biological entities that evolved with bodies that need to operate in the physical and social worlds to get things done. Language is a tool that helps people do that.

GPT-3 is an artificial software system that predicts the next word. It does not need to get anything done with those predictions in the real world.



It takes a body to understand the world – why ChatGPT and other language AIs don’t know what they’re saying

Published: April 6, 2023 2.09pm CEST

Words have meaning for people because we use them to make sense of the world. RyanJLane/E+ via Getty Images

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When we asked GPT-3, an extremely powerful and popular artificial intelligence language system, whether you’d be more likely to use a paper map or a stone to fan life into coals for a barbecue, it preferred the stone.

To smooth your wrinkled skirt, would you grab a warm thermos or a hairpin? GPT-3 suggested the hairpin.

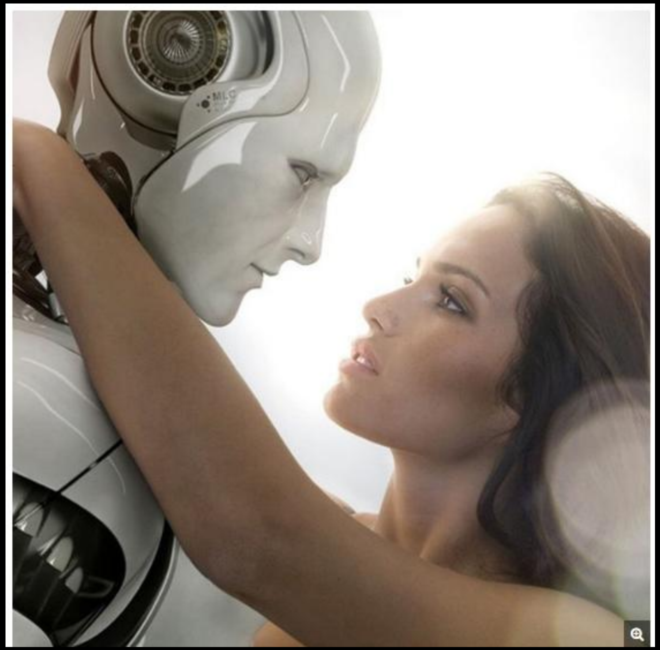
Authors

Arthur Glenberg
Emeritus Professor of Psychology, Arizona State University

Cameron Robert Jones
Doctoral Student in Cognitive Science, University of California, San Diego

Interagerende robotter

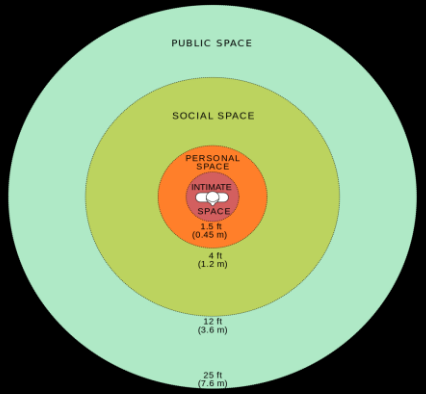
- Human Computer Interaction (HCI) er blevet stadig mere sofistikeret (ChatGPT)
- Meningsfuld interaktion mellem mennesker og kropsliggjorte agenter har endnu ikke vist sig at være mulig
- Men når dette problem er løst, vil kropsliggjorte agenter trænge ind i hverdagen



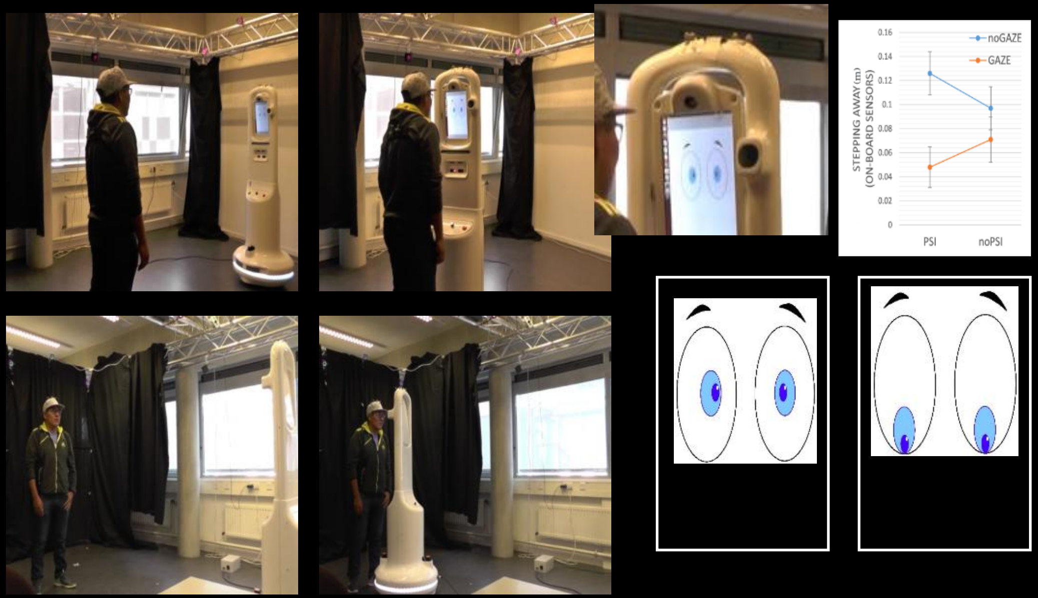
126

Video: Water delivery

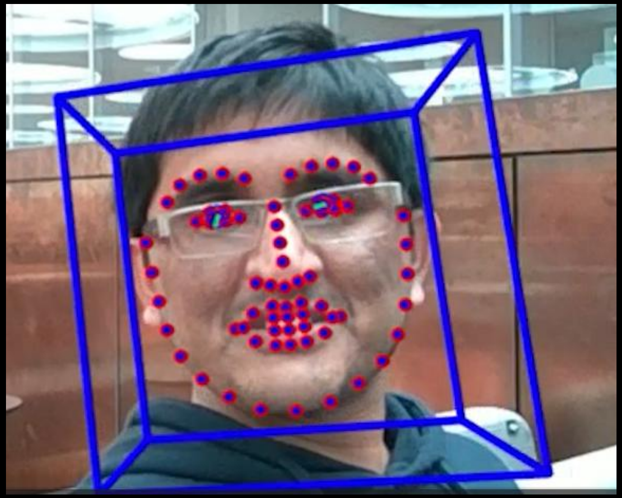
Problemer i HRI



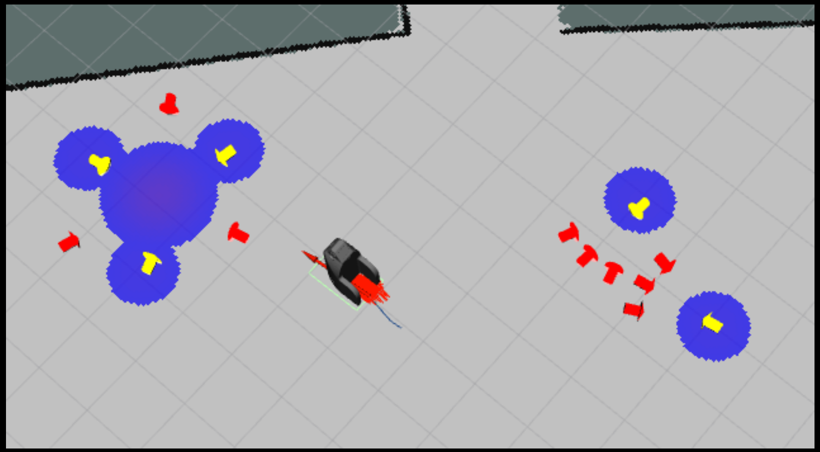
Håndtering af menneskers ”personlige rum”



Sociale Signaler



Opmærksomhed



Socially aware navigation

Personal assistants (with embodyment)

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3 Dec 2018 | 19:45 GMT

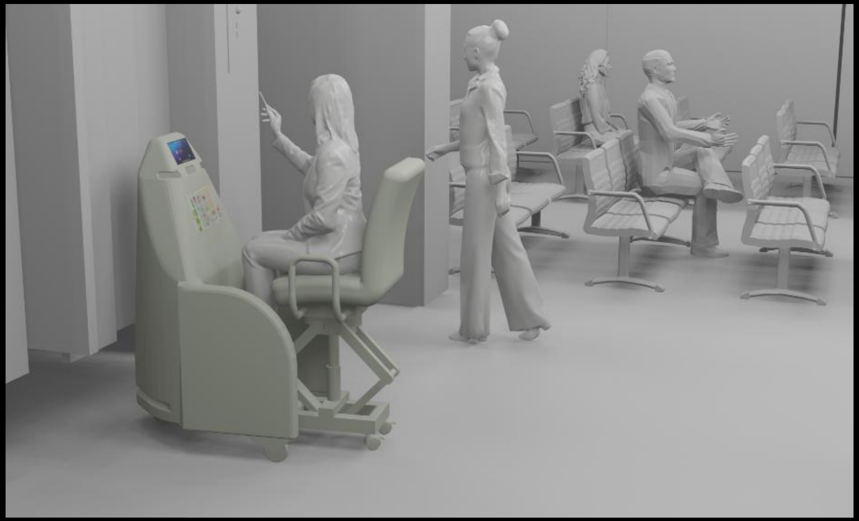
Jibo Is Probably Totally Dead Now

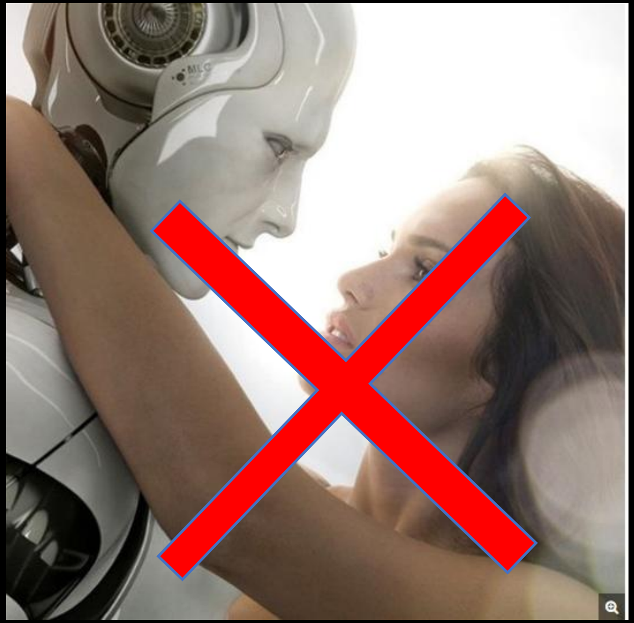
The pioneering social robot company has sold its IP and assets, meaning that this is pretty much the end

By **Evan Ackerman**

https://www.youtube.com/watch?v=1jTs9Rml6YQ

Et mere realistisk billede af fremtiden





Interagerende robotter i den virkelige verden






Robociety: Interacting Robots in Everyday Life and the Transformation of Society

DIAS Wicked Problems

The robots are coming - how do we make them a gift for society?

The Robociety project seeks to deepen our understanding of both robots and humans, with a strong focus on developing predictive models of human behavior to enable seamless interactions and foster positive societal acceptance.

Explore Robociety

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Superintelligens

En superintelligens er en hypotetisk agent, der besidder intelligens, der langt overgår de mest begavede menneskelige sind.

- I en undersøgelse af de 100 mest citerede forfattere inden for AI (pr. maj 2013 ifølge Microsofts akademiske søgning), var median-året, hvormed respondenterne forventede maskiner "der kan udføre de fleste menneskelige erhverv mindst lige så godt som et typisk menneske" (forudsat at der ikke sker nogen global katastrofe):
  - med 10% sikkerhed 2024
  - med 50% sikkerhed 2050
  - med 90% sikkerhed 2070.

https://en.wikipedia.org/wiki/Superintelligence

https://www.kdnuggets.com/beyond-human-boundaries-the-rise-of-superintelligence

Opgaven

- Læs of forstå teksten
  - Kun indledningen og bidragene af Animesh Garg og Aude Billard
- Alle hjælpermidler tilladt (men ikke i MCQ-testen)

SCIENCE ROBOTICS | VIEWPOINT

ARTIFICIAL INTELLIGENCE

"Data will solve robotics and automation: True or false?": A debate

Nancy M. Amato[1], Seth Hutchinson[2], Animesh Garg[3], Aude Billard[4], Daniela Rus[5], Russ Tedrake[5], Frank Park[6], Ken Goldberg[7]*

Leading researchers debate the long-term influence of model-free methods that use large sets of demonstration data to train numerical generative models to control robots.

INTRODUCTION

Nancy M. Amato, Seth Hutchinson, and Ken Goldberg

The robotics and automation research community is experiencing what historian of science Thomas Kuhn would call a "paradigm shift" (*1*). This was the subject of a featured debate at the IEEE International Conference on Robotics and Automation (ICRA), the largest research conference for the field, where many presentations celebrated algorithmic ("model-based") breakthroughs over the decades. However, since 2012, advances in deep "neural" networks that combine unprecedented amounts of example data, advances in stochastic gradient descent techniques, and advances in computing, in particular GPUs (graphics processing units), have produced remarkable results in computer vision and speech recognition. And with the emergence of ChatGPT and associated large vision language models (VLMs) in 2022, with associated advances in natural language processing and denoising diffusion methods, the paradigm of "end-to-end" (also known as "model-

As cochairs of the 2025 IEEE International Conference on Robotics and Automation (ICRA), held in Atlanta GA, USA, 26 to 29 May 2025, N.M.A. and S.H. decided to host a conference-wide keynote debate in the last session of the conference. To moderate the debate, we invited K.G., who has been very active in the past 40 years of model-based robotics and automation research and also active in robot learning (recently elected president of the Robot Learning Foundation). K.G. formulated a catchy title and short description: "Will the future of robotics and automation be written in code or in data? Is the *Handbook of Robotics* obsolete? Are home humanoids overhyped? Join us for a high-voltage debate about physics vs pixels, theory vs terabytes."

The three of us then worked to identify a panel of six debaters, inviting three to represent each side in the spirit of the classic Oxford debates. Six highly respected researchers accepted our invitation: A.G., D.R., and R.T. advocating for "true," and A.B., Leslie Kaebling (of MIT), and F.P. advocating for

to discuss how such massive datasets can be collected and processed.

Interestingly, most of the debaters conveyed a shared respect for hybrid methods that ultimately combine model-free and model-based methods. Ultimately, a show of hands indicated that not many audience members were persuaded to change their initial positions, but everyone was pushed to think more deeply about the new data paradigm.

DATA ARE NOT MERELY BENEFICIAL; RATHER, THEY ARE INDISPENSABLE AND FOUNDATIONAL

Animesh Garg

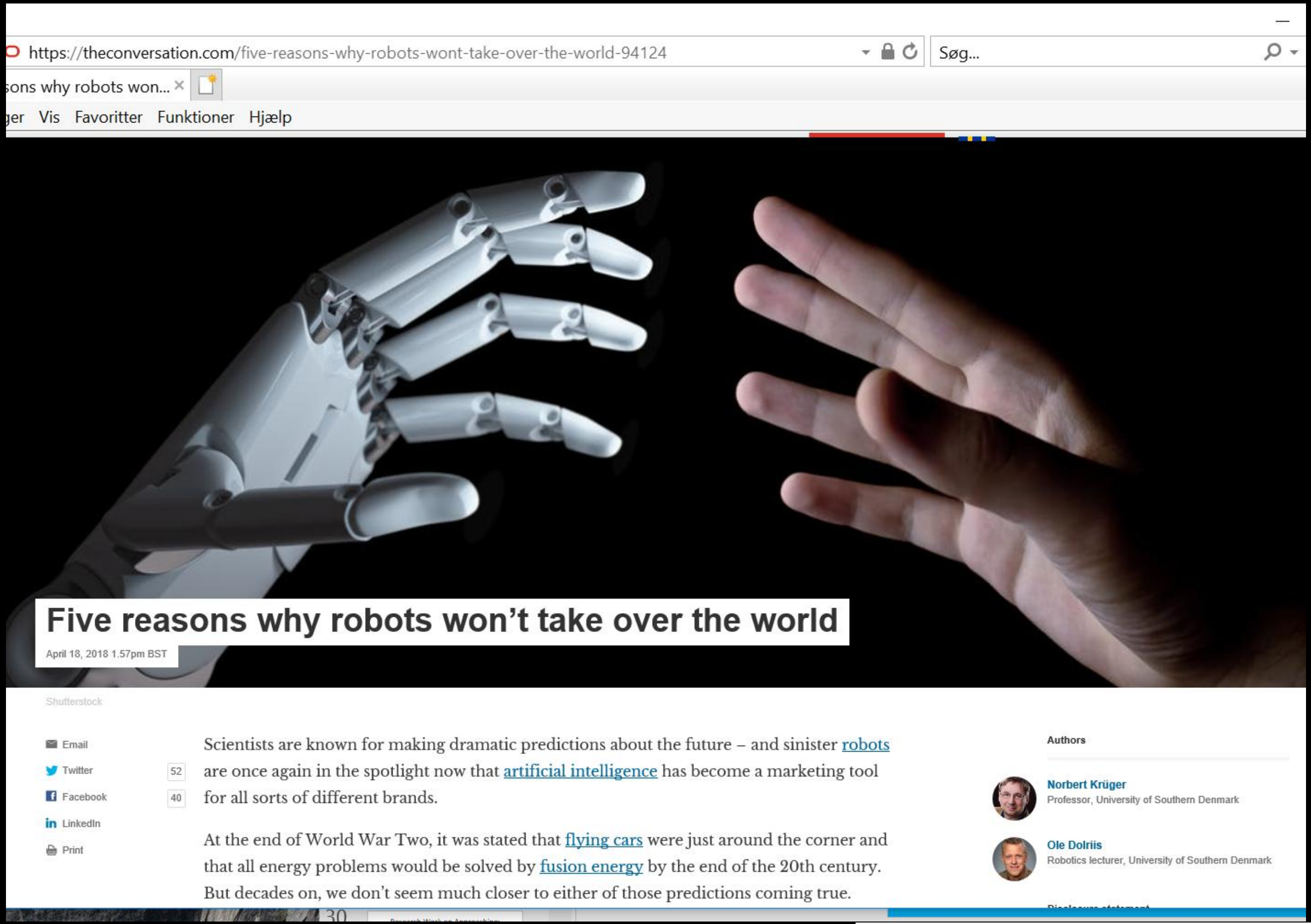
Humanity has frequently developed "know-how" before the "know-why." We engineered technologies mainly using observational understanding—such as metalworking, sailing ships, internal combustion engines, and airplanes (all foundational to civilization)—yet preceding a thorough understanding of their underlying sciences.

We are in the midst of building next-

Five reasons why robots won’t take over the world

theconversation.com (18.4.2018)

- Mangler menneskelignende hænder
- Ingen pålidelig taktil perception
- Kompleksitet i kontrol ved manipulation
- Kompleksitet i menneske-robot-interaction
- Mennesker kan altid beslutte sig for ikke at bruge en bestemt teknologi



https://theconversation.com/five-reasons-why-robots-wont-take-over-the-world-94124